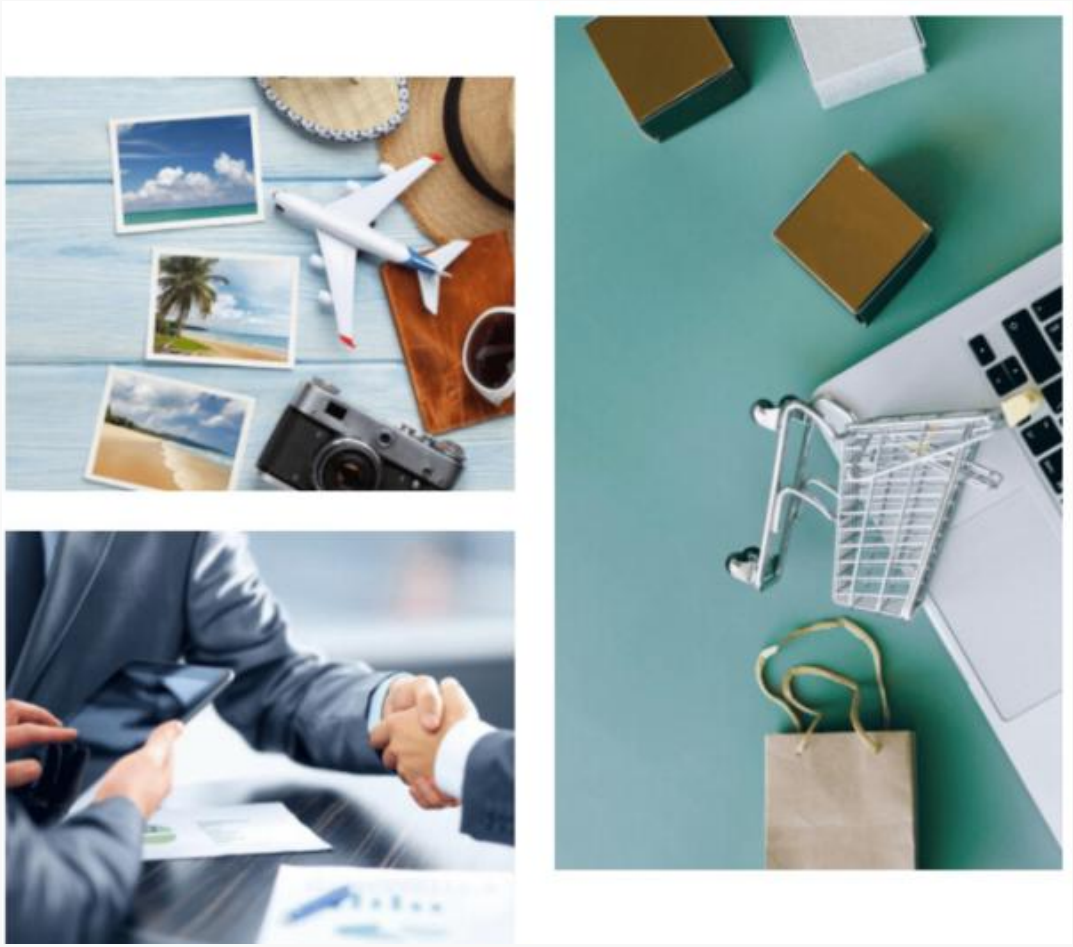


Brazilian E-Commerce, OLIT Sales & Customer Insights Analysis

Divyam Raj

Business Context

- Olist is one of Brazil's largest e-commerce marketplaces, connecting thousands of independent sellers with customers nationwide. As the business scales, managing performance across sales, customers, sellers, and delivery becomes increasingly critical
- The client has provided us with sales data from 2016 to 2018, spanning across 9 tables providing data of 99,441 customers, 3095 sellers & 32,951 products
- We are expected to provide data driven analysis of key metrics at product level, customer level, sales level, supplier level, etc.

Problem Statement

The client is facing problems seeing valuable insights which are hidden in the interconnected tables. They need our expertise in the following:

- Understand sales performance
- Improve customer satisfaction
- Monitor seller performance
- Reduce delivery delays
- Identify profitable categories
- Support data driven decisions

Objectives

- Analyze sales, customer behavior, seller performance, and delivery operations.
- Identify trends in orders, revenue, and product categories.
- Evaluate customer satisfaction using reviews, delivery time, and payment methods.
- Analyze regional sales performance across states and cities.
- Perform data cleaning and EDA
- Build interactive Power BI dashboards for business insights
- Develop a predictive model to forecast customer review scores.
- Provide actionable recommendations to improve business performance.

Tools & Technologies used

- **SQL Server**: Data Importing, Data Audit, Data Consistency Checks & Analysis.
- **Python**: Descriptive, Diagnostic & Predictive Analytics, Streamlit applications.
- **Power BI Desktop:** Visualization and Interactive Dashboard Development.
- **Power BI Service**: Publishing Dashboards for stakeholder access.
- **PowerPoint**: Presentation of findings, insights, and recommendations.

Review data sources (1/3)

Table Name	Total columns	No. of rows	Granularity	Primary key	Foreign key	Details
Customer	5	99,441	Customer level data	customer id	NA	• customer_id, customer_unique_id, , customer_city, customer_zip_code_prefix, customer_state • Total data given for 96,096 customers.
Orders	8	99,441	Order level data	order Id	customer id	• order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date • Total orders provided for 99,441 orders
Order payments	5	1,03,886	Order level data	NA	order Id	• order_id, payment_sequential, payment_type, payment_installments, payment_value • Total data provided for 1,03,886 orders, with 5 payment types

Review data sources (2/3)

Table Name	Total columns	No. of rows	Granularity	Primary key	Foreign key	Details
Geolocation	5	10,00,163	Location level data	geolocation zip code prefix	NA	• geolocation_zip_code_prefix, geolocation_lat, geolocation_lng, geolocation_state, geolocation_city • Total locations provided for 19,015 locations, 8,010 cities & 27 states
Orders Items	7	1,12,650	Order level data	NA	order id, product Id, Seller id	• order_id, order_item_id, product_id, seller_id, shipping_limit_date, price, freight_value • Total orders provided for 98,666 order Ids, 3,095 sellers & 32,951 products
Order Reviews	7	99,224	Order level data	review id	order Id	• review id ,order_id, review_score, review_comment_title, review_comment_message, review_creation_date, review_answer_timestamp • Total review ids are 98,410 giving rating between 1-5.

Review data sources (3/3)

Table Name	Total columns	No. of rows	Granularity	Primary key	Foreign key	Details
Product category name translation	2	72	Product level data	NA	NA	• product_category_name, product_category_name_english • Total name translations provided for 72 products from Latin to English
Products	9	32,951	Product level data	product_id	NA	• product_id, category_name, product description length, product photos quantity, product weight & product dimensions • Products in total, 32,951.
Sellers	4	3,095	Seller level data	seller id	NA	• seller_id, seller_zip_code_prefix, seller_city, seller_state • Total 3,095 sellers from 611 cities & 23 states.



Data Dictionary

Data Dictionary (Customer)

Column Name	Data Type	Field Length	Constraint	Description
customer_id	nvarchar	50	Primary Key	Customer id generated for each order. Auto-generated
customer_unique_id	nvarchar	50	Not Null	Unique customer id provided to each customer at the time of account creation. One unique id per customer.
customer_zip_code_prefix	int	-	Not Null	Zip code for customer city
customer_city	nvarchar	50	Not Null	Customer City in Brazil
customer_state	nvarchar	50	Not Null	Customer state in Brazil

Data Dictionary (Geolocation)

Column Name	Data Type	Field Length	Constraint	Description
geolocation_zip_code_prefix	int	-	Primary Key	Zip code for cities.
geolocation_lat	float	-	-	Location Latitude
geolocation_lng	int	-	-	Location Longitude
geolocation_city	nvarchar	50	Not Null	City names in Brazil
geolocation_state	nvarchar	50	Not Null	State names in Brazil

Data Dictionary (order_items)

Column Name	Data Type	Field Length	Constraint	Description
order_id	nvarchar	50	Foreign Key	Order id for items that have been bought per customer.
order_item_id	tinyint	-	Not Null	Sequential line number of an item within a single order.
product_id	nvarchar	50	Foreign Key	Unique product id for each product.
seller_id	nvarchar	50	Foreign Key	Unique seller id for each seller.
shipping_limit_date	datetime	-	Not Null	Maximum date before which the order should be shipped.
price	float	-	Not Null	Price per item in Brazilian Real
freight_value	float	-	Not Null	Shipping or delivery cost per item/product

Data Dictionary (order_payments)

Column Name	Data Type	Field Length	Constraint	Description
order_id	nvarchar	50	Foreign Key	Order id linking back to orders table providing the unique id for each order.
payment_sequential	tinyint	-	Not Null	Sequential line number for a payment per order.
payment_type	nvarchar	50	Not Null	Payment method used for making the purchase.
payment_instalments	tinyint	-	Not Null	Number of parts in which a payment was made.
payment_value	float	-	Not Null	Total amount received per order.

Data Disctionary (order_reviews)

Column Name	Data Type	Field Length	Constraint	Description
review_id	nvarchar	50	Primary Key	Unique id for each review.
order_id	nvarchar	50	Foreign Key	Unique id per order.
review_score	tinyint	-	Not Null	Customer rating between 1-5, 1 being lowest score and 5 being highest.
review_comment_title	nvarchar	50	-	Review title for the feedback of the order.
review_comment_message	nvarchar	250	-	Detailed review provided by the customer.
review_creation_date	datetime	-	Not Null	Date on which the company created the review link.
review_answer_timestamp	datetime	-	Not Null	Date & time when the customer answered the review.

Data Dictionary (orders)

Column Name	Data Type	Field Length	Constraint	Description
order_id	nvarchar	50	Primary Key	Order id for items that have been bought per customer.
customer_id	nvarchar	50	Foreign Key	Unique customer id per order for each customer.
order_status	nvarchar	50	Not Null	Order status about the current position of the order.
order_purchase_timestamp	datetime	-	Not Null	Date and time on which the order was placed.
order_approved_at	datetime	-	-	Date & time when the order got approved by the seller.
order_delivered_carrier_date	datetime	-	-	Date & time when order got delivered to carrier
Order_delivered_customer_date	datetime	-	-	Date & time when customer received the order.
order_estimated_delivery_date	datetime	-	-	Date & time of expected delivery date for the order.

Data Dictionary (product_category_name_translation)

Column Name	Data Type	Field Length	Constraint	Description
product_category_name	nvarchar	50	-	Product category name in Latin.
product_category_name_english	nvarchar	50	-	Product category name in English.

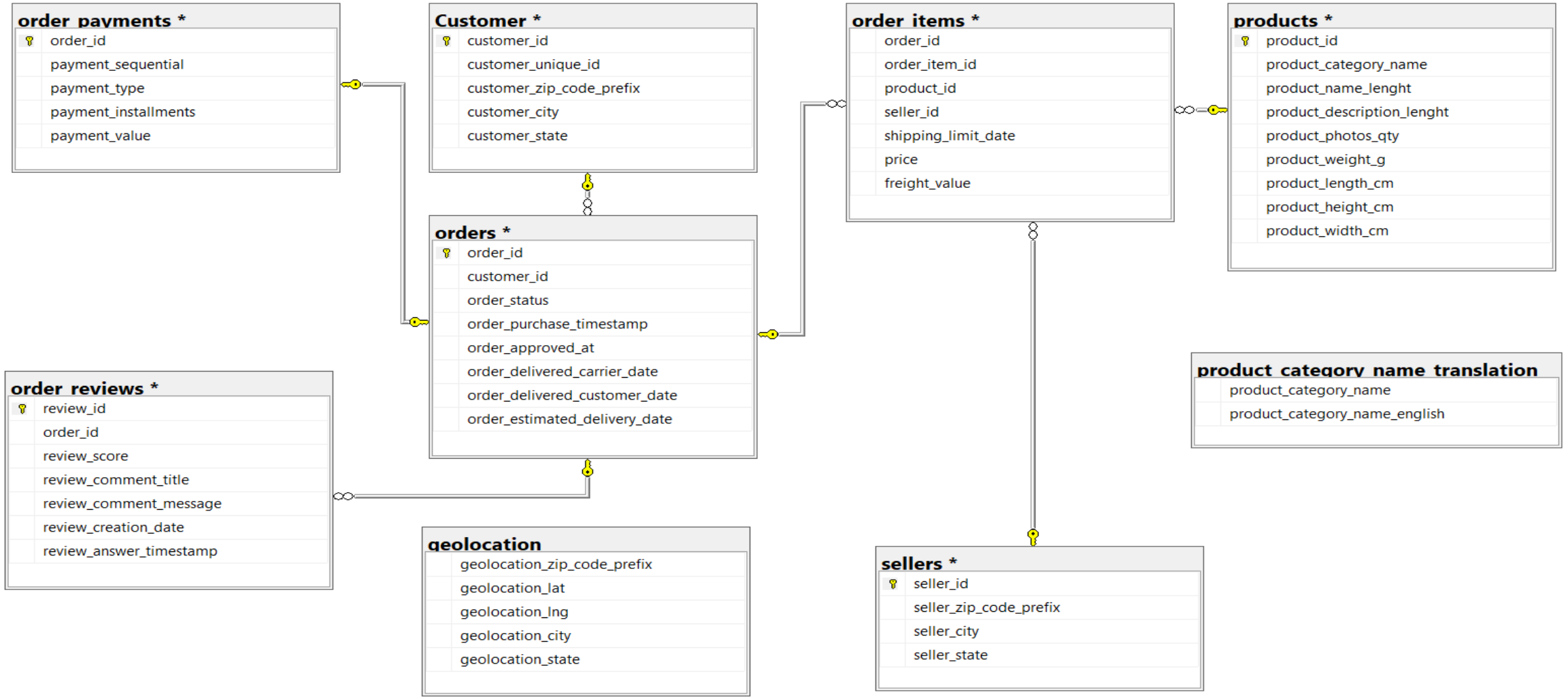
Data Dictionary (products)

Column Name	Data Type	Field Length	Constraint	Description
product_id	nvarchar	50	Primary Key	Unique id per product.
product_category_name	nvarchar	50	-	Product category name per product id
product_name_length	tinyint	-	-	Number of character in the name title of the product
product_description_length	int	-	-	Number of characters in the product description.
products_photos_qty	int	-	-	Number of photos uploaded for each product.
product_weight_g	int	-	-	Weight of product in grams.
product_length_cm	tinyint	-	-	Length of the products in centimeters.
product_height_cm	tinyint	-	-	Height of the products in centimeters.
product_width_cm	tinyint	-	-	Width of the product in centimeters.

Data Dictionary (sellers)

Column Name	Data Type	Field Length	Constraint	Description
seller_id	nvarchar	50	Primary Key	Unique identification key for each seller.
seller_zip_code_prefix	int	-	Not Null	Zip code from where each seller is.
seller_city	nvarchar	50	-	City name of the seller.
seller_state	nvarchar	50	-	State name of the seller.

Entity relationship diagram



Initial Observations

- One customer has multiple orders
- One order id has multiple payment methods (Credit card & Vouchers)
- One order has multiple products under the same order id
- Certain city names are repeated with different spellings each in Latin and English
- Single order id has multiple seller ids, indicating different products are being sold to one customer through multiple sellers.
- Each customer has a unique customer id which is created at the time of account creation, and customer_ids are created for each order.

Data discrepancies (1/7)

❖ Data discrepancies in “order_payments” Table

	order_id	payment_sequential	payment_type	payment_installments	payment_value
1	0016dfedd97fc2950e388d2971d718c7	1	credit_card	5	52.6300010681152
2	0016dfedd97fc2950e388d2971d718c7	2	voucher	1	17.9200000762939
3	002f19a65a2ddd70a090297872e6d64e	1	voucher	1	44.1100006103516
4	002f19a65a2ddd70a090297872e6d64e	2	voucher	1	33.1800003051758
5	0071ee2429bc1efdc43aa3e073a5290e	1	voucher	1	100
6	0071ee2429bc1efdc43aa3e073a5290e	2	voucher	1	92.4400024414063

- Same order_id has multiple payments with different payment types and values
- There are payment_type which are mentioned as “not_defined”

Data discrepancies (2/7)

❖ Data discrepancies in “geolocation” Table

	geolocation_zip_code_prefix	geolocation_lat	geolocation_lng	geolocation_city	geolocation_state
22	1001	-23.5497798919678	-46.6339569091797	sao paulo	SP
23	1001	-23.5492916107178	-46.6335601806641	sao paulo	SP
24	1001	-23.5499515533447	-46.6340255737305	são paulo	SP
25	1001	-23.5492916107178	-46.6335601806641	sao paulo	SP
26	1001	-23.5492916107178	-46.6335601806641	sao paulo	SP
27	1002	-23.5485515594482	-46.6350708007813	sao paulo	SP
28	1002	-23.544641494751	-46.6331787109375	são paulo	SP
29	1002	-23.5485515594482	-46.6350708007813	sao paulo	SP
30	1002	-23.5485515594482	-46.6350708007813	sao paulo	SP

- Same geolocation_zip_code_prefix is mentioned for same geological_city & geological_state with slight variation in latitudes and longitudes.
- Some geological_city have spelling variation while still representing the same city

Data discrepancies (3/7)

❖ Data discrepancies in “products” Table

	product_id	product_category_name	product_name_lenght	product_description_lenght	product_photos_qty
1	a41e356c76fab66334f36de622ecbd3a	NULL	NULL	NULL	NULL
2	d8dee61c2034d6d075997acef1870e9b	NULL	NULL	NULL	NULL
3	56139431d72cd51f19eb9f7dae4d1617	NULL	NULL	NULL	NULL
4	46b48281eb6d663ced748f324108c733	NULL	NULL	NULL	NULL
5	5fb61f482620cb672f5e586bb132eae9	NULL	NULL	NULL	NULL
6	e10758160da97891c2fdcbc35f0f031d	NULL	NULL	NULL	NULL
7	39e3b9b12cd0bf8ee681bbc1c130feb5	NULL	NULL	NULL	NULL

- 610 product_ids do not have product_category_name & product descriptions.
- There are no translations available for the same 610 records.

Data discrepancies (4/7)

❖ Data discrepancies in “order_reviews” Table

	review_id	order_id	review_score
7	017f0e1ea6386de662cbeba299c59ad1	02355020fd0a40a0d56df9f6ff060413	1
8	0c8e7347f1cdd2aede37371543e3d163	02355020fd0a40a0d56df9f6ff060413	3
9	61fe4e7d1ae801bbe169eb67b86c6eda	029863af4b968de1e5d6a82782e662f5	4
10	04d945e95c788a3aa1ffbee42105637b	029863af4b968de1e5d6a82782e662f5	5

- There are around 1089 entries where one order_id has multiple review_ids with different review_scores.
- For majority of reviews, there is no review comment title & no review comment description.

Data discrepancies (5/7)

❖ Data discrepancies in “orders” Table

	order_id	customer_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date	order_delivered_customer_date	order_estimated_delivery_date
1	e04abd8149ef81b95221e88f6ed9ab6a	2127dc6603ac33544953ef05ec155771	delivered	2017-02-18 14:40:00.0000000	NULL	2017-02-23 12:04:00.0000000	2017-03-01 13:25:00.0000000	2017-03-17 00:00:00.0000000
2	8a9adc69528e1001fc68dd0aaebbb54a	4c1ccc74e00993733742a3c786dc3c1f	delivered	2017-02-18 12:45:00.0000000	NULL	2017-02-23 09:01:00.0000000	2017-03-02 10:05:00.0000000	2017-03-21 00:00:00.0000000
3	7013bcfc1c97fe719a7b5e05e61c12db	2941af76d38100e0f8740a374f1a5dc3	delivered	2017-02-18 13:29:00.0000000	NULL	2017-02-22 16:25:00.0000000	2017-03-01 08:07:00.0000000	2017-03-17 00:00:00.0000000
4	5cf925b116421afa85ee25e99b4c34fb	29c35fc91fc13fb5073c8f30505d860d	delivered	2017-02-18 16:48:00.0000000	NULL	2017-02-22 11:23:00.0000000	2017-03-09 07:28:00.0000000	2017-03-31 00:00:00.0000000
5	12a95a3c06dbaec84bcfb0e2da5d228a	1e101e0daffaddce8159d25a8e53f2b2	delivered	2017-02-17 13:05:00.0000000	NULL	2017-02-22 11:23:00.0000000	2017-03-02 11:09:00.0000000	2017-03-20 00:00:00.0000000

	order_id	customer_id	order_purchase_timestamp	order_delivered_carrier_date	order_delivered_customer_date
1	a1abeb653a4d4cd1e142ccb8c82cd069	5f50465da00b7fed5dd1239f4ecf6e2c	2017-07-20 11:20:00.0000000	2017-07-28 16:57:00.0000000	2017-07-25 19:32:00.0000000
2	383aa8b2724fe452d9ccd9934a8c628b	b1cb2f9d7a19480f3749e248db14d58f	2017-07-02 20:58:00.0000000	2017-07-07 17:22:00.0000000	2017-07-06 14:27:00.0000000
3	cb1134f9010d242e9515ad1c78ec0c39	2fd33ac77677bd214b1882868317eeed	2017-07-16 12:35:00.0000000	2017-07-20 19:22:00.0000000	2017-07-19 14:13:00.0000000
4	dceb62e8fa94b46006c9554fed743df0	2721900eb4e0f1cc2c836dd7bc1b1e11	2017-07-20 20:58:00.0000000	2017-08-01 18:23:00.0000000	2017-07-26 18:09:00.0000000
5	5f9d46795c3126674e52becb3a1a517f	79287bcaafdde5c793b996fc40bb7d9f	2017-07-18 11:48:00.0000000	2017-07-20 23:03:00.0000000	2017-07-20 18:52:00.0000000
6	8c78d01de3a9009e23d6877a7cc9be20	6cd7106899e59a1fbd0622d5f1efedf4	2016-10-08 15:36:00.0000000	2016-10-26 11:41:00.0000000	2016-10-25 17:51:00.0000000
7	b27af682321527a6349f1761eb3f360c	9859dd92e872dbaa60ca3cd5f0d7ad07	2017-06-14 20:17:00.0000000	2017-06-27 14:51:00.0000000	2017-06-26 15:45:00.0000000

- 14 records in orders table have missing “order_approved_at” despite being delivered or shipped.
- Few orders have been delivered to customers before it got delivered to the carrier. There are 23 such records.
- There are null values present in “order_delivered_customer_date” and “order_delivered_carrier_date”.

Data discrepancies (6/7)

	order_id	customer_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date
1	2aa91108853cecb43c84a5dc5b277475	afeb16c7f46396c0ed54acb45ccaaa40	delivered	2017-09-29 08:52:00.0000000	2017-09-29 09:07:00.0000000	NULL
2	2d858f451373b04fb5c984a1cc2defaf	e08caf668d499a6d643dafd7c5cc498a	delivered	2017-05-25 23:22:00.0000000	2017-05-25 23:30:00.0000000	NULL

- There are two records which have missing “order_delivered_carrier_date” despite being delivered.

	order_id	customer_id	order_status	order_purchase_timestamp	order_delivered_customer_date
1	2d1e2d5bf4dc7227b3bfebb81328c15f	ec05a6d8558c6455f0cbbd8a420ad34f	delivered	2017-11-28 17:44:00.0000000	NULL
2	f5dd62b788049ad9fc0526e3ad11a097	5e89028e024b381dc84a13a3570decb4	delivered	2018-06-20 06:58:00.0000000	NULL
3	2ebdfc4f15f23b91474edf87475f108e	29f0540231702fda0cfdee0a310f11aa	delivered	2018-07-01 17:05:00.0000000	NULL
4	e69f75a717d64fc5ecdfae42b2e8e086	cfda40ca8dd0a5d486a9635b611b398a	delivered	2018-07-01 22:05:00.0000000	NULL
5	0d3268bad9b086af767785e3f0fc0133	4f1d63d35fb7c8999853b2699f5c7649	delivered	2018-07-01 21:14:00.0000000	NULL
6	2d858f451373b04fb5c984a1cc2defaf	e08caf668d499a6d643dafd7c5cc498a	delivered	2017-05-25 23:22:00.0000000	NULL
7	ab7c89dc1bf4a1ead9d6ec1ec8968a84	dd1b84a7286eb4524d52af4256c0ba24	delivered	2018-06-08 12:09:00.0000000	NULL
8	20edc82cf5400ce95e1afacc25798b31	28c37425f1127d887d7337f284080a0f	delivered	2018-06-27 16:09:00.0000000	NULL

- 8 records have null “order_delivered_customer_date” despite being delivered.

- There are total 1381 records which have date time inconsistency where the predecessor event’s date is greater than the successor event. This is roughly 1.38% of total records.

Data discrepancies (7/7)

❖ Data discrepancies in “geolocation” Table

- There are 1,28,270 entries which are exact duplicates.
- Same “geolocation_zip_code” has multiple latitudes & longitudes with minor variations which would create multiple extra rows if a join is performed without cleaning.
- “geolocation_city” has different spellings for the same city in Latin or English.
- Date time values need to be converted to proper formats.



Data Cleaning

Data cleaning recommendations (1/6)

- “order_payments” Table
  - Convert “payment_type” column to wide format to accommodate multiple payments into one row, this will remove duplicate “order_ids” and provide better insights.
  - Group by “order_id” and aggregate by sum of payment_value.
  - Cleaned “order_payments” table

	order_id	tot_orders	Total_installments	credit_card	boleto	voucher	debit_card	others	total_amount
1	00010242fe8c5a6d1ba2dd792cb16214	1	2	72.19	0	0	0	0	72.19
2	00018f77f2f0320c557190d7a144bdd3	1	3	259.83	0	0	0	0	259.83
3	000229ec398224ef6ca0657da4fc703e	1	5	216.87	0	0	0	0	216.87
4	00024acbcdf0a6daa1e931b038114c75	1	2	25.78	0	0	0	0	25.78
5	00042b26cf59d7ce69dfabb4e55b4fd9	1	3	218.04	0	0	0	0	218.04
6	00048cc3ae777c65dbb7d2a0634bc1ea	1	1	0	34.59	0	0	0	34.59
7	00054e8431b9d7675808bcb819fb4a32	1	1	31.75	0	0	0	0	31.75
8	000576fe39319847cbb9d288c5617fa6	1	10	880.75	0	0	0	0	880.75

Data cleaning recommendations (2/6)

❖ “orders” Table

- Create a inconsistent flag for date time values which are in the wrong order and filter them out during analysis or do the next mentioned cleaning in separate table.

- It is normal to have null values for "order_delivered_carrier_date","order_delivered_customer_date" and "order_approved_at" for cancelled orders and no imputation is needed.

- For "shipped" order_status, it is expected to have null "order_delivered_customer_date". We can segregate them into a different table for analysis purpose. It is advised to keep the null values as it shows valuable insights.

- Orders that are "unavailable","processing" and "invoiced" will have null value for "order_delivered_carrier_date“ and "order_delivered_customer_date". We should keep them as null for analysis of KPIs and trends related to incomplete or cancelled orders.

- Calculate median time in hours it takes for an order to get approved, then add that to the "order_purchase_timestamp" to impute missing values in “order_approved_at”. And do the same for the rest date time inconsistencies.

Data cleaning recommendations (3/6)

- for orders which have delivery date before approved date, we need to confirm with the client for correct data or assuming these values got exchanged, we can interchange these values to get them in the correct orders.

- For orders missing “order_delivered_carrier_date”, we will calculate average time it takes from an order to be delivered to the carrier and add that to our “order_approved_at”

- Same approach will be used for imputing “order_delivered_customer_date”. We will calculate average time in hours or days it takes for an order to be delivered to a customer after being delivered to a carrier grouped by state or seller then add it to the carrier delivered date.

Data cleaning recommendations (4/6)

❖ “geolocation” Table

- Average “geolocation_lat” & “geolocation_lng” and group them on the zip codes.
- Use max() window function to group rows by “geological_zip_code_prefix” to have consistent spellings for “geological_city” which gives record of 19,015 zip_codes with one spelling for each “geological_city”.
- Change the spellings to English for better understanding.
- Cleaned “geolocation” table

	geolocation_zip_code_prefix	geolocation_city	geolocation_state	geolocation_lat	geolocation_lng
0	1001	sao paulo	SP	-23.550271	-46.634047
1	1002	sao paulo	SP	-23.547657	-46.634991
2	1003	sao paulo	SP	-23.548991	-46.635653
3	1004	sao paulo	SP	-23.549829	-46.634792
4	1005	sao paulo	SP	-23.549487	-46.636650

Data cleaning recommendations (5/6)

❖ “order_reviews” Table

- For “order_id” which have multiple reviews. It is better to consider the latest review since it offers better understanding of customer review.
- For null values in "review_comment_title" & "review_comment_message" we can simply replace them with "not_mentioned" or "N/A" since it is common for customers to simply rate and not provide any comment.
- For our current analysis we can even remove these two columns as they do not provide any insight apart from sentiment analysis.
- Cleaned “order_reviews” table

	order_id	review_id	review_score	review_creation_date	review_answer_timestamp
1	00010242fe8c5a6d1ba2dd792cb16214	97ca439bc427b48bc1cd7177abe71365	5	2017-09-21 00:00:00.0000000	2017-09-22 10:57:00.0000000
2	00018f77f2f0320c557190d7a144bdd3	7b07bacd811c4117b742569b04ce3580	4	2017-05-13 00:00:00.0000000	2017-05-15 11:34:00.0000000
3	000229ec398224ef6ca0657da4fc703e	0c5b33dea94867d1ac402749e5438e8b	5	2018-01-23 00:00:00.0000000	2018-01-23 16:06:00.0000000
4	00024acbcdf0a6daa1e931b038114c75	f4028d019cb58564807486a6aaf33817	4	2018-08-15 00:00:00.0000000	2018-08-15 16:39:00.0000000
5	00042b26cf59d7ce69dfabb4e55b4fd9	940144190dcba6351888cafa43f3a3a5	5	2017-03-02 00:00:00.0000000	2017-03-03 10:54:00.0000000
6	00048cc3ae777c65dbb7d2a0634bc1ea	5e4e50af3b7960b7a10d86ec869509e8	4	2017-05-23 00:00:00.0000000	2017-05-24 19:00:00.0000000
7	00054e8431b9d7675808bcb819fb4a32	0381de7572d99d75230ce91207807265	4	2017-12-19 00:00:00.0000000	2017-12-29 02:40:00.0000000

Data cleaning recommendations (6/6)

❖ “products” Table

- There are 610 "product_ids" which do not have "product_category_name" & three other product descriptions. Removing these products which have no “product_category_name” would affect "orders" table and other connected tables, generating more null values when joins are made for our analysis, we would change it to “unknown”

- For the purpose of analysis and business insights, product name descriptions do not serve much purpose, we can take them out and just keep "product_id","product_category_name" ,"product_photos_qty“ & other product dimensions and weight.

- If we want to still keep them then it is better to change the null values to "unknown".

- For imputing "product_photos_qty", replace null values with 0.

	product_id	product_category_name	product_photos_qty	product_weight_g	product_length_cm	product_height_cm	product_width_cm
1	a41e356c76fab66334f36de622ecbd3a	unknown	0	650	17	14	12
2	d8dee61c2034d6d075997acef1870e9b	unknown	0	300	16	7	20
3	56139431d72cd51f19eb9f7dae4d1617	unknown	0	200	20	20	20
4	46b48281eb6d663ced748f324108c733	unknown	0	18500	41	30	41
5	5fb61f482620cb672f5e586bb132eae9	unknown	0	300	35	7	12



EDA Plans

Exploratory Data Analysis Plan (1/2)

- **Expected KPIs** – Total customers, Total Cities, Total States, Total Sellers, Total Products, Avg. Delivery time, Avg. Delivery to carrier, Avg. Rating, Avg. Review answered time, Total revenue, Customer retention rate.
- **Expected KRAs** – Customer retention rate, Customer acquisition, Gross Sales revenue, Avg. Transaction value, Gross and net revenue, Shrinkage rate due to cancelled/returned orders, Customer Basket size, Customer lifetime value, Churn rate, Order delivery lead time etc.
- **EDA Plan:**

1. Month wise analysis of total revenue/orders
2. Revenue generated per state
3. Orders per payment type.
4. Count of orders per order status
5. Total orders per state.
6. Average instalment per payment type.
7. Month wise customer acquisition.
8. Monthly revenue generated by new customers.
9. Classify customers based on order volume in low, medium high

Exploratory Data Analysis Plan (2/2)

- **EDA Plan:**

10. RFM segmentation of customers.
11. Behaviour of one-time buyers based on state.
12. Customer preference for product category and payment method.
13. Category penetration of products
14. Top 10 highest ordered products
15. Sales contribution per each category
16. Most and least profitable/revenue generating product categories.
17. Category wise sales (Pareto analysis)
18. Product wise average rating score.

Etc.



Predictive Analysis plan

Predictive analysis (1/2)

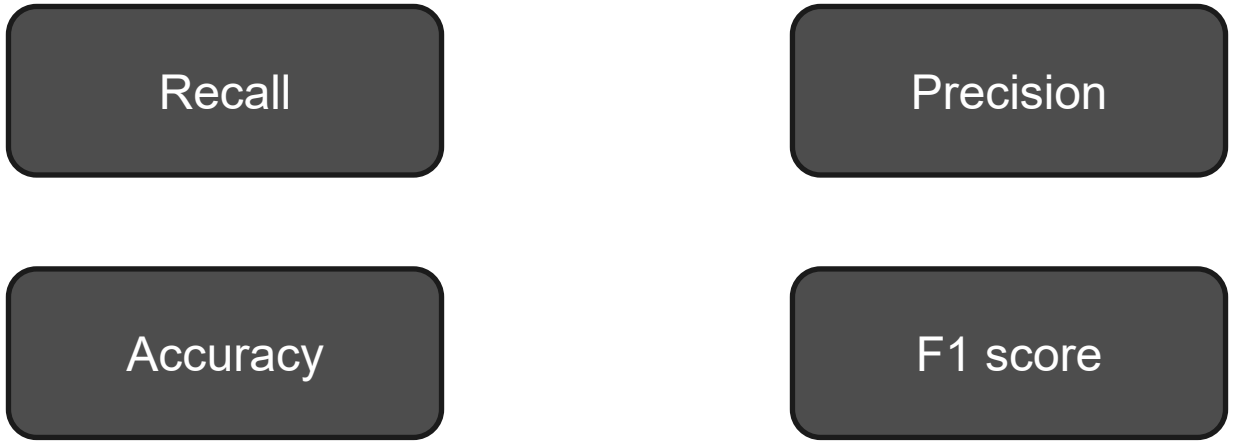
- For this analysis, the best predictive model is a rating classification model using **Logistic Regression**.
- We will predict customer ratings between 1 to 5 based on certain features.

Feature selection: We will select features which directly affect customer ratings. For example,

- **Delivery time**
- **Order processing time**
- **Price**
- **Freight value**
- **Payment type**
- **Order dimensions**
- **Payment instalments**
- **Payment sequential**

Predictive analysis (2/2)

- **Evaluation metrics:**



- We will create a classification report and confusion matrix to evaluate the accuracy of our model in predicting the ratings.

- We can also do segmentation using ML models, classifying customers into different segments based on their revenue generated or orders placed.



Exploratory Data Analysis

KPIs

Total customers: 96,096
Total customer cities: 4,119
Total customer states: 27

Average rating: 4
Average review response time:3.22 Days
Average orders per customer: 1.08

Total sellers: 3095
Total seller cities: 611
Total seller cities: 23

Total geolocation cities: 5,756
Total zip codes: 19,015

Average product price: BRL 120.65
Average freight value: BRL 19.90
Total revenue: BRL 1,60,08,872.12

Total orders: 99,441
Total payment methods: 4

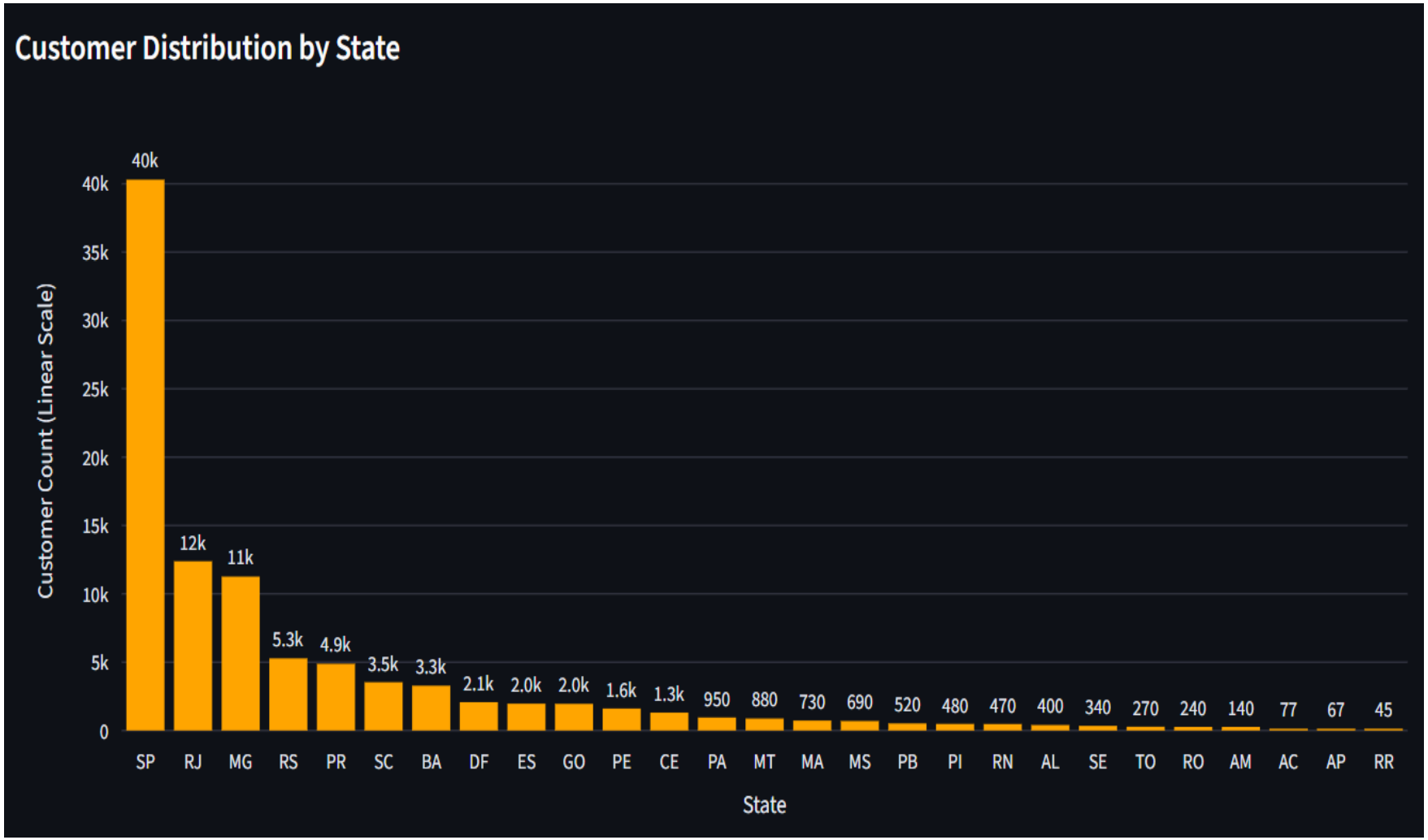
Total products: 32,951
Total product categories: 73

Median approval hours: 0.33 Hours
Median carrier delivery: 1 Day
Median customer delivery: 7 days



Exploratory Data Analysis Customer level

Customers per state



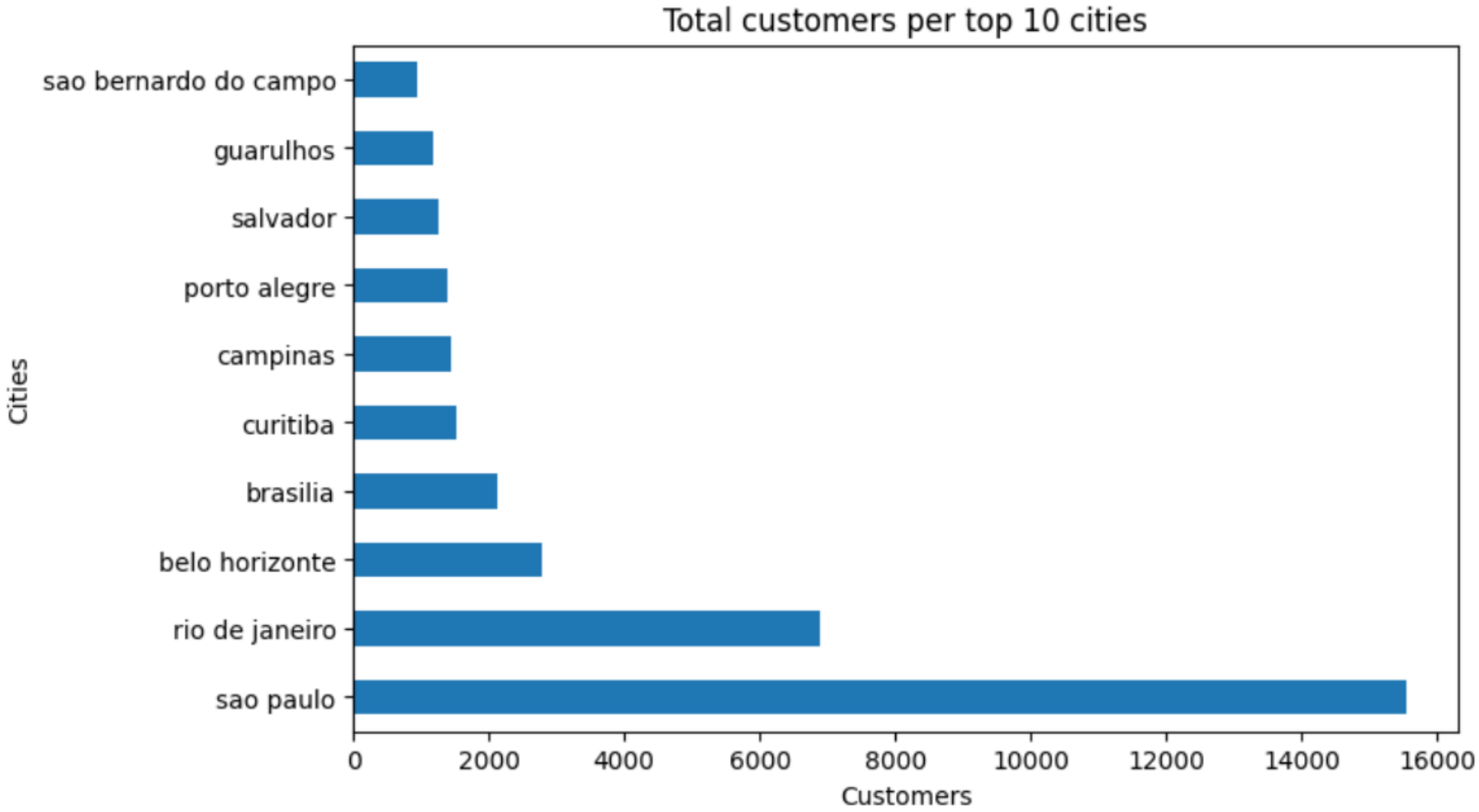
Observations:

- Majority of customer are from SP state (over 40K).
- States AM, AC, AP & RR have less than 200 customers per state

Recommendations:

- Check population density of states with low customers and try to isolate if low density is the reason for low customer count.
- Check for any local competitions present in such cities and introduce targeted offers for such states.

Customers per city (Top 10 cities)



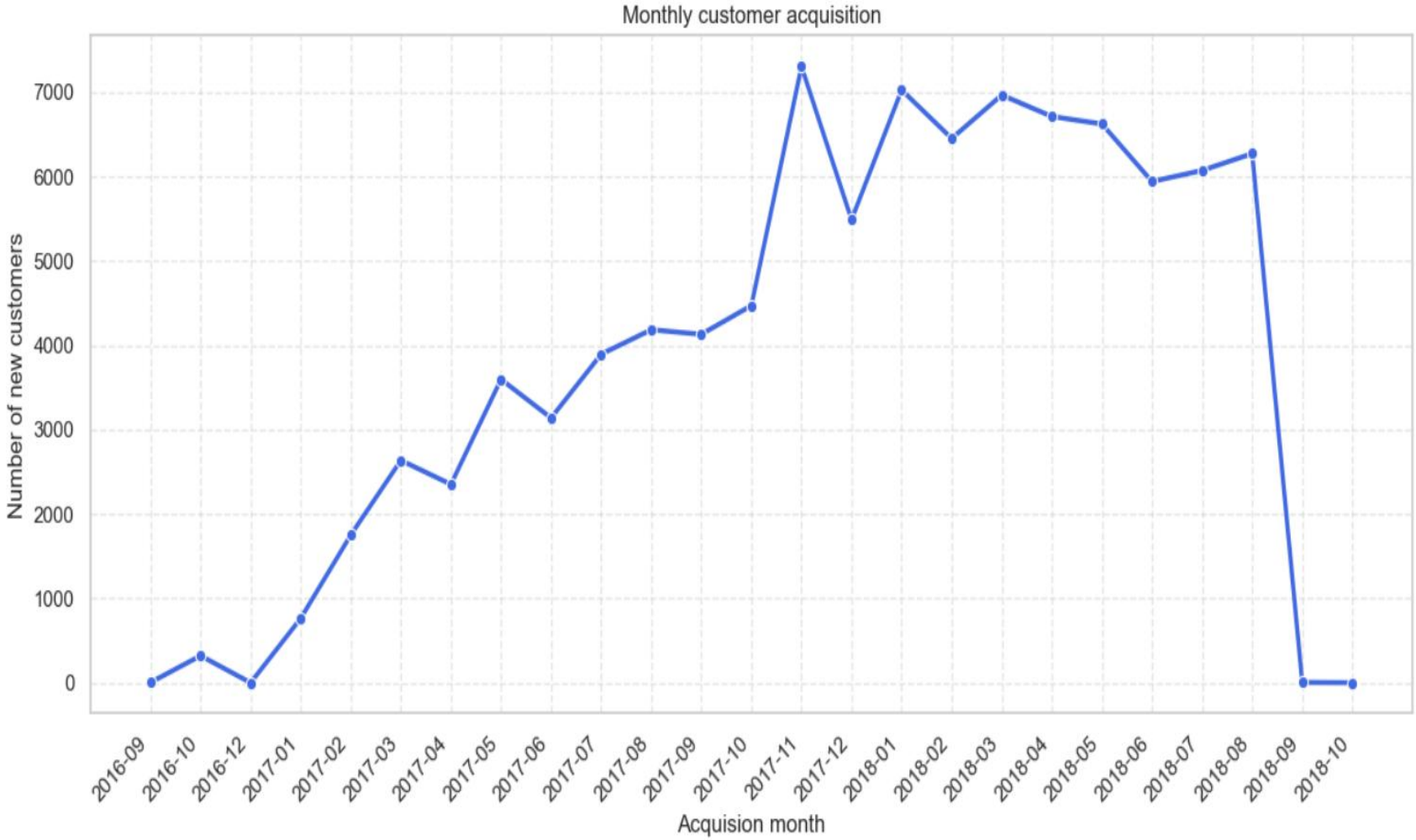
Observations:

- More than 15K customers belong to Sao Paulo city alone followed by Rio De Janario.
- More than 4100 cities have less than 2000 customers.

Recommendations:

- Same approach as for the states should be implemented for such cities having low customers.
- Check how many sellers belong to these states and see if low count of sellers is the primary reason for less customers ordering from these cities.

Monthly Customer acquisition



Observations:

- We observe a positive trend in customer acquisition till march 2018, after which it falls till June 2018 and again picks up a positive pace.

Recommendations:

- Identify the reason for this fall. A slight fall is seen every 6 months. Sales can be introduced during such months offering better prices.

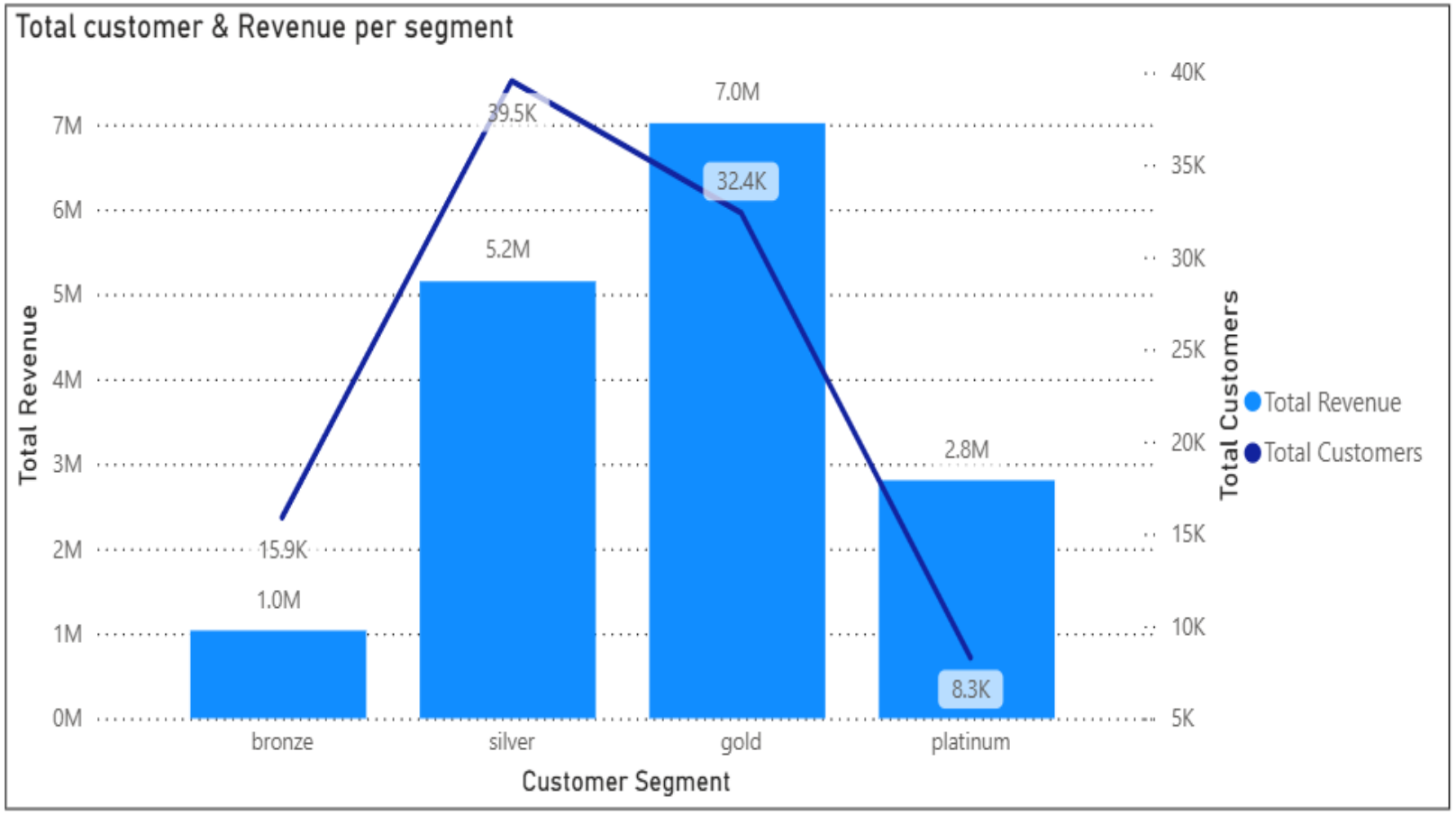
RFM Segmentation Analysis

- **Recency (R):** How recently a purchase was made by customer.
- **Frequency (F):** Over a certain period, how many times does a customer make a purchase.
- **Monetary (M):** How much money does a customer spend on all purchases.

- Each unique customer was assigned a score for **Recency, Frequency & Monetary** from 1 to 5 using pandas **qcut** function in python. Having the maximum score of 15 (5+5+5) and minimum score of 3 (1+1+1).
- **Recency_score:** More recent purchases get higher score.
- **Frequency_score:** More frequently ordering customers get higher score.
- **Monetary_score:** High spending customers get a higher score.

- Based on these scores customers were segmented into group of 4, Bronze, Silver, Gold and platinum. Where bronze customers have lower RFM score and platinum customers have higher RFM scores.

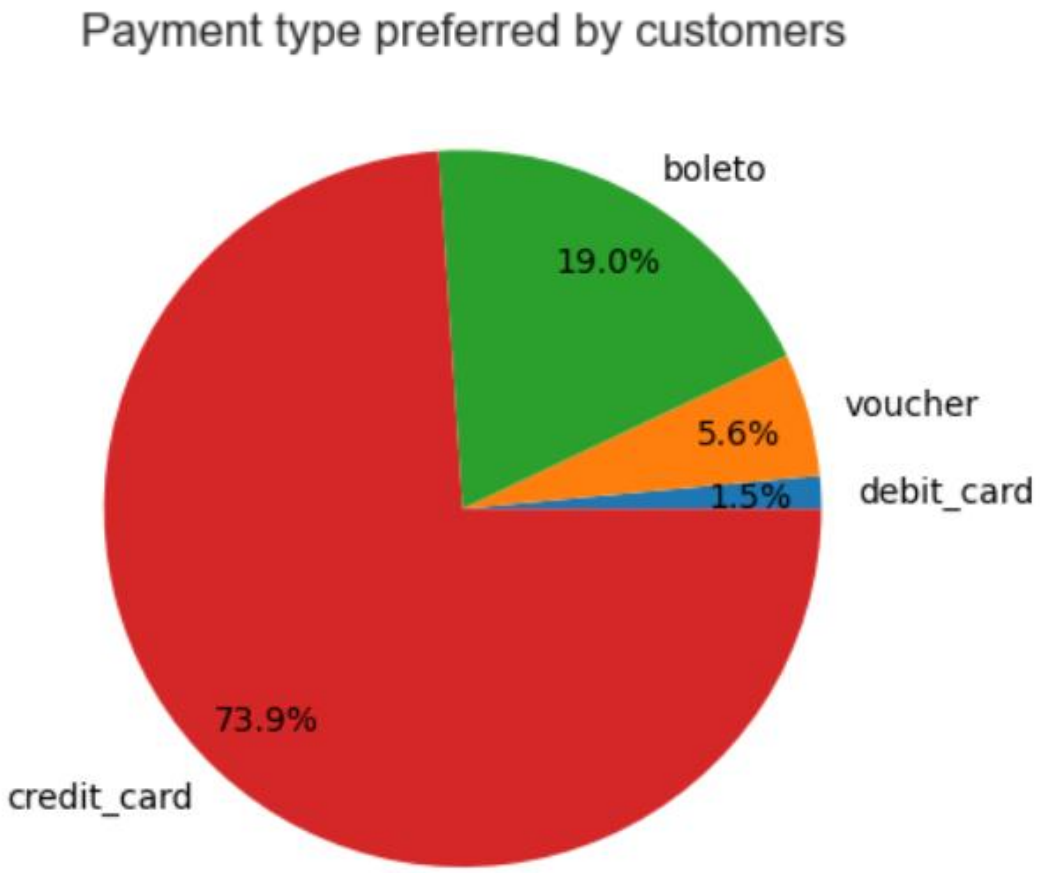
Customer segmentation



	customer segment	total_sales	total_customers
0	bronze	1035611.84	15883
1	gold	7015466.36	32394
2	platinum	2806316.90	8279
3	silver	5151477.02	39539

Customer Classification:
Platinum: RFM score above 12
Gold: RFM score between 10 & 12
Silver: RFM score between 7 & 9
Bronze: RFM score below 7

Customer preference



Total Customers: 96,096

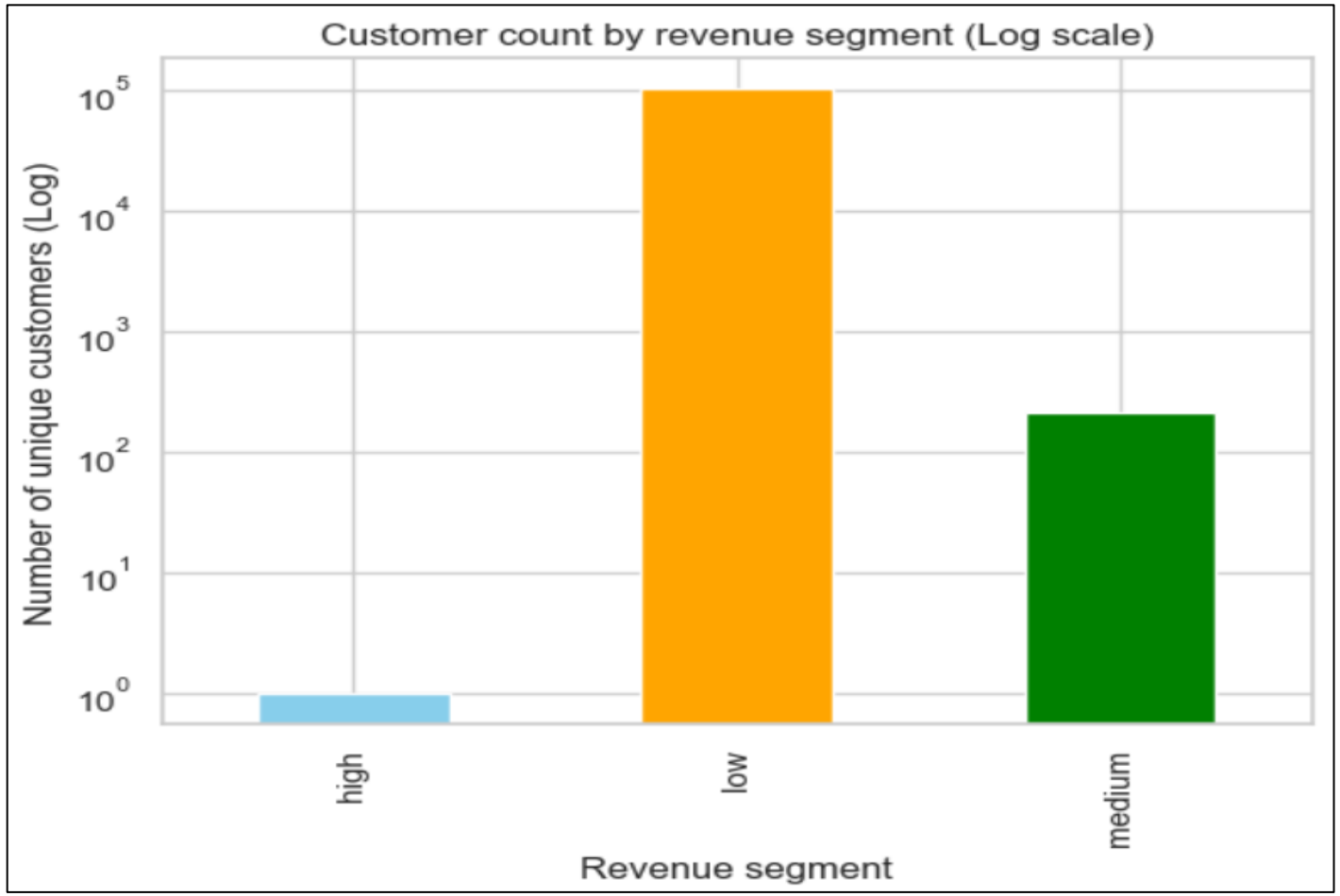
Customers preferring Credit card: 76,795

Customers preferring Debit card: 1,529

Customers preferring Boleto: 19,784

Customers preferring Voucher: 5,775

Customer revenue segmentation



Total Customers: 96,096

Low revenue customers: 103671

Medium revenue customers: 214

High revenue customers: 1

Customer behavior observations

- One-time buyers: 93,098
- One-time buyers percentage: 96.88%
- Repeat buyers: 2,997
- Repeat buyers percentage: 3.12%
- Revenue generated by one-time customers: BRL 1,50,64,849.41
- Revenue generated by repeat customers: BRL 9,44,022.71
- One time buyers contribute to almost 97% orders and nearly 94% revenue.
- Majority of customers lie in Silver segment and least in Platinum.
- Majority of revenue is being generated by Gold segment customers and least by Bronze.
- Most customers belong to low revenue segment.

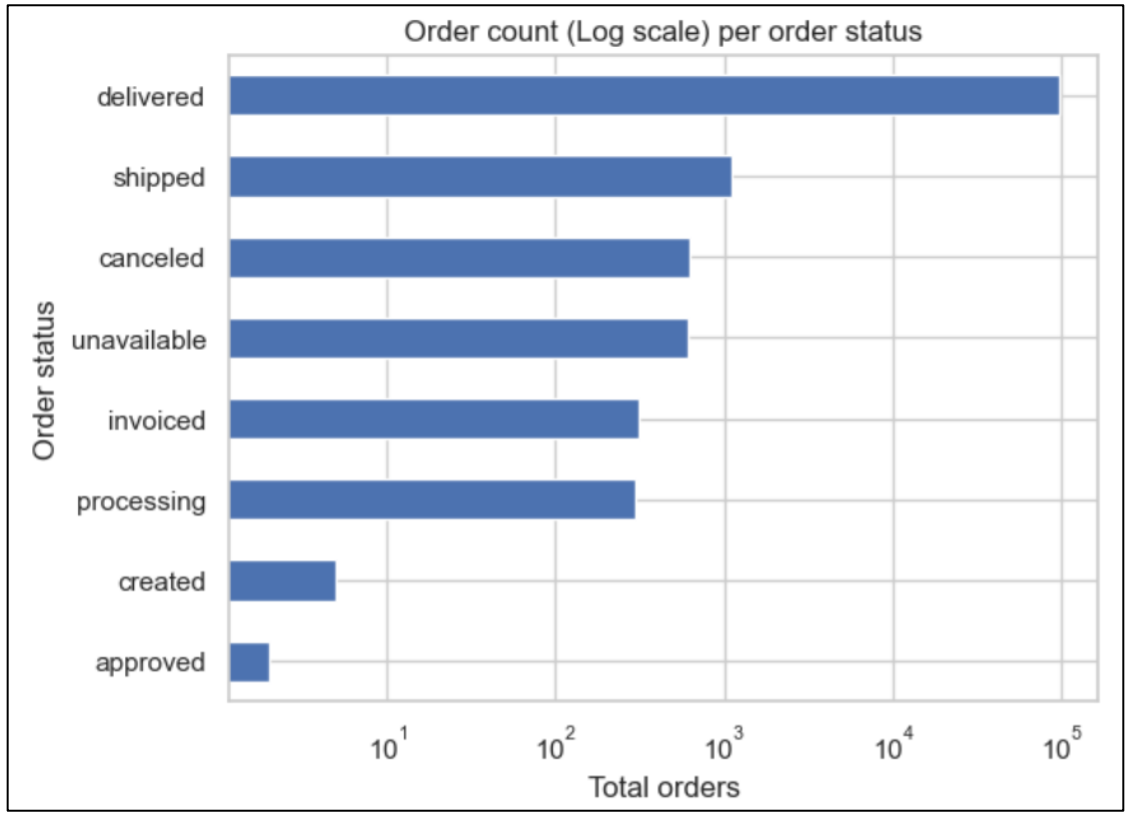
Customer analysis Recommendations

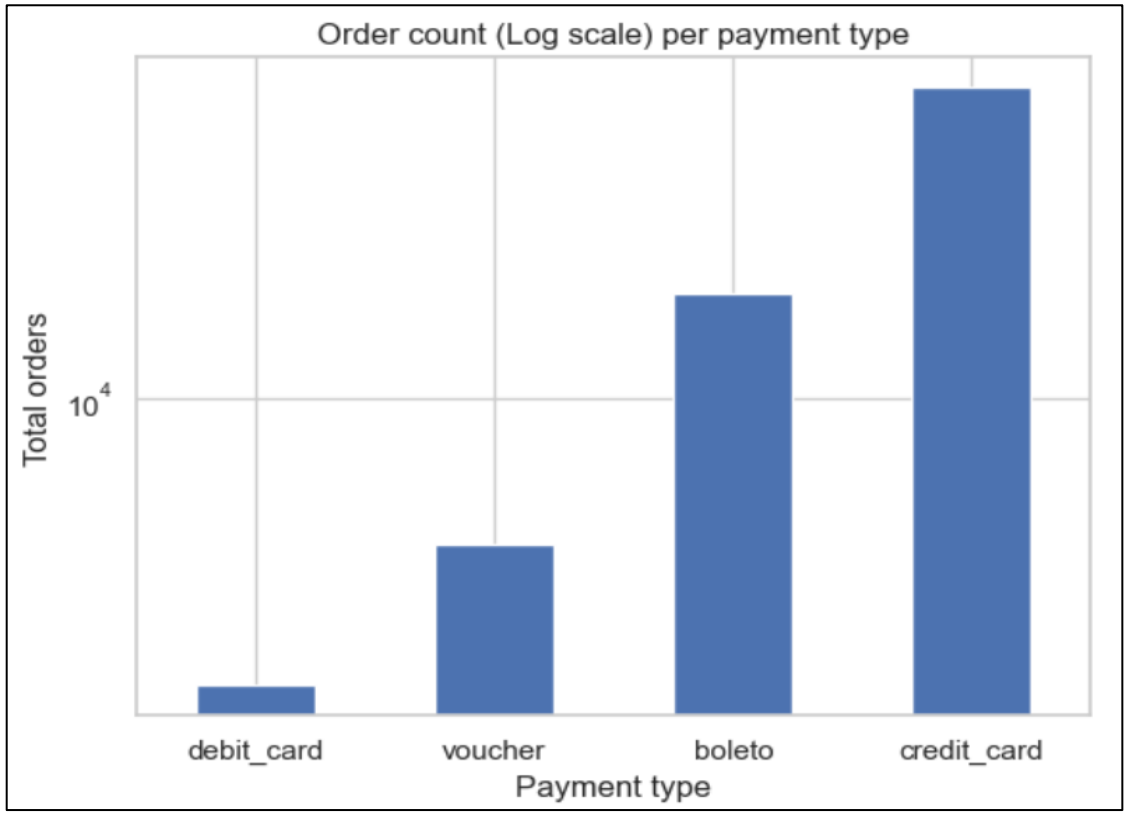
- Nearly 97% customers buy only once and never return. It is recommended to create loyalty programs and customer specific promotions that target them individually and promote future ordering. Also, retention programs for repeat buyers should be introduced along with specific offers that increase their retention.
- Majority of revenue comes from these customers, therefore their conversion to repeat buyers would be beneficial for the company which would increase the profit by many folds.
- Few states and few cities contribute to majority of customers. An analysis should be made if population density is the reason of others being less popular or is there any competition present in the respective city or state.
- RFM segmentation shows that silver tier customers are the majority. Personalized offers should be made to improve their RFM score and enhance their tiers.
- Gold customers generate maximum revenue, they should be rewarded with cyclic offers and membership benefits meanwhile other less revenue generating customers should be checked and personalized offers should be given to them to promote their orders.
- Customer acquisition saw a steady increase up until August 2018, after which there was a sharp decline. Investigate the reason and eliminate it so that a steady pace can be maintained again. Check if this is a cyclic trend or just one time thing.
- Majority of customers belong to low revenue generators and far lower belong to others. Measures should be taken to retain medium revenue customers and provide them options to enourage them into shifting to high revenue customers.



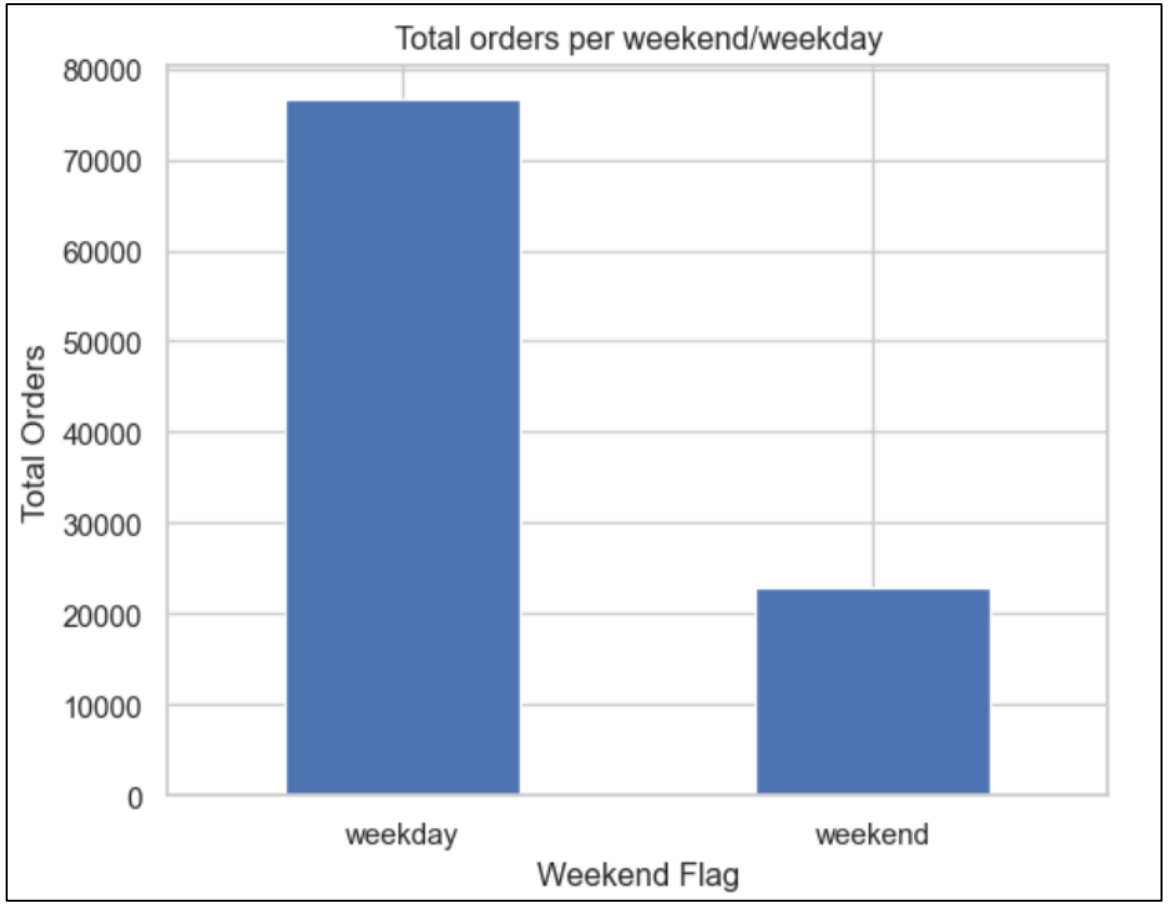
Exploratory Data Analysis Orders & Sales Trends

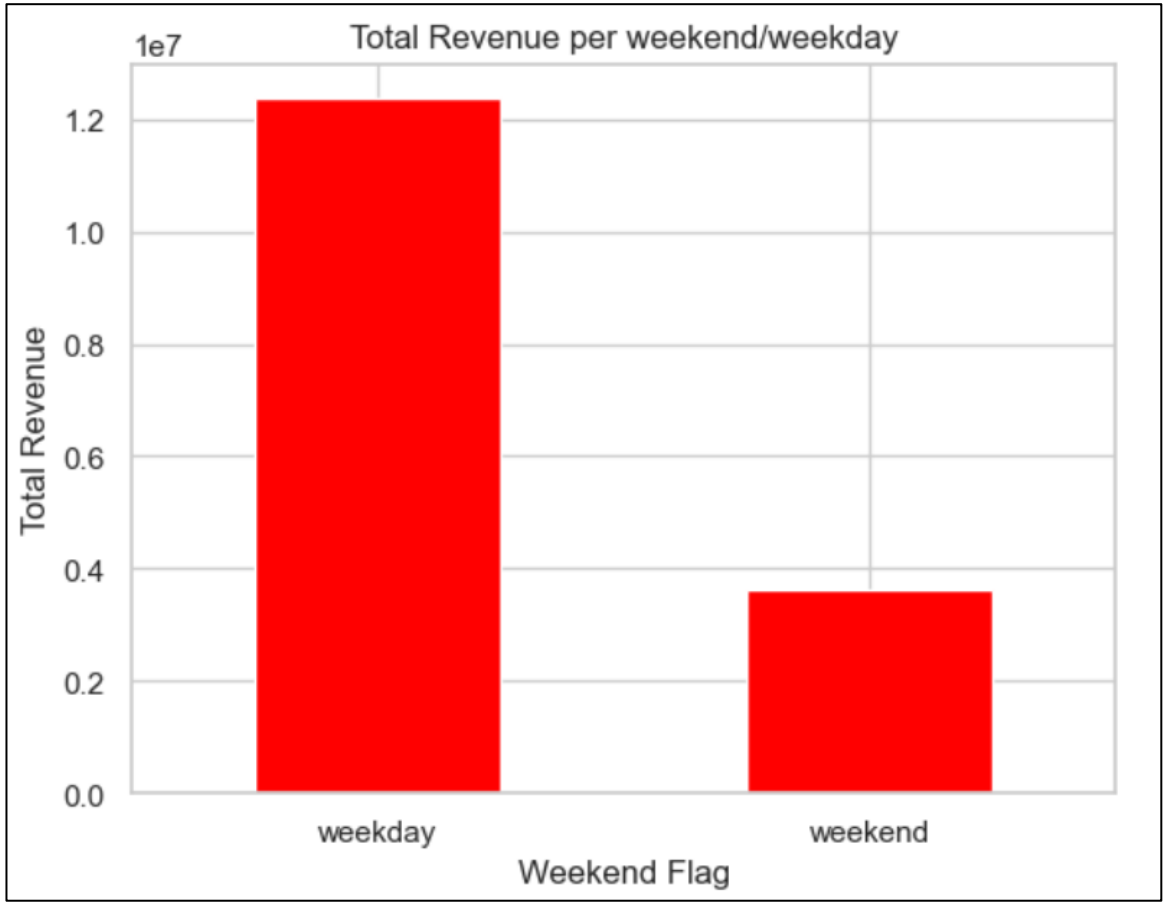
Orders data analysis



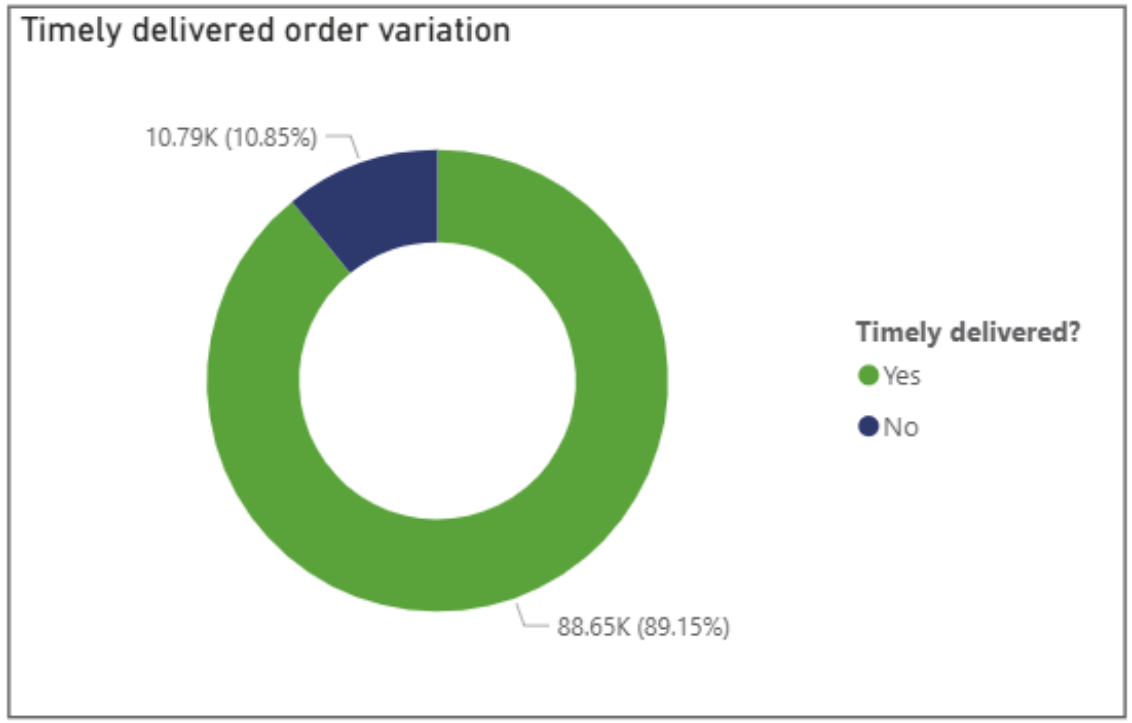


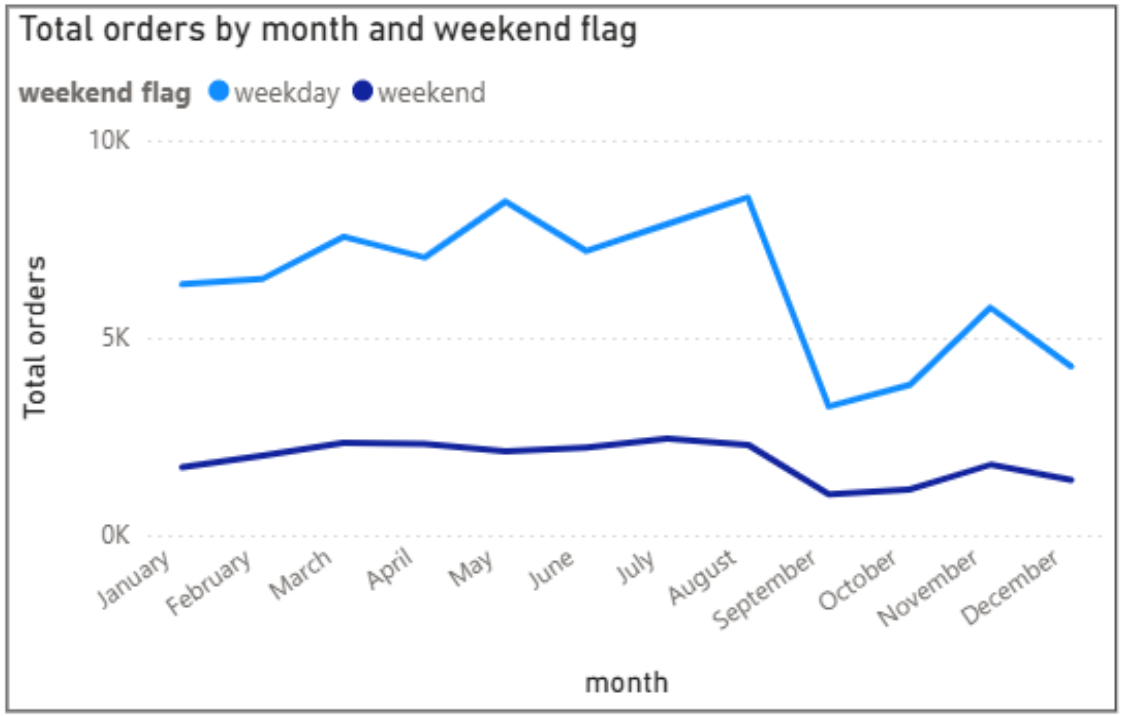
Orders & Revenue per weekday flag



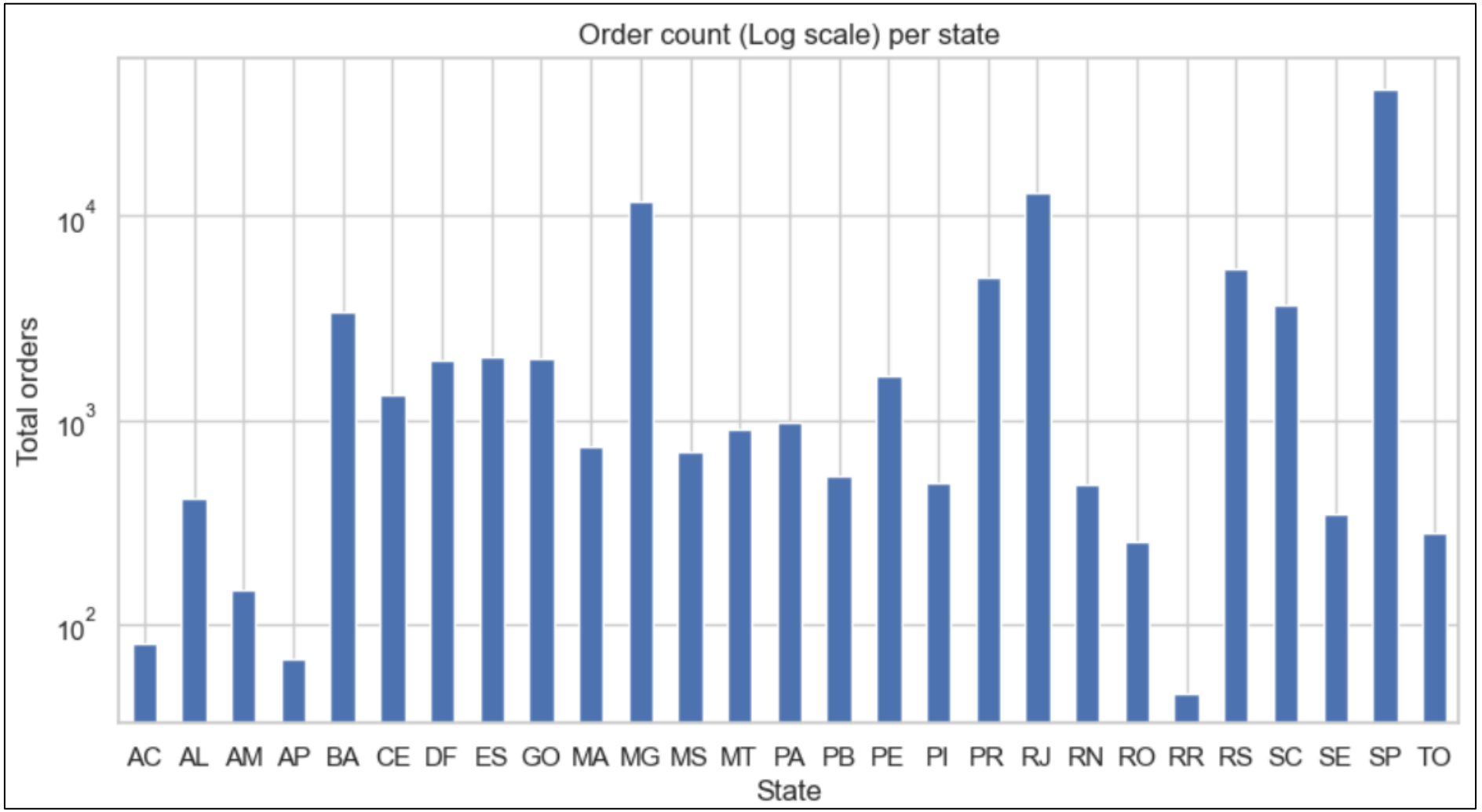


Orders analysis

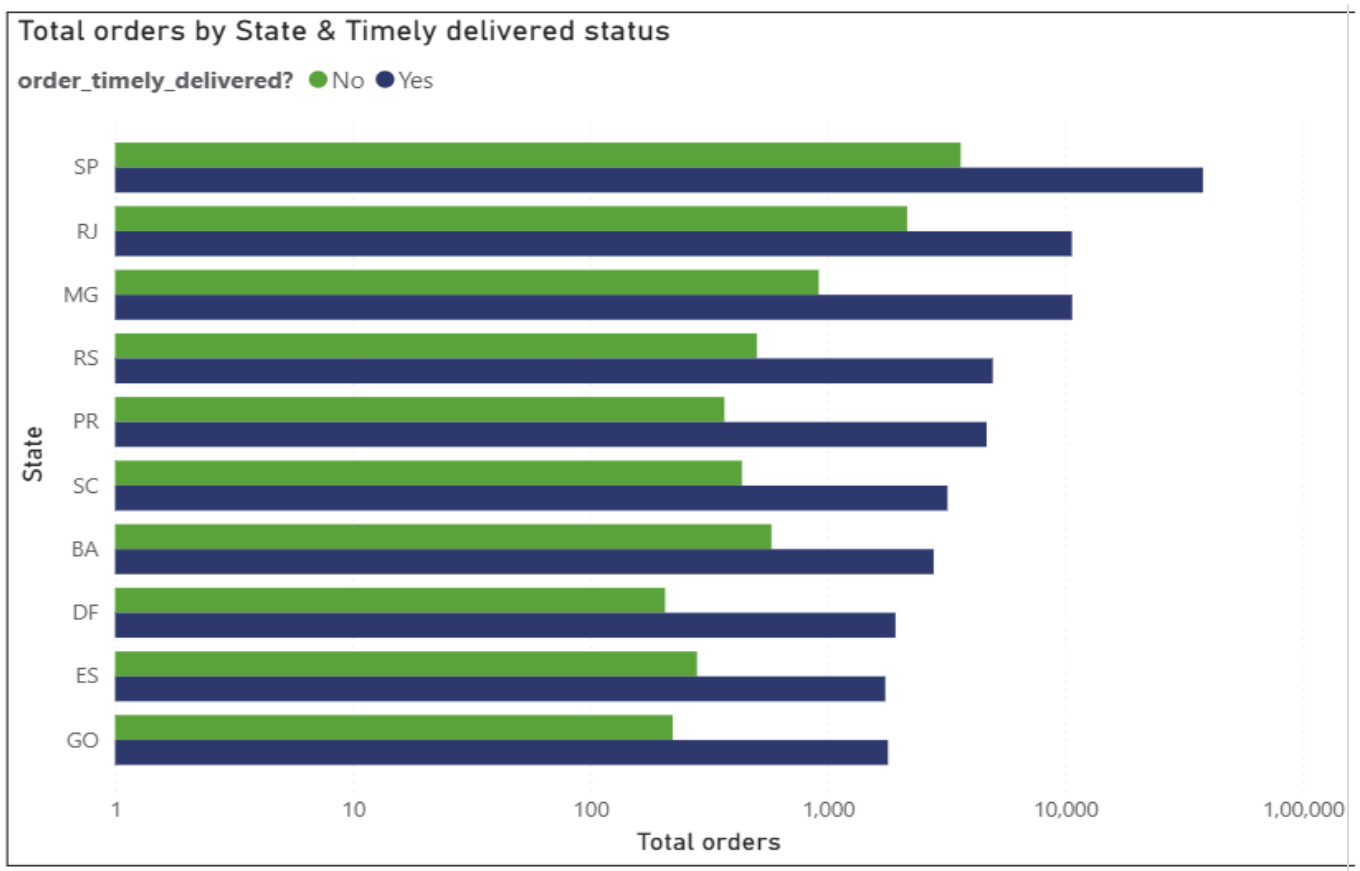




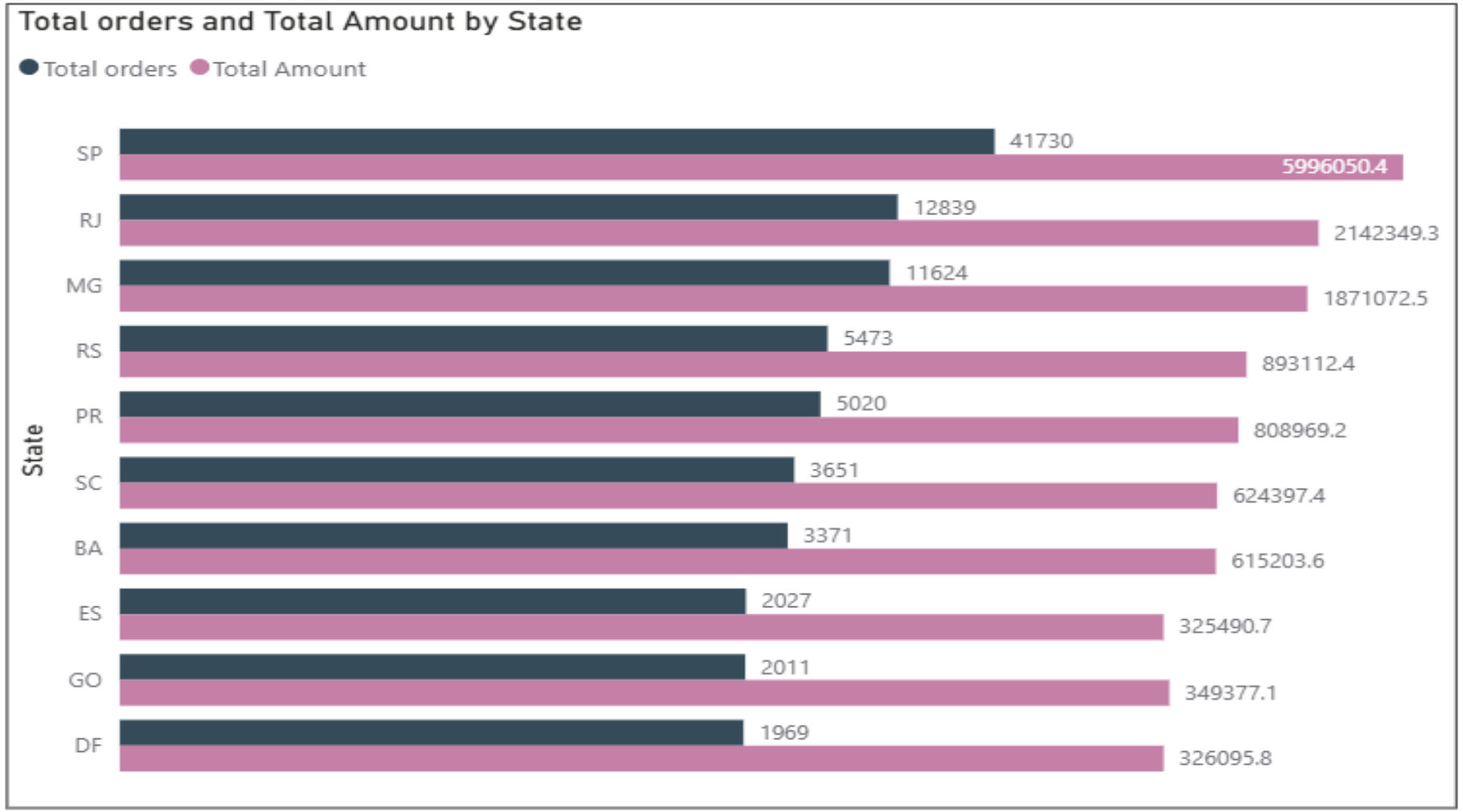
Orders demographics analysis



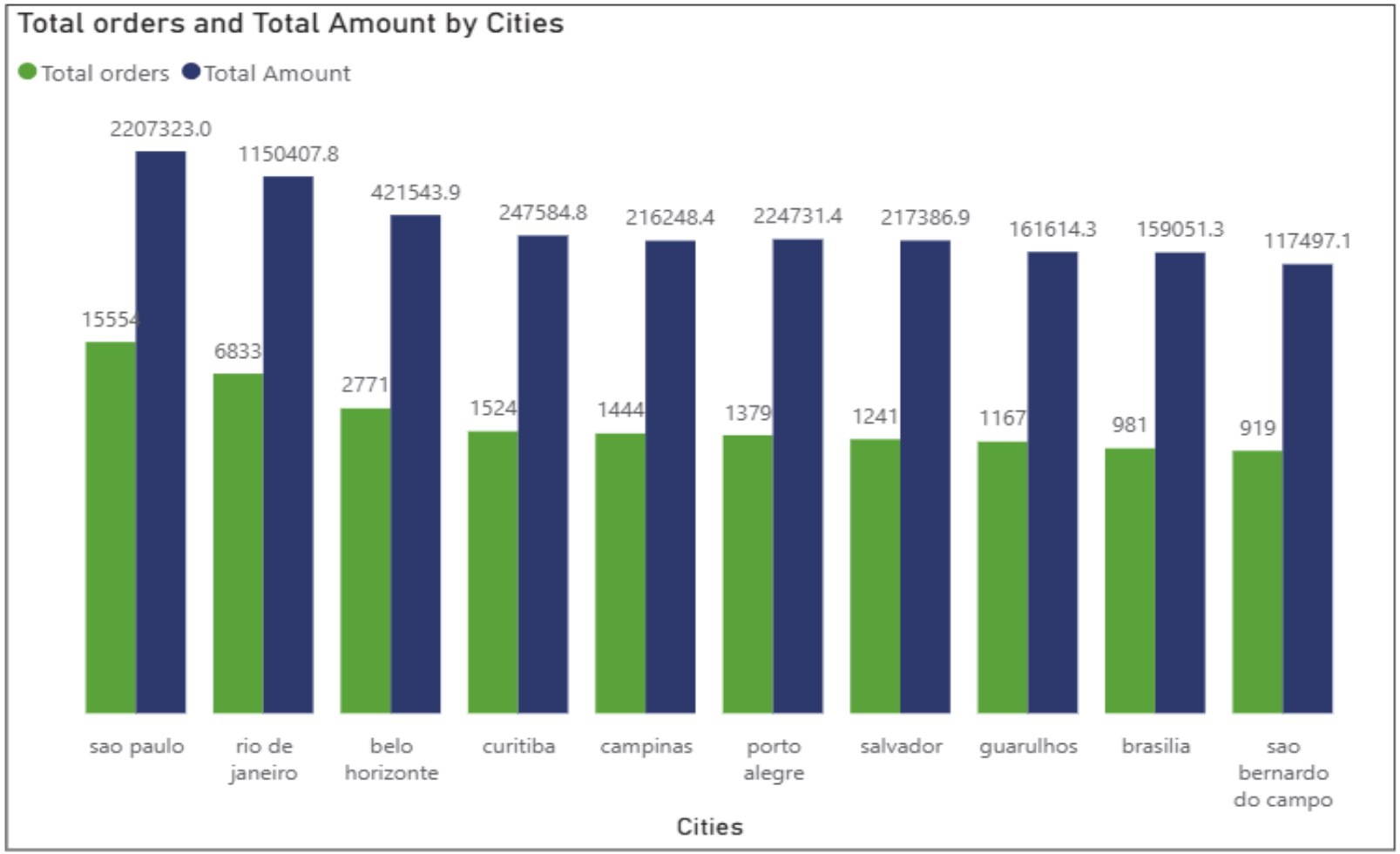
State wise timely delivered status



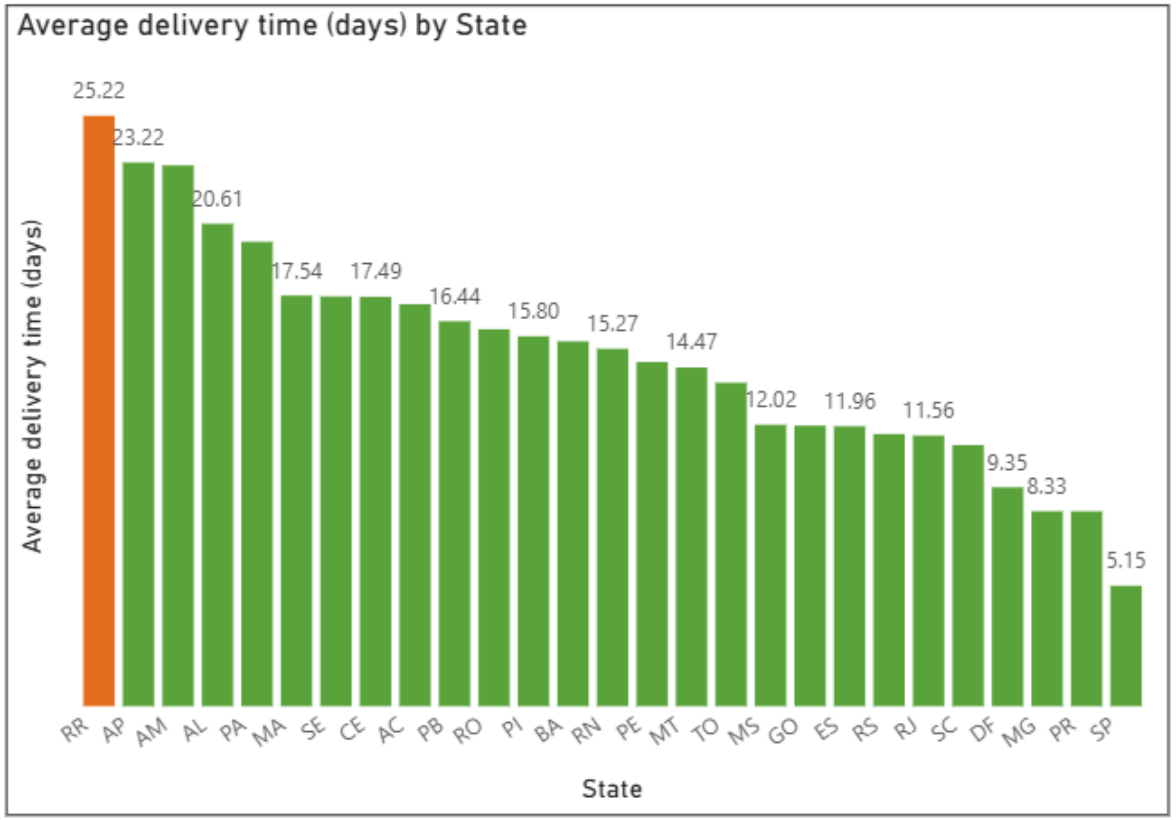
Order & Revenue demographics analysis (Top 10 States)

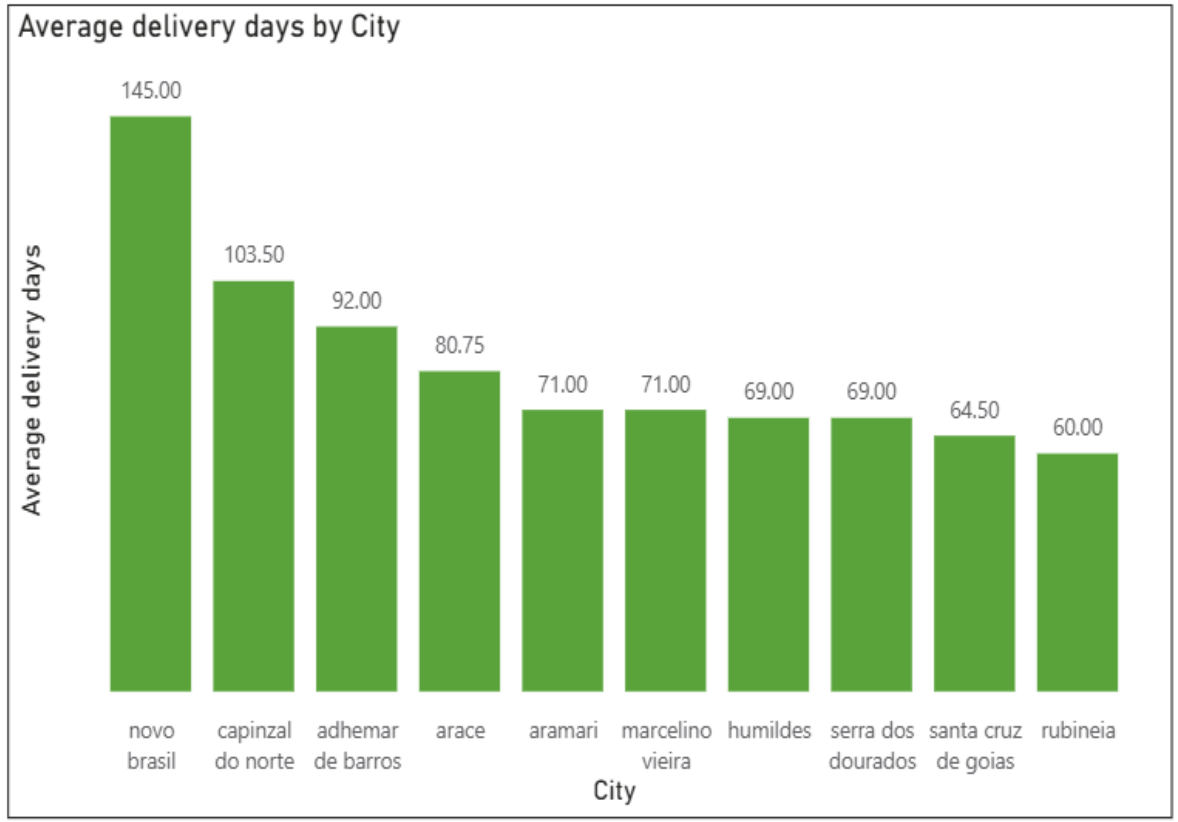


Order & Revenue demographics analysis (Top 10 Cities)

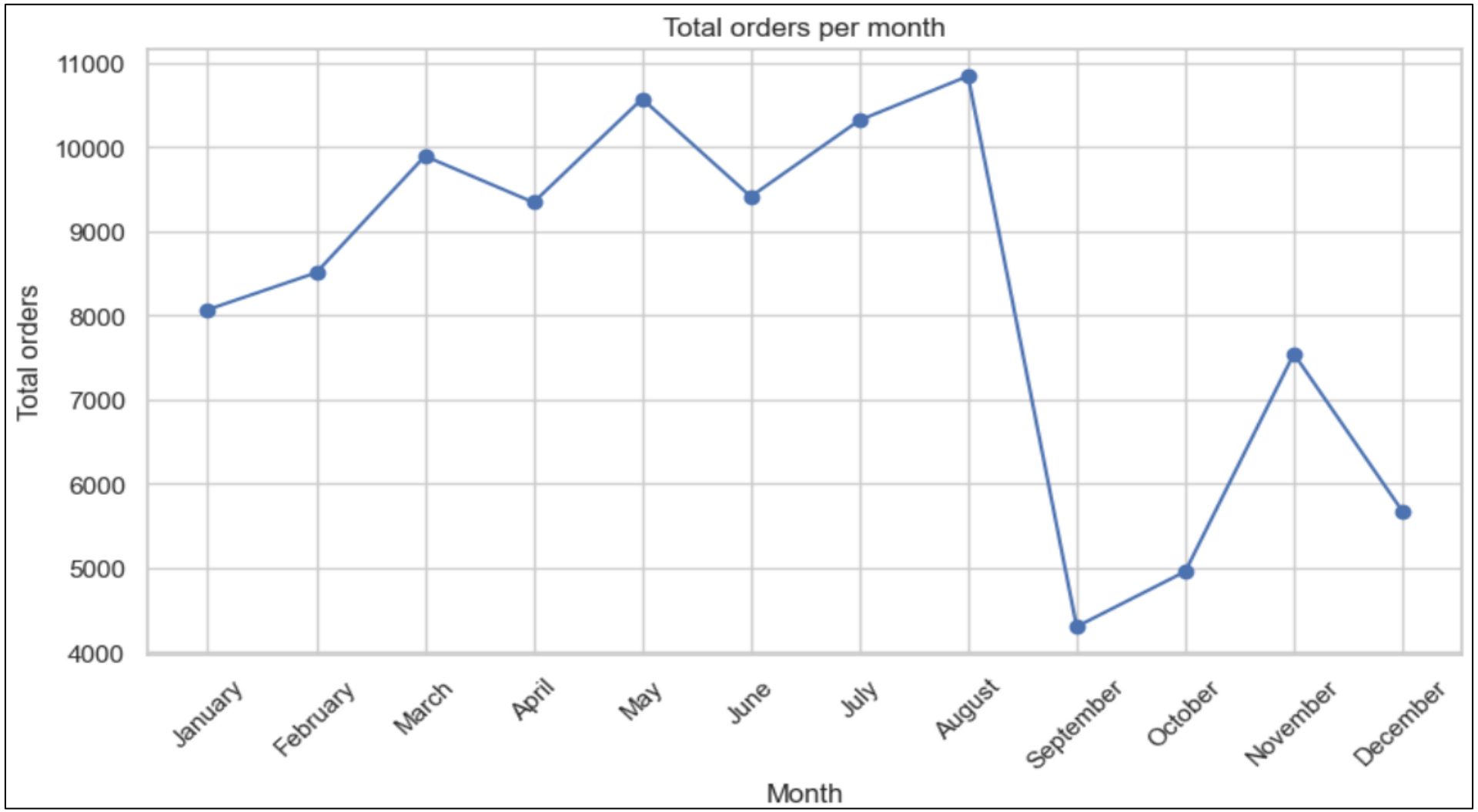


Delivery days by state and city

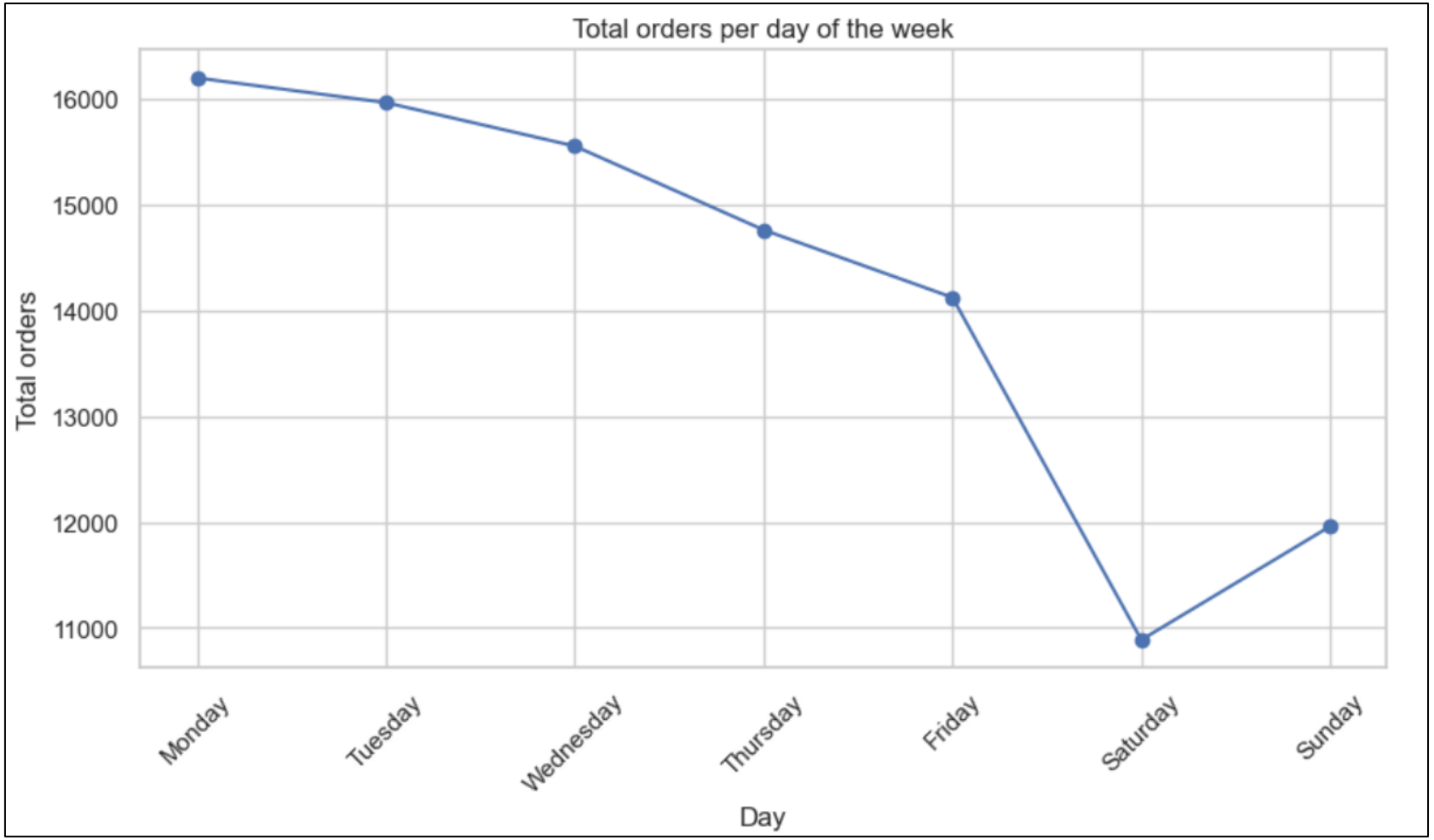




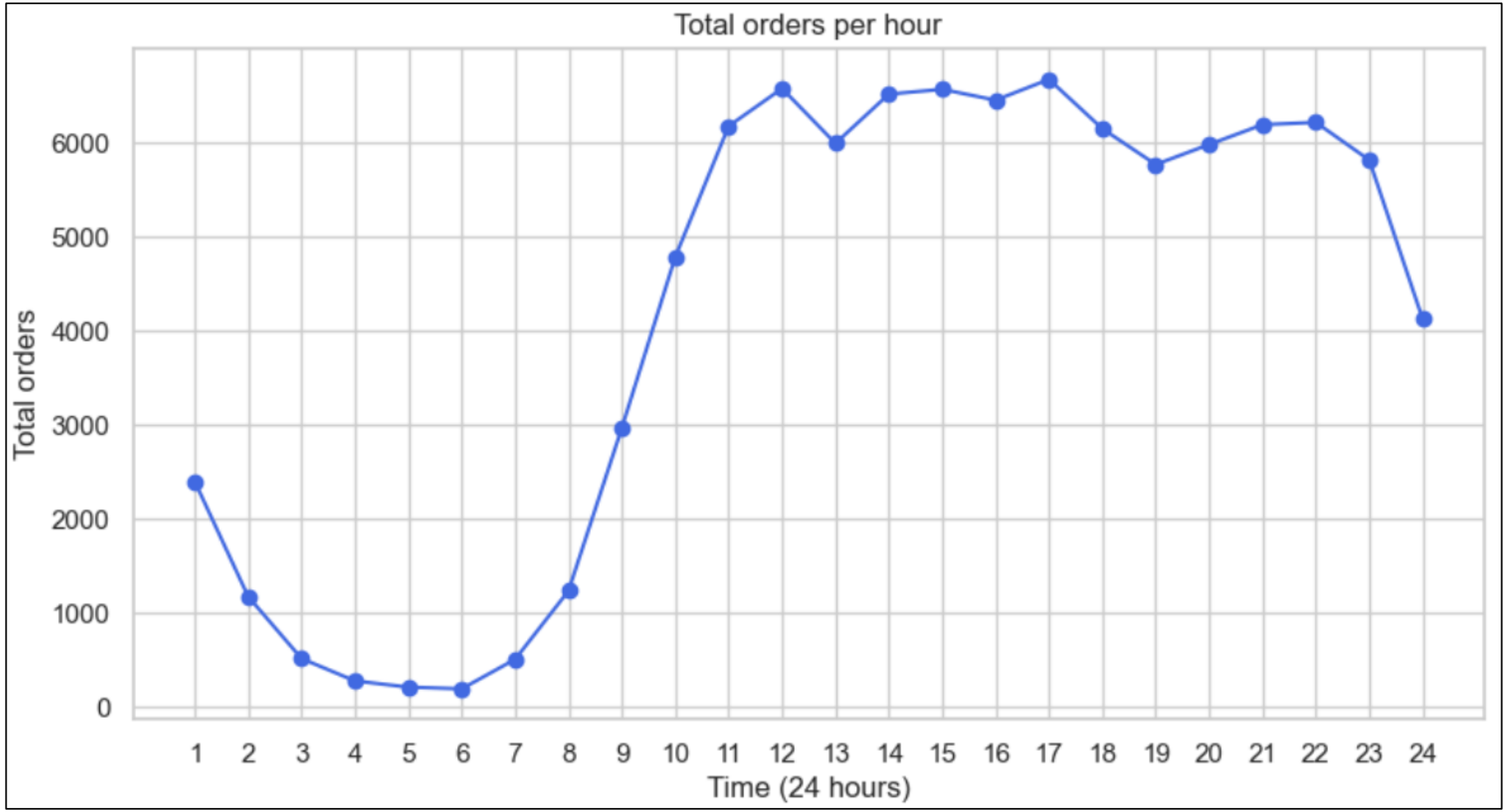
Total orders per month



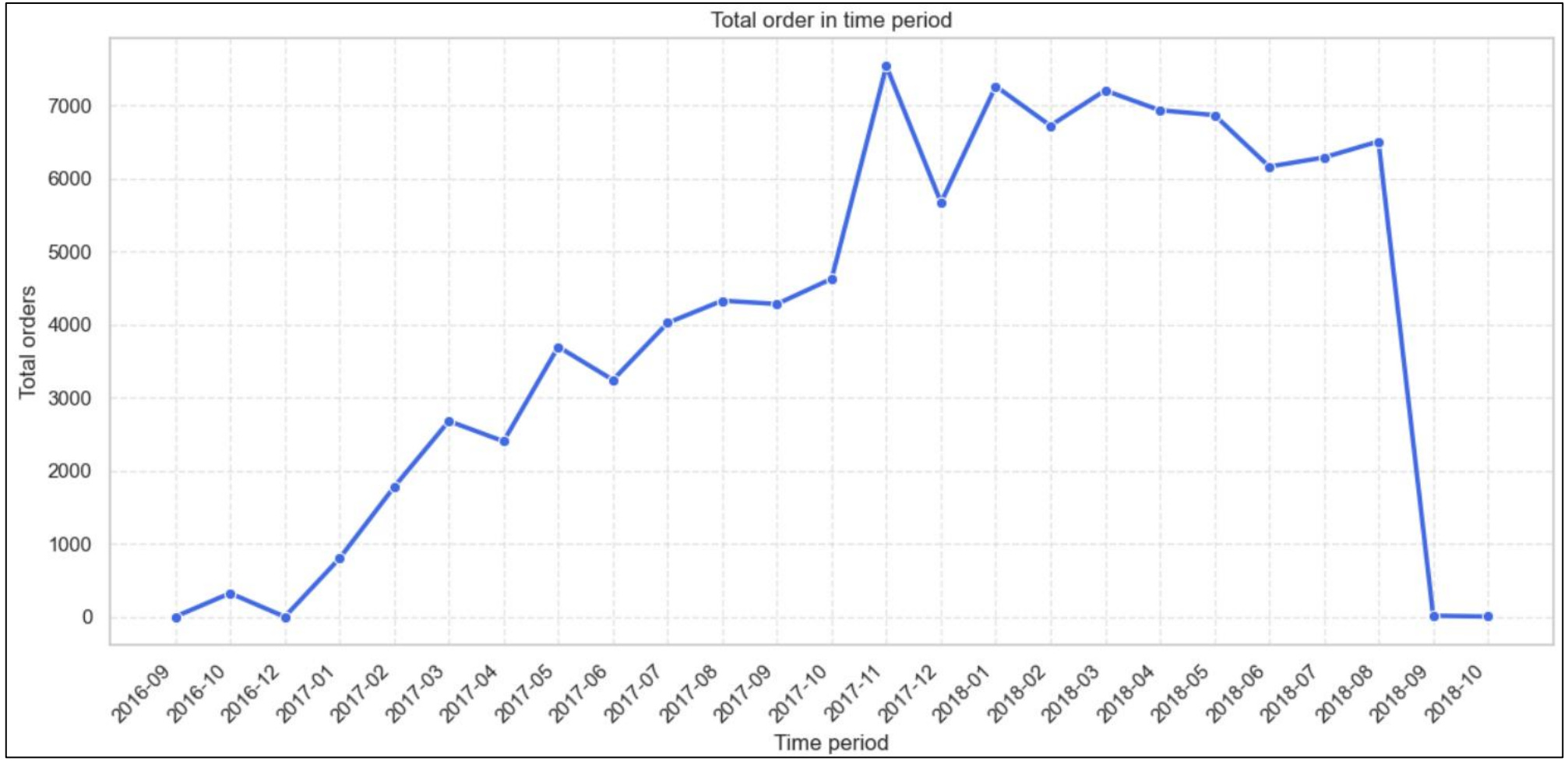
Total orders per day of the week



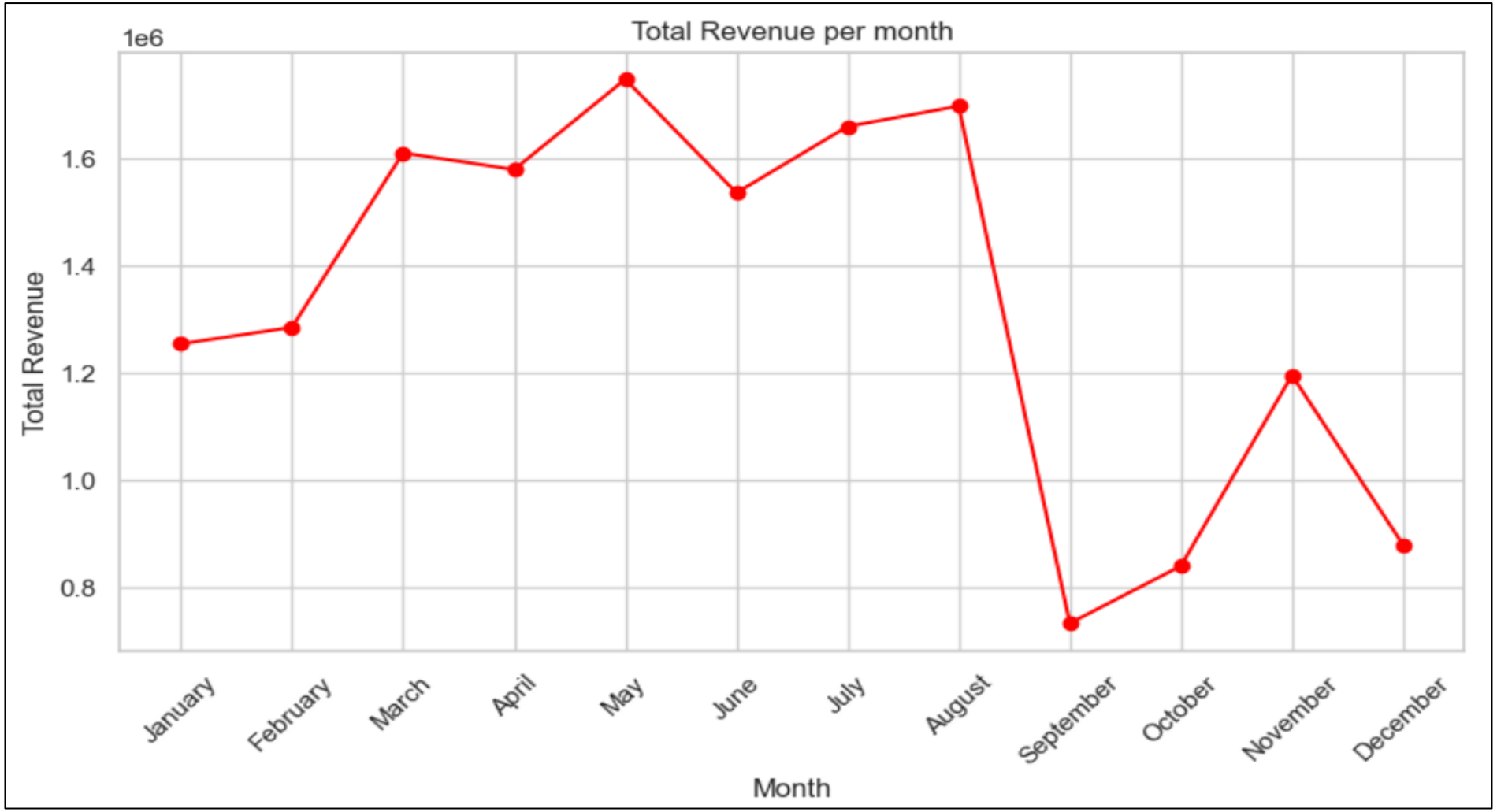
Orders variation by hour



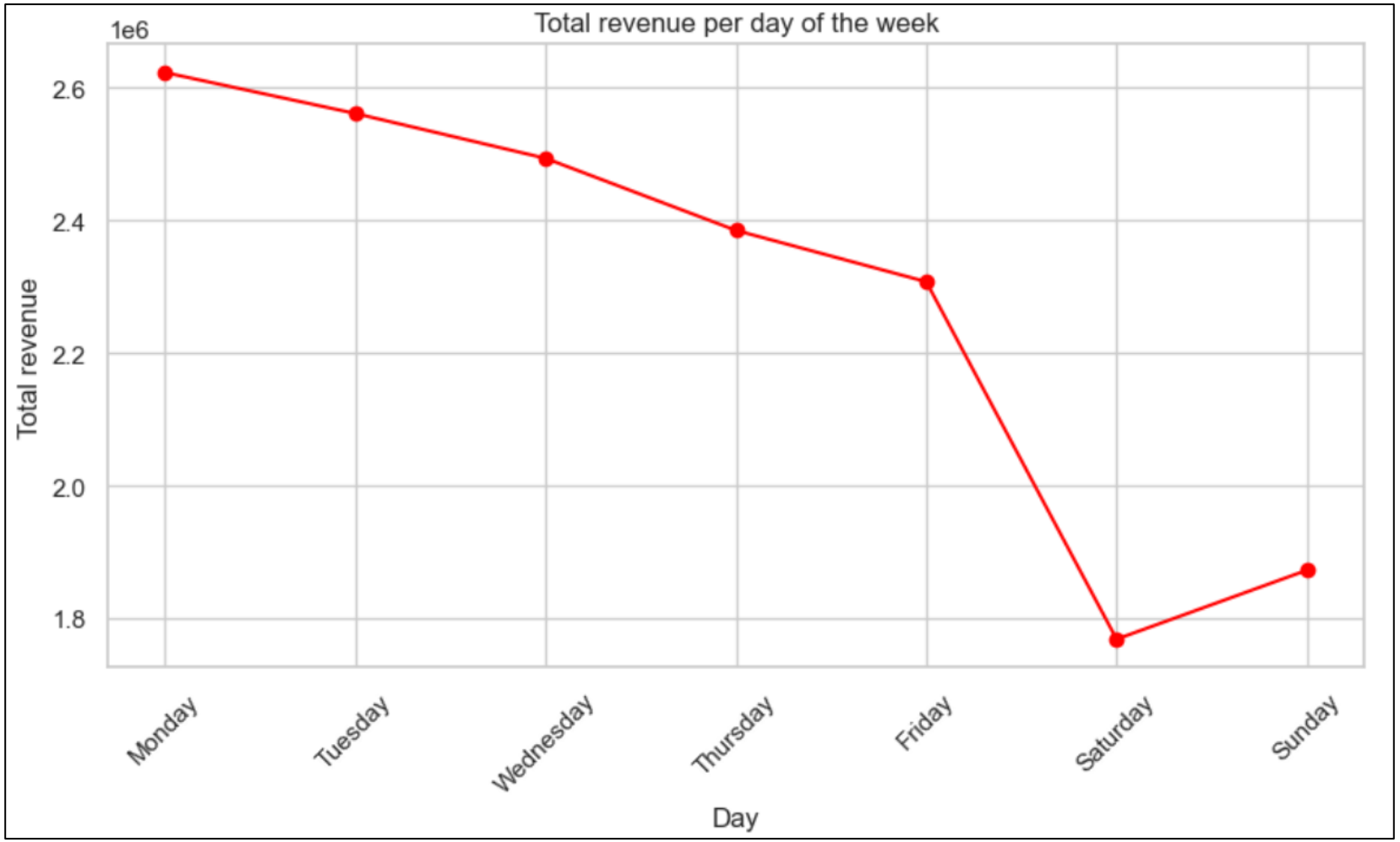
Total orders trend



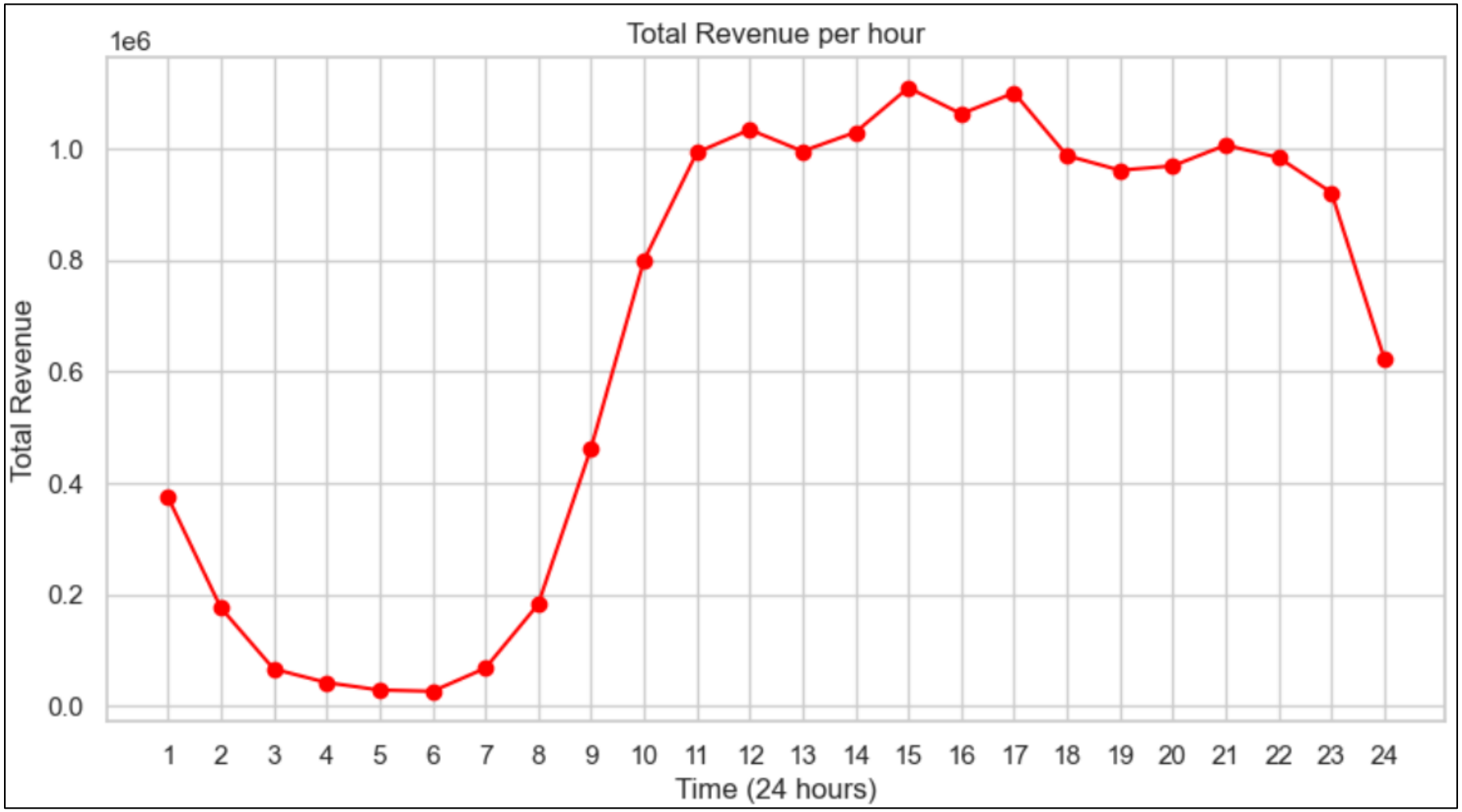
Revenue per month



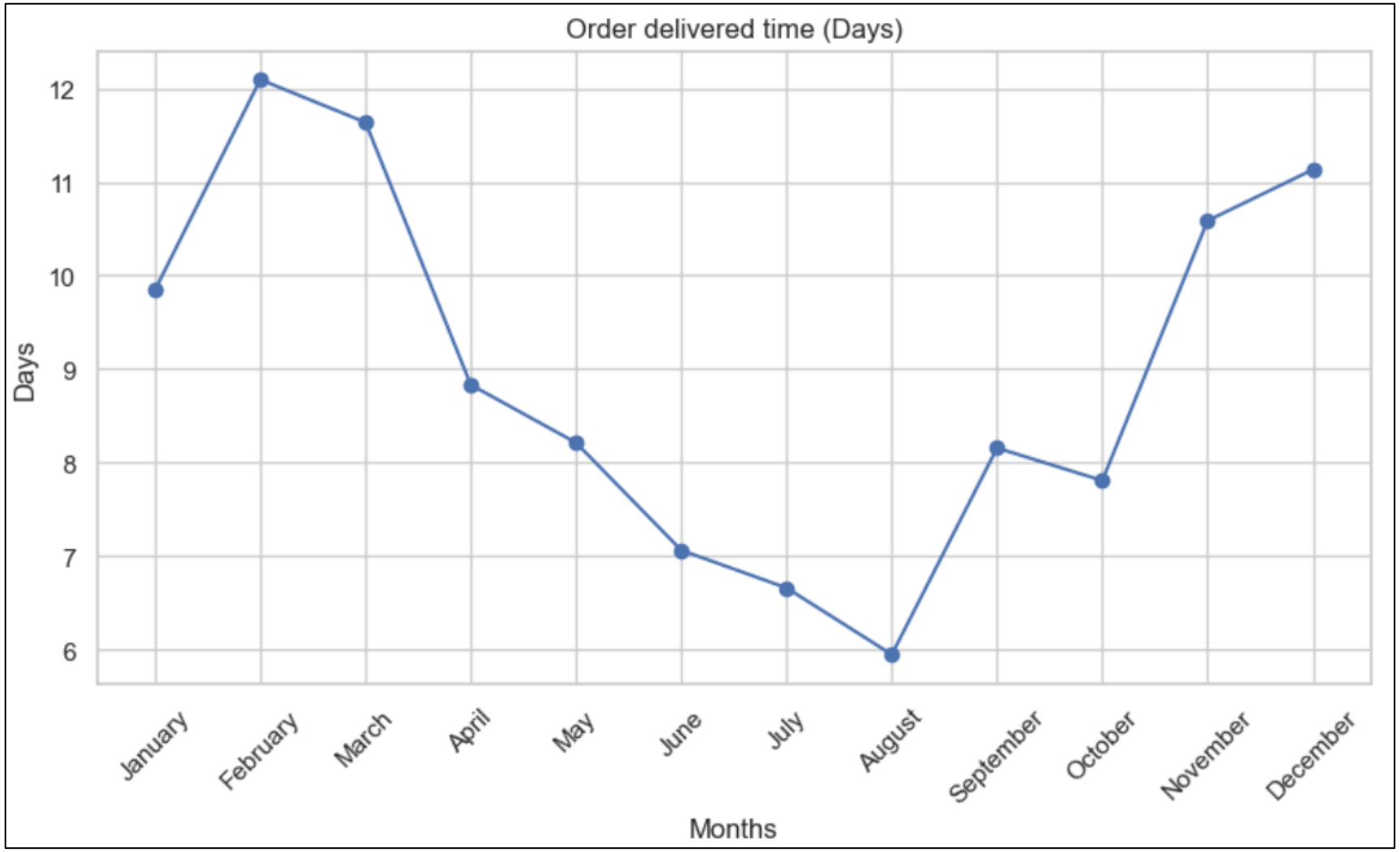
Total revenue per day of the week



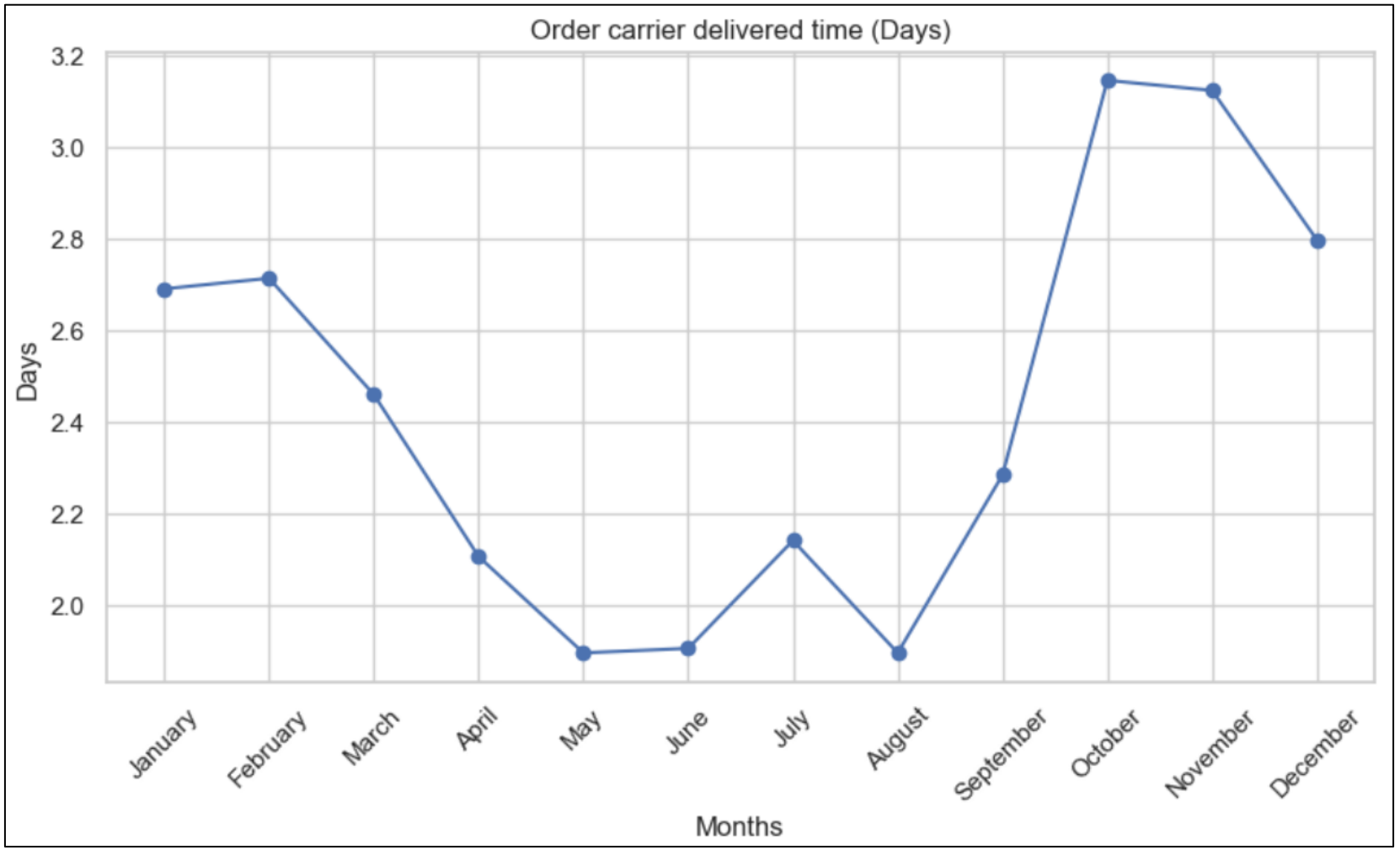
Revenue variation by hour



Average order delivery time per month



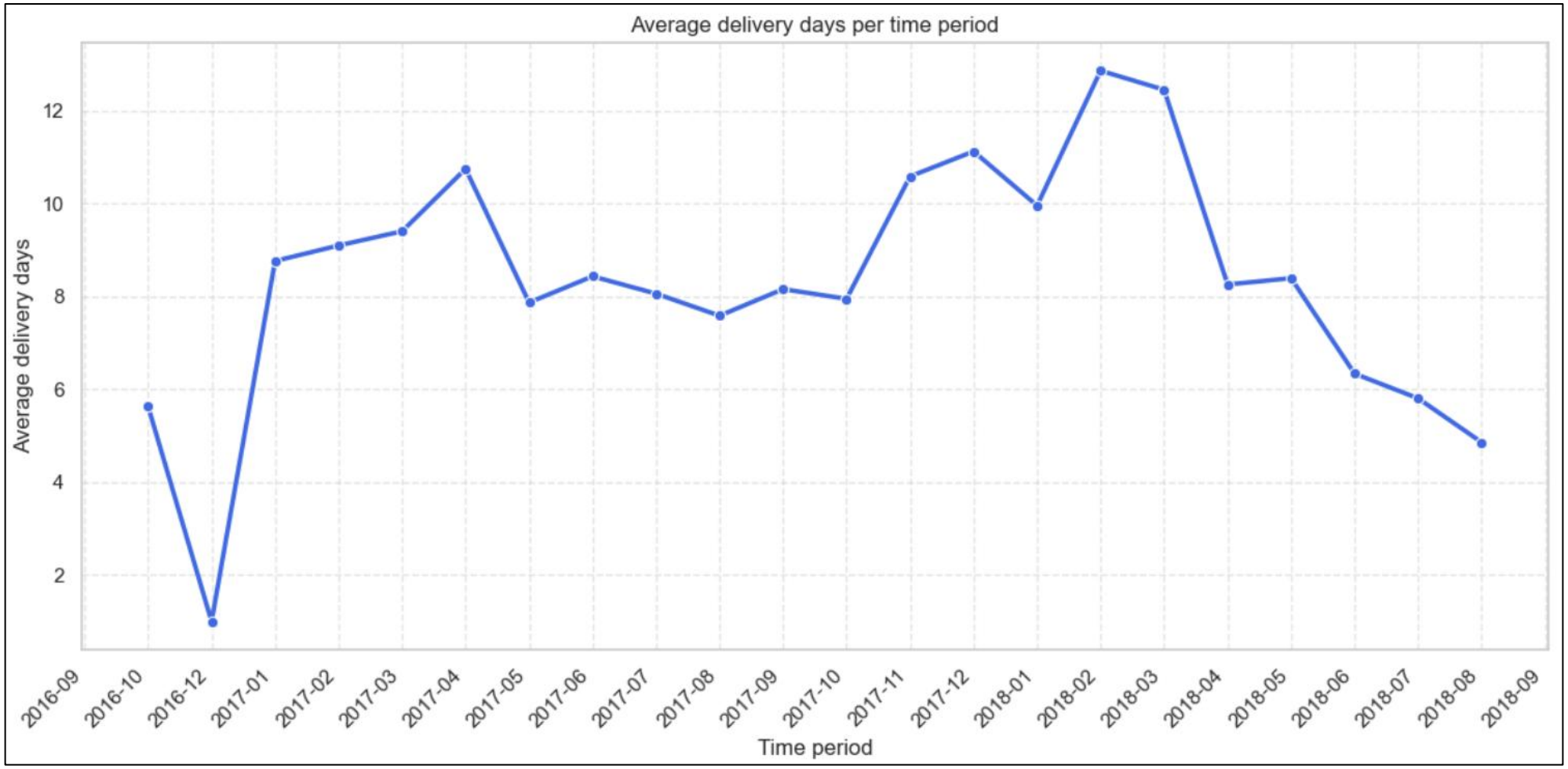
Average order carrier delivered time (days)



Total revenue trend



Average delivery time per time period



Order analysis observations

- Weekday sales and generated revenue surpass weekend sales and revenue by a big margin. Weekend & weekday sales both decline sharply after August and keeps falling until December.
- More than 89% orders are timely delivered.
- State SP contributes to the highest number of orders and highest revenue generated, while state RR contributes to the lowest number of orders and revenue.
- State RR has the highest average delivery time while SP has the lowest delivery time.
- Certain cities like Novo Brazil, Arace, Aramari etc. have an average delivery time crossing over 2 months with maximum being over 3 months.
- Total order and Total revenue trend shows a steady upward trend up until August, then had a deadly dip in September then again following the same trend.
- It was observed that both sales and revenue dropped as the weekend approaches with them being lowest on Saturday and Sunday.
- Majority of orders are placed between 11 AM and 9 PM.
- Total voucher amount: 379436.87 (Potential revenue)

Order Analysis recommendations

- **Logistics & Regional Delivery:** Proactively monitor fulfilment bottlenecks for remote states experiencing extended delivery windows (up to 2 months). Prioritize carrier contract renegotiations in high-delay regions to protect customer satisfaction.

- **Holiday Seasonality:** Average delivery times spike during the Q4 peak holiday season (Black Friday, Christmas and New Year). Mitigate carrier strain by onboarding temporary logistics staff and introducing performance incentives ahead of Q4.

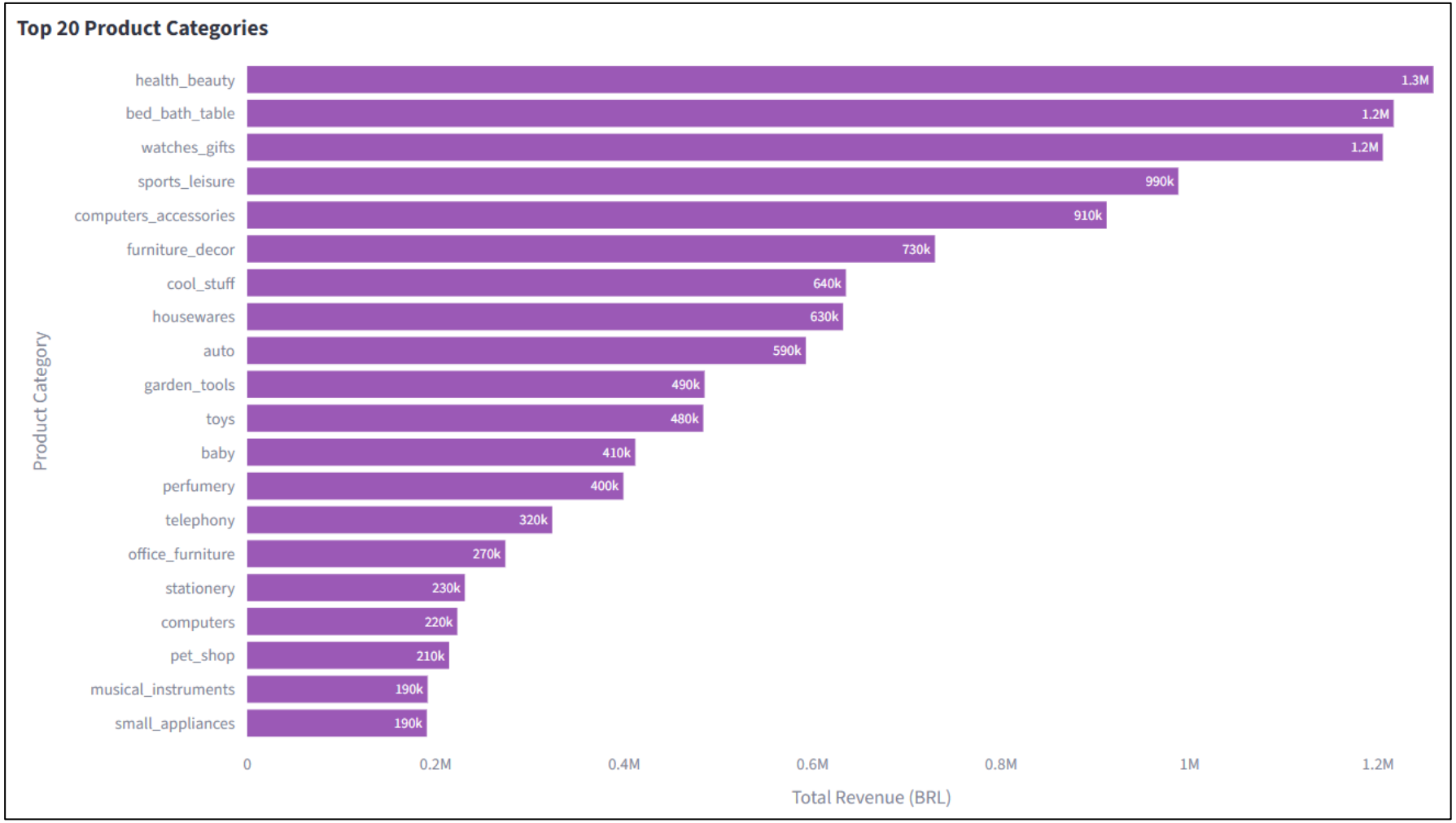
- **Targeted Marketing & Acquisition:** Deploy targeted digital acquisition and competitive first-order incentives in underperforming regions to stimulate localized demand.

- **Demand & Inventory Planning:** Capitalize on early-week shopping momentum by ensuring high-demand inventory is fully restocked by Monday morning, supported by weekend promotional push notifications.

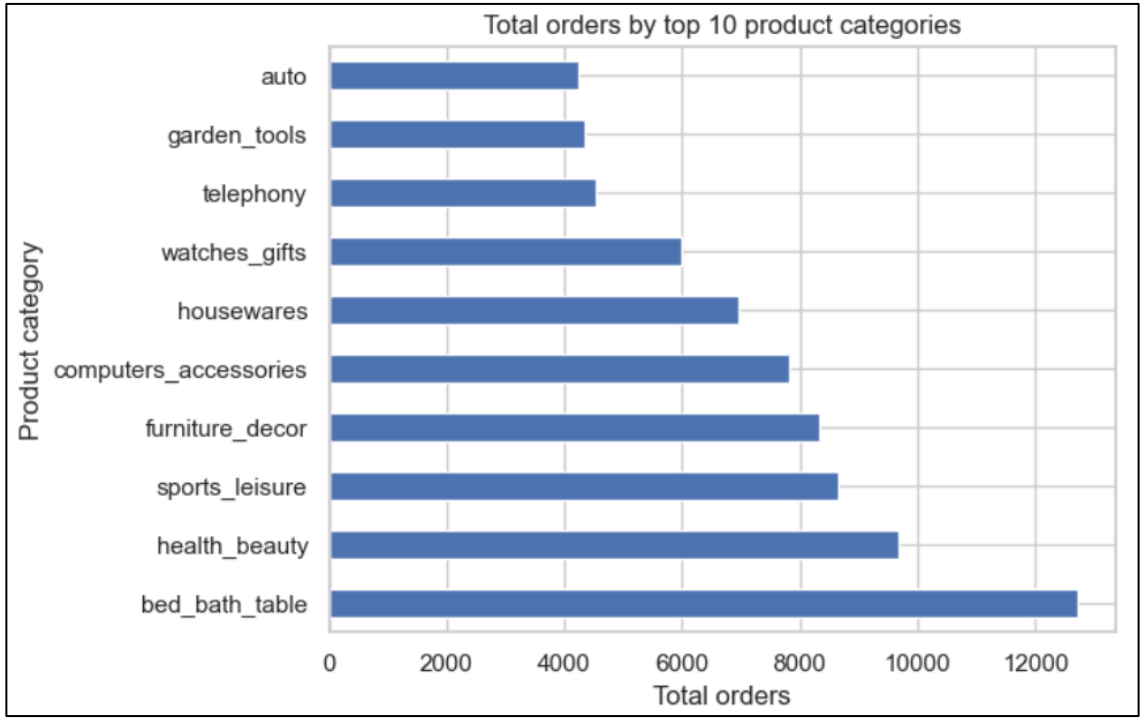


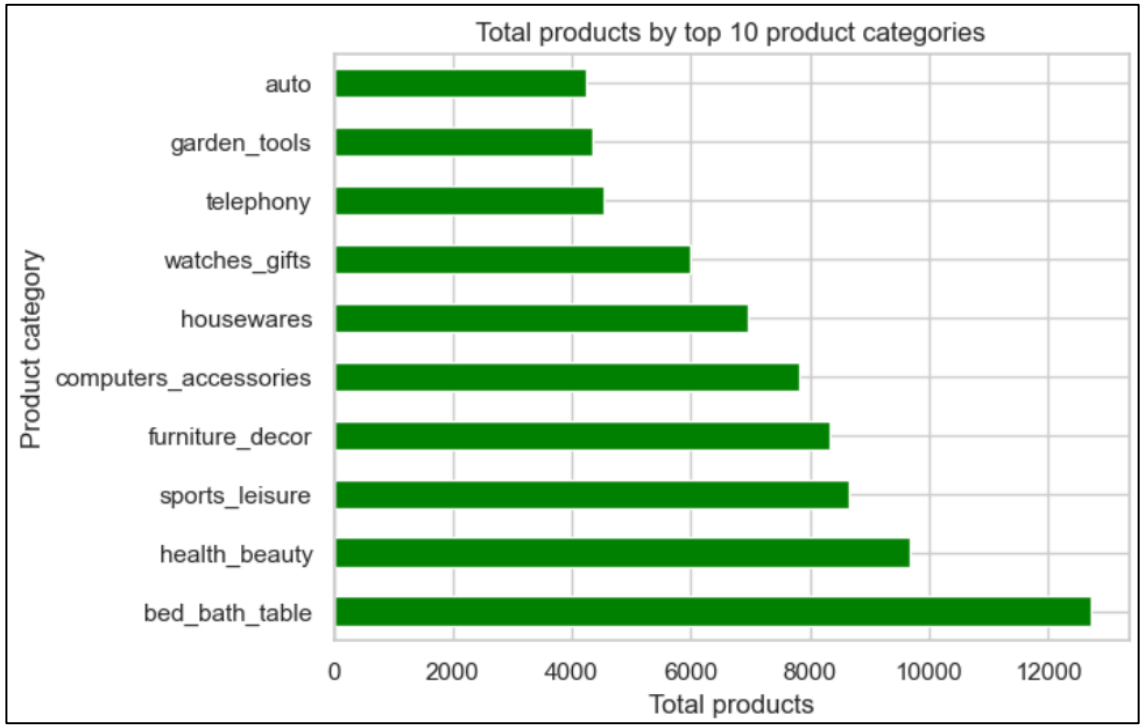
Exploratory Data Analysis Products level

Top 20 Product categories by total revenue

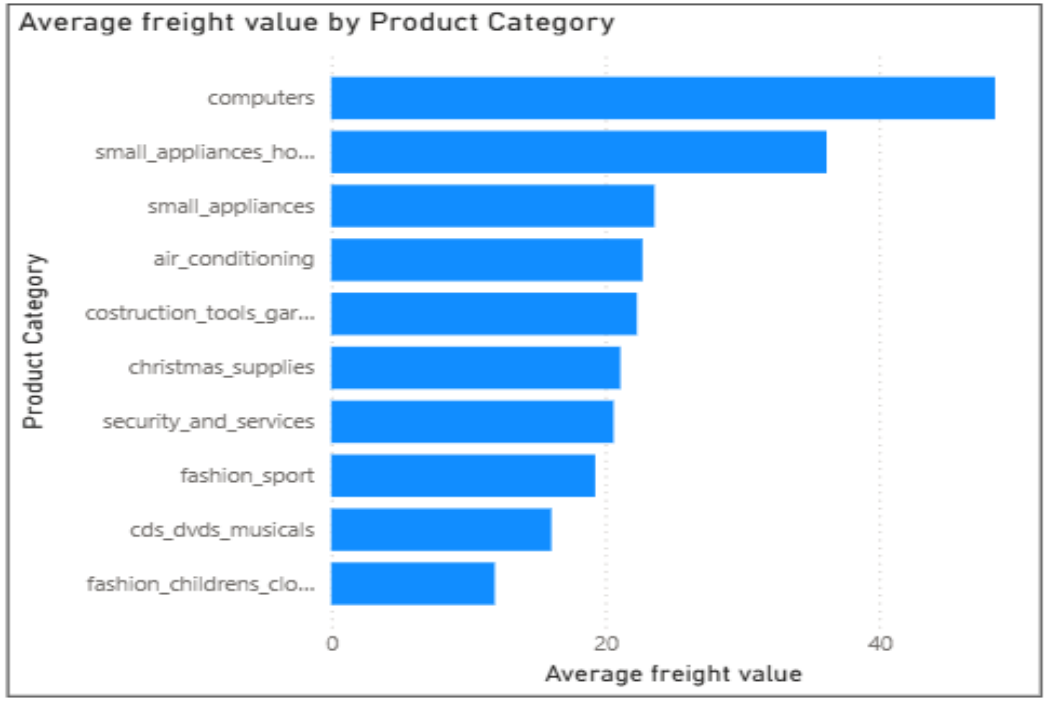


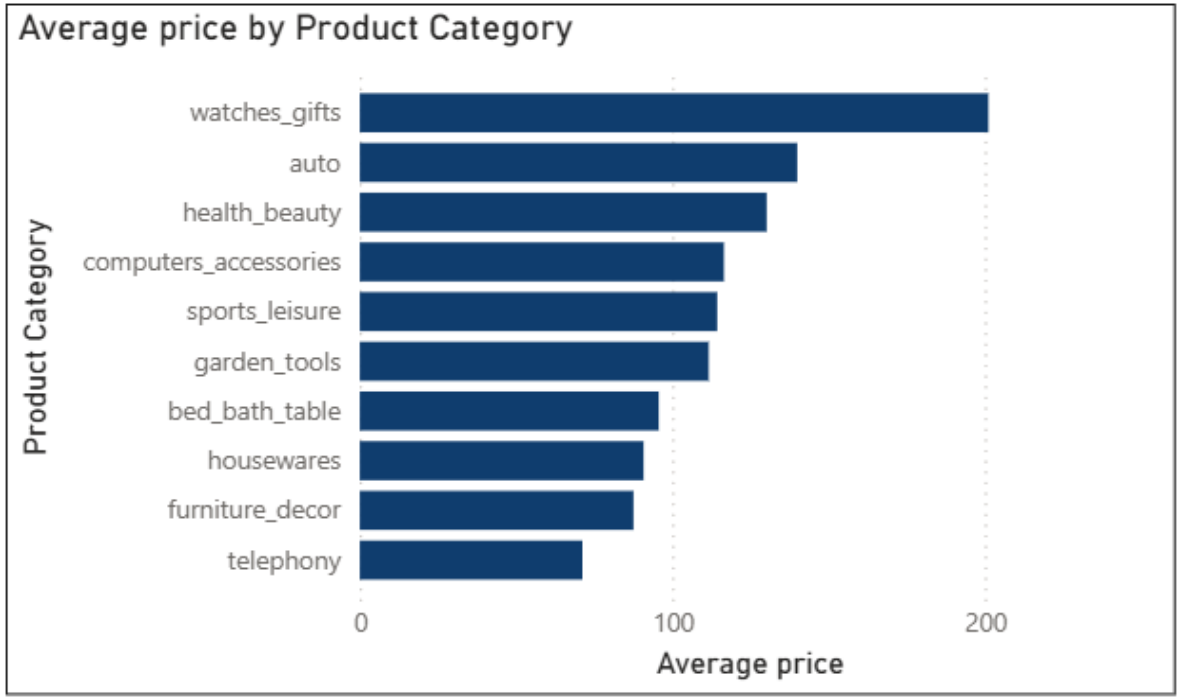
Top 10 Product categories





Product Categories (Top 10)

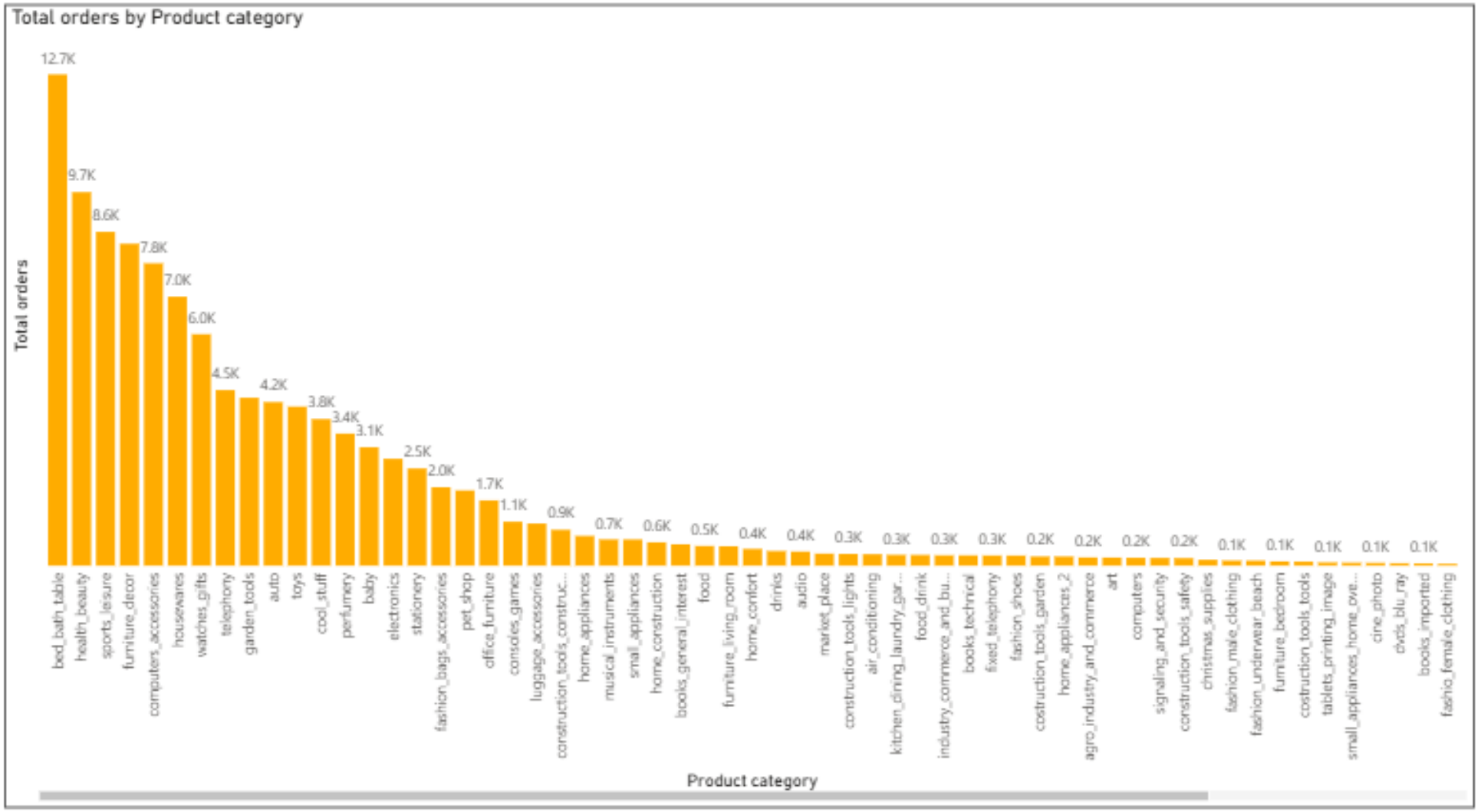




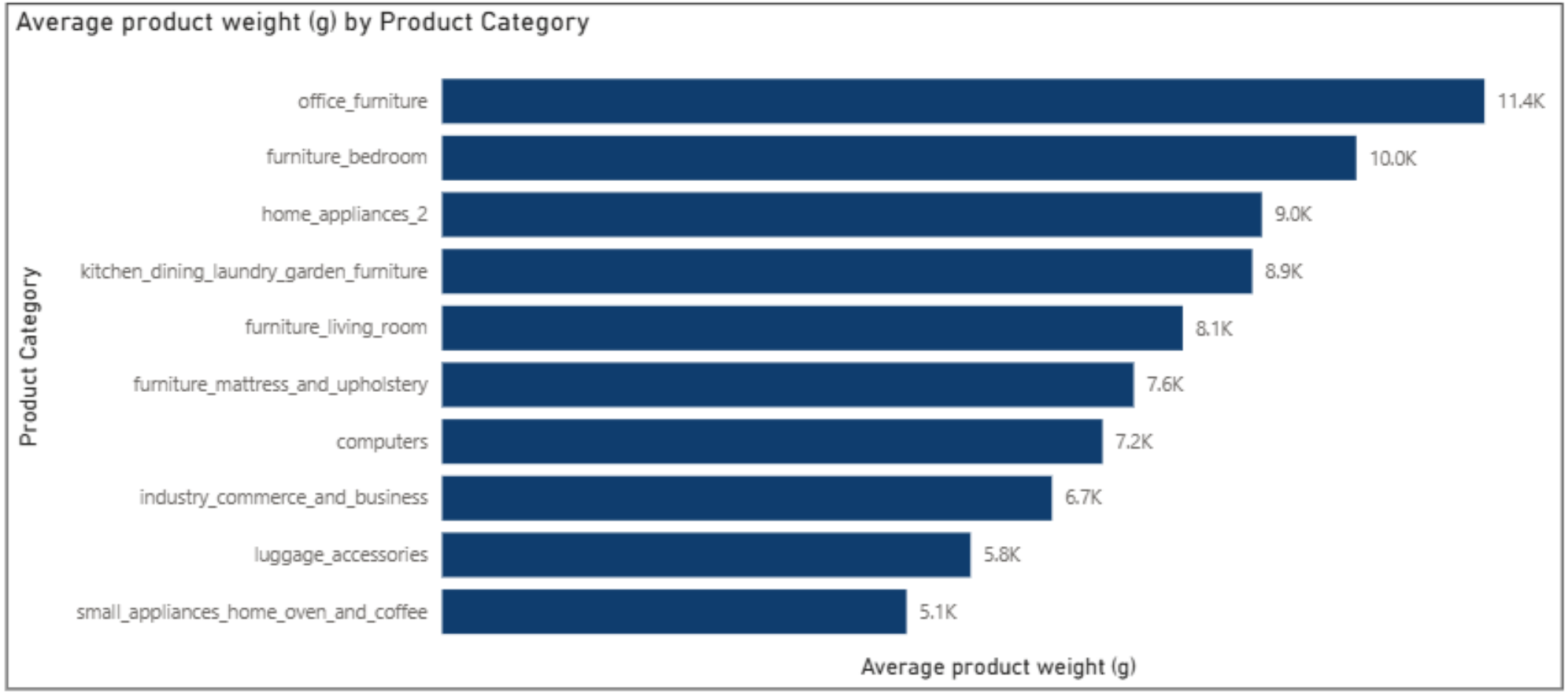
Popular products state wise

seller_state

- [ ] (Blank)
- [ ] AC
- [ ] AM
- [ ] BA
- [ ] CE
- [ ] DF
- [ ] ES
- [ ] GO
- [ ] MA
- [ ] MG
- [ ] MS
- [ ] MT
- [ ] PA
- [ ] PB
- [ ] PE
- [ ] PI
- [ ] PR
- [ ] RJ
- [ ] RN
- [ ] RO
- [ ] RS
- [ ] SC
- [ ] SE
- [ ] SP



Product Categories (Top 10)

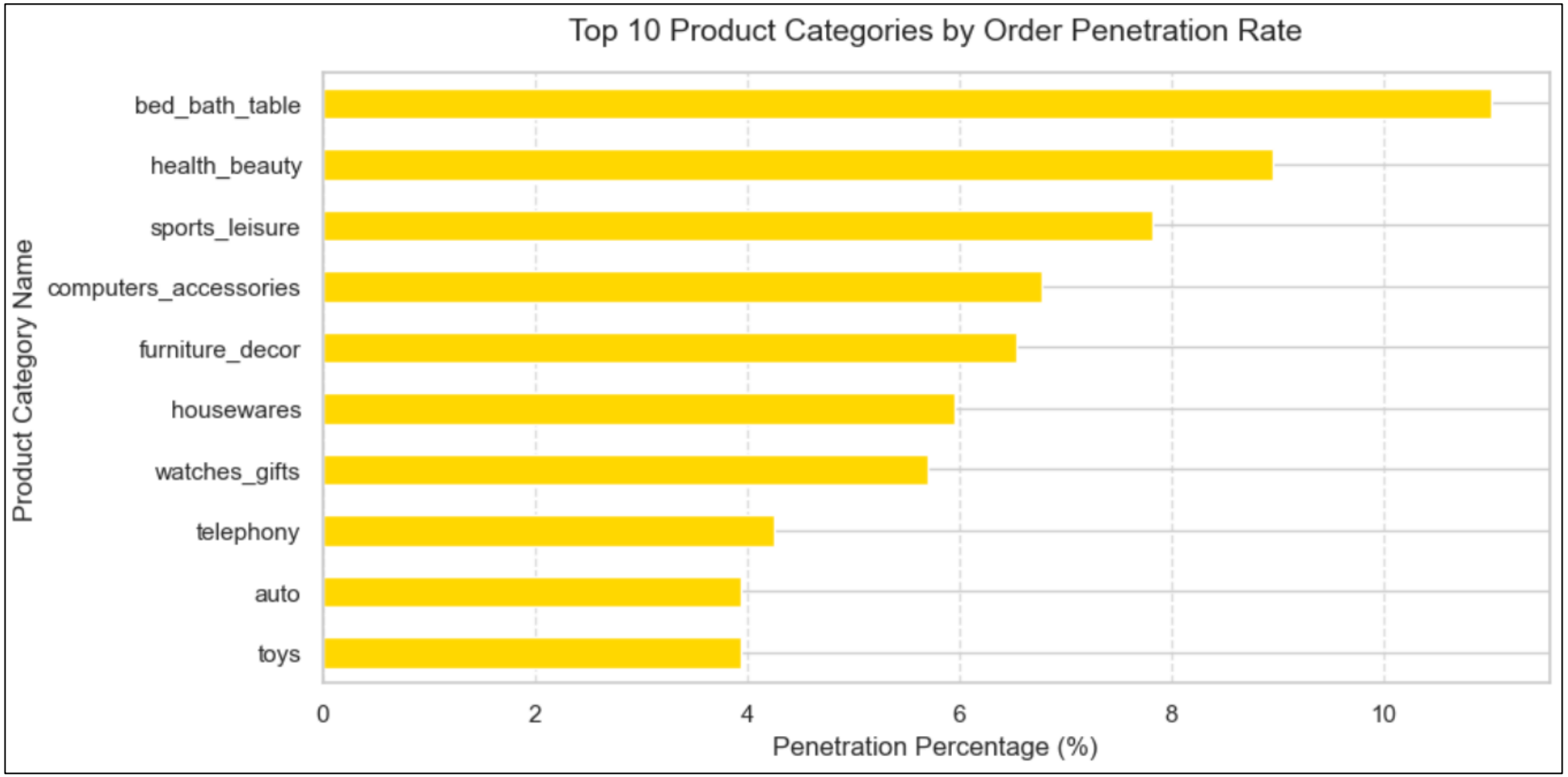


Product category penetration

	product_category_name_english	unique_orders	penetration_percentage
0	bed_bath_table	10860	11.006831
1	health_beauty	8836	8.955466
2	sports_leisure	7720	7.824377
3	computers_accessories	6689	6.779438
4	furniture_decor	6449	6.536193
5	housewares	5884	5.963554
6	watches_gifts	5624	5.700039
7	telephony	4199	4.255772
8	auto	3897	3.949689
9	toys	3886	3.938540

$$\text{Category Penetration (\%)} = \frac{\text{No. of orders having that category}}{\text{Total number of orders}} \times 100$$

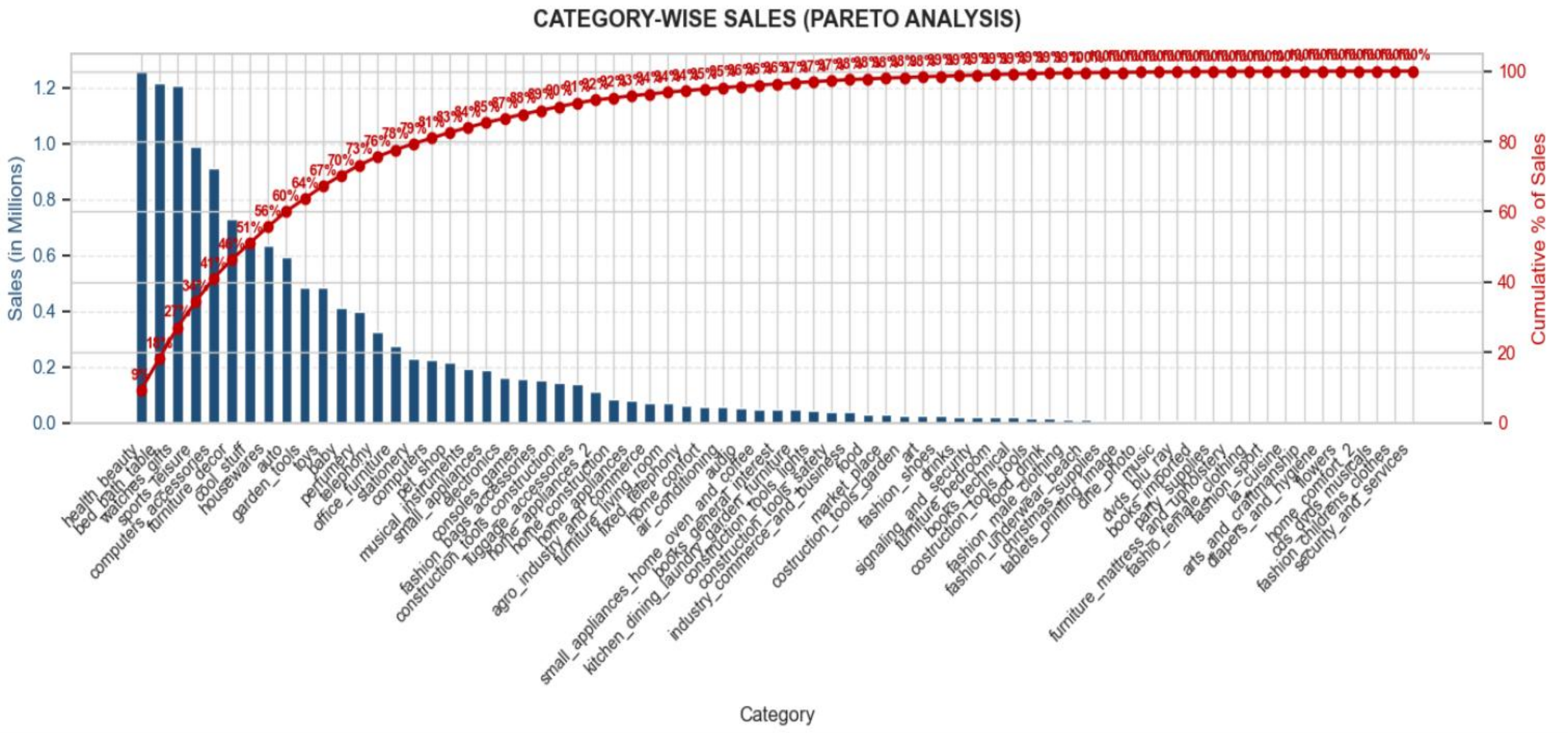
Product category penetration



Pareto Analysis

- Pareto Analysis: Pareto analysis, also known as the 80-20 analysis rule.
- It states that 80% of the effect comes from 20% of the causes.
- In this analysis we observed that 20% of categories contribute to roughly 78% of the sales over the time period of 2 years

Pareto analysis



Product price and freight value relationship



Cross selling products (with respect to bed_bath_table)

	product_category_name_english	bed_bath_table
0	bed_bath_table	1610
1	furniture_decor	73
2	home_confort	43
3	housewares	34
4	baby	17
5	health_beauty	13

Product analysis observations

- Health beauty category is the highest revenue generating category followed by bed_bath_table and watches_gift. Health beauty generates 1.22M in revenue.
- Bed bath table category dominates in number of orders and is also the category with most number of products (over 12000) across all states.
- Watches gift category has the most expensive products with an average price of BRL 200.
- Office and household furniture constitute the largest orders in terms of average dimensions and weight, whereas garden supplies have the highest average freight value.
- Bed bath table have the highest category penetration with roughly 11% products per orders belonging to bed bath table.
- Pareto analysis showed that roughly 20% of categories contribute to nearly 78% of all orders.
- Considering the highest selling category (bed_bath_table), the highest cross selling products are **furniture_decor, home_comfort,homewares, baby & health_beauty.**

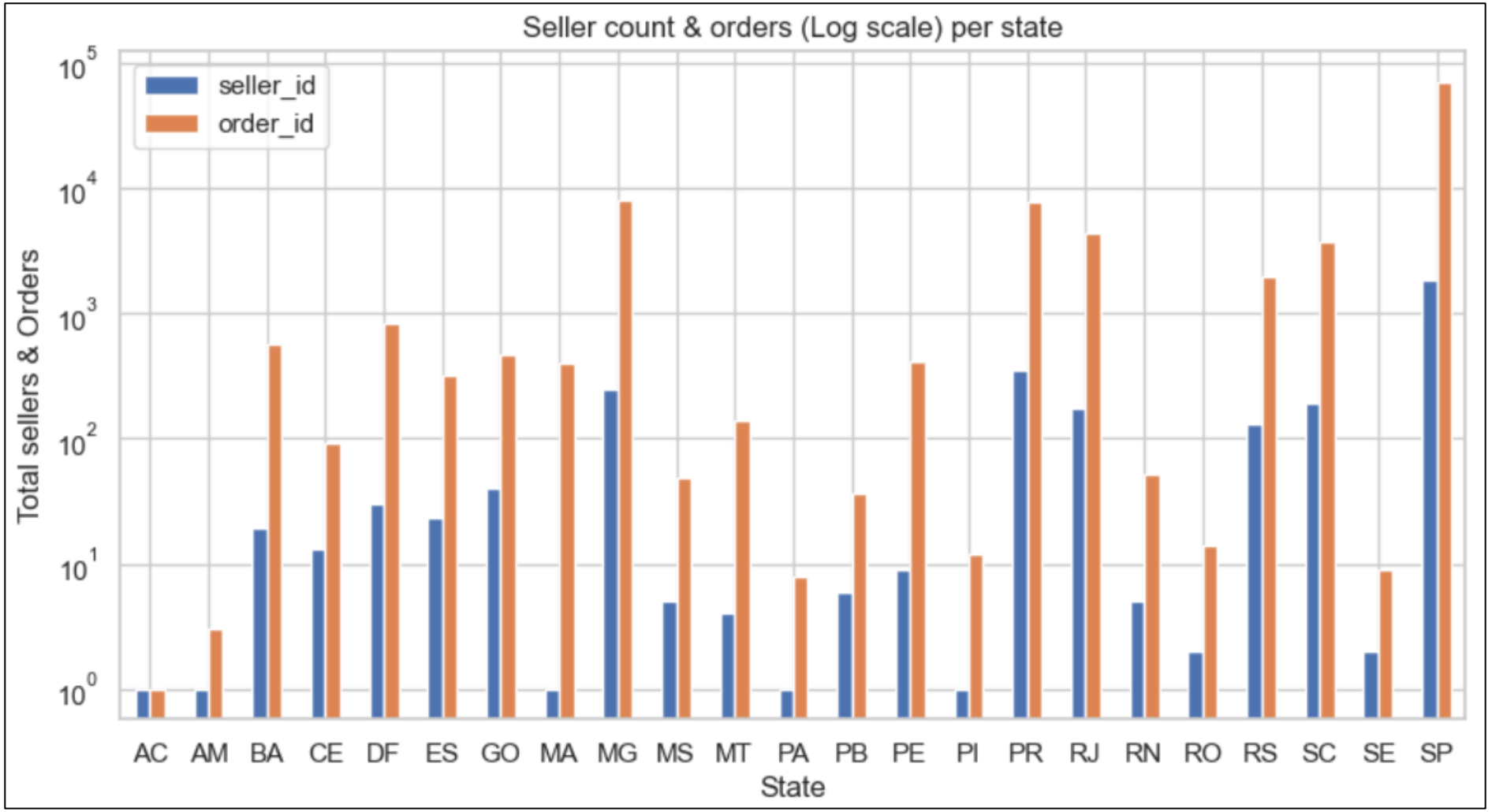
Product analysis recommendations

- **Concentrate on Class A Categories:** Focus promotional spend and seller acquisition programs on the top Pareto categories that drive the majority of platform revenue, ensuring high stock availability for high-demand items.
- **Leverage High-Penetration Anchors:** Use high-penetration categories (such as Bed, Bath & Table and Health & Beauty) as traffic drivers on landing pages, cross-promoting complementary lower-penetration items to lift overall catalogue visibility.
- **Mitigate Regional Shipping Drag:** Address disproportionately high freight-to-price ratios in remote states by partnering with regional carriers or offering localized warehousing options for bulky inventory to protect conversion rates.
- **Proactive Q4 Capacity Planning:** Average delivery times spike during peak holiday windows due to high order volume. Combat seasonal fulfilment delays by securing advance carrier agreements and onboarding temporary logistics personnel ahead of Q4.
- **Automated "Frequently Bought Together" Bundles:** Implement data-driven cross-selling modules at checkout based on co-occurrence and association rules (e.g. pairing home decor with furnishing items) to increase Average Order Value (AOV).
- **Strategic Product Bundling:** Create bundled pricing packages for low-cost, high-freight items where shipping costs otherwise deter single-unit purchases.

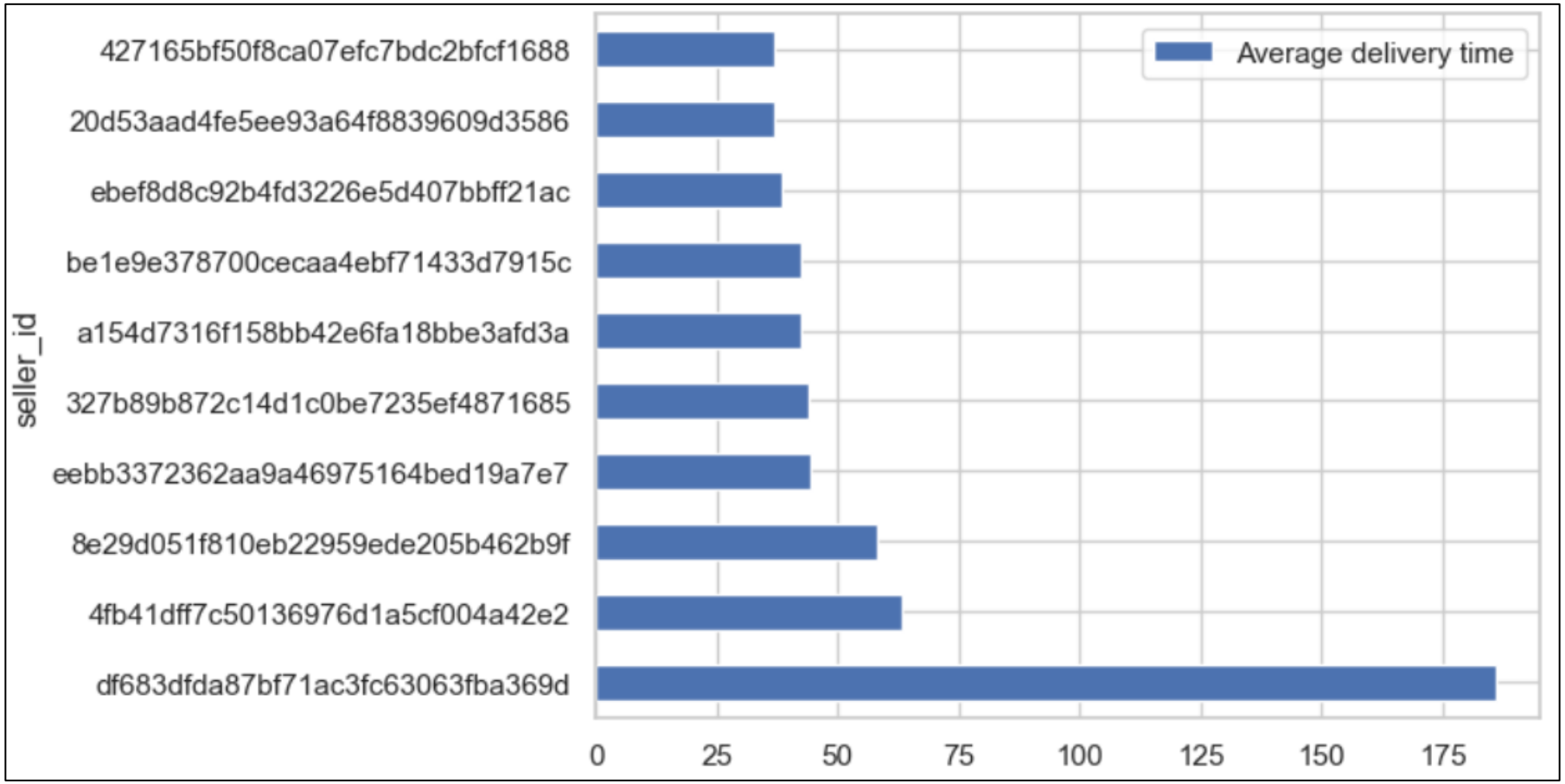


Exploratory Data Analysis Seller level

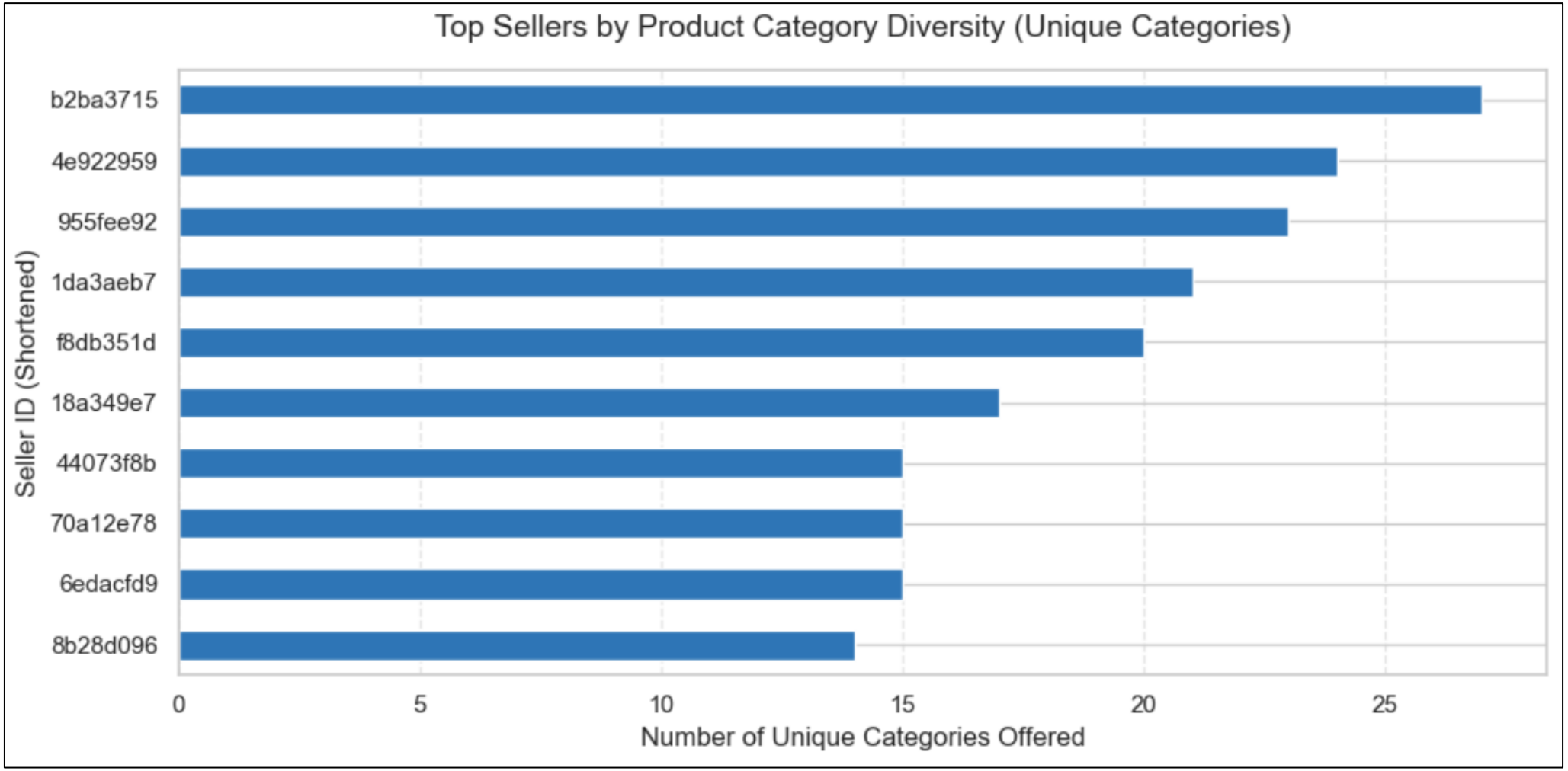
Seller level analysis



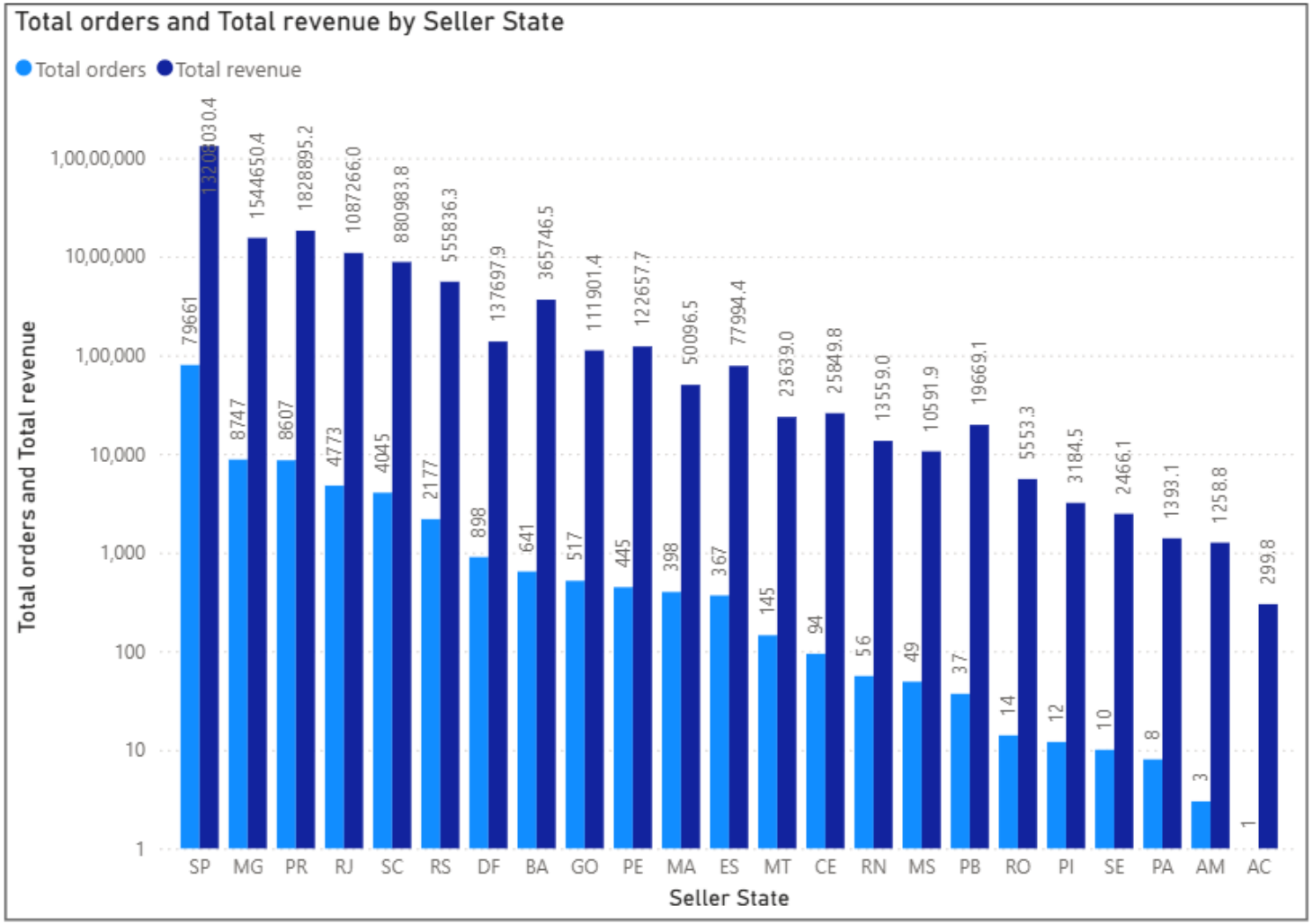
Worst performing sellers in terms of Delivery times



Seller product diversity



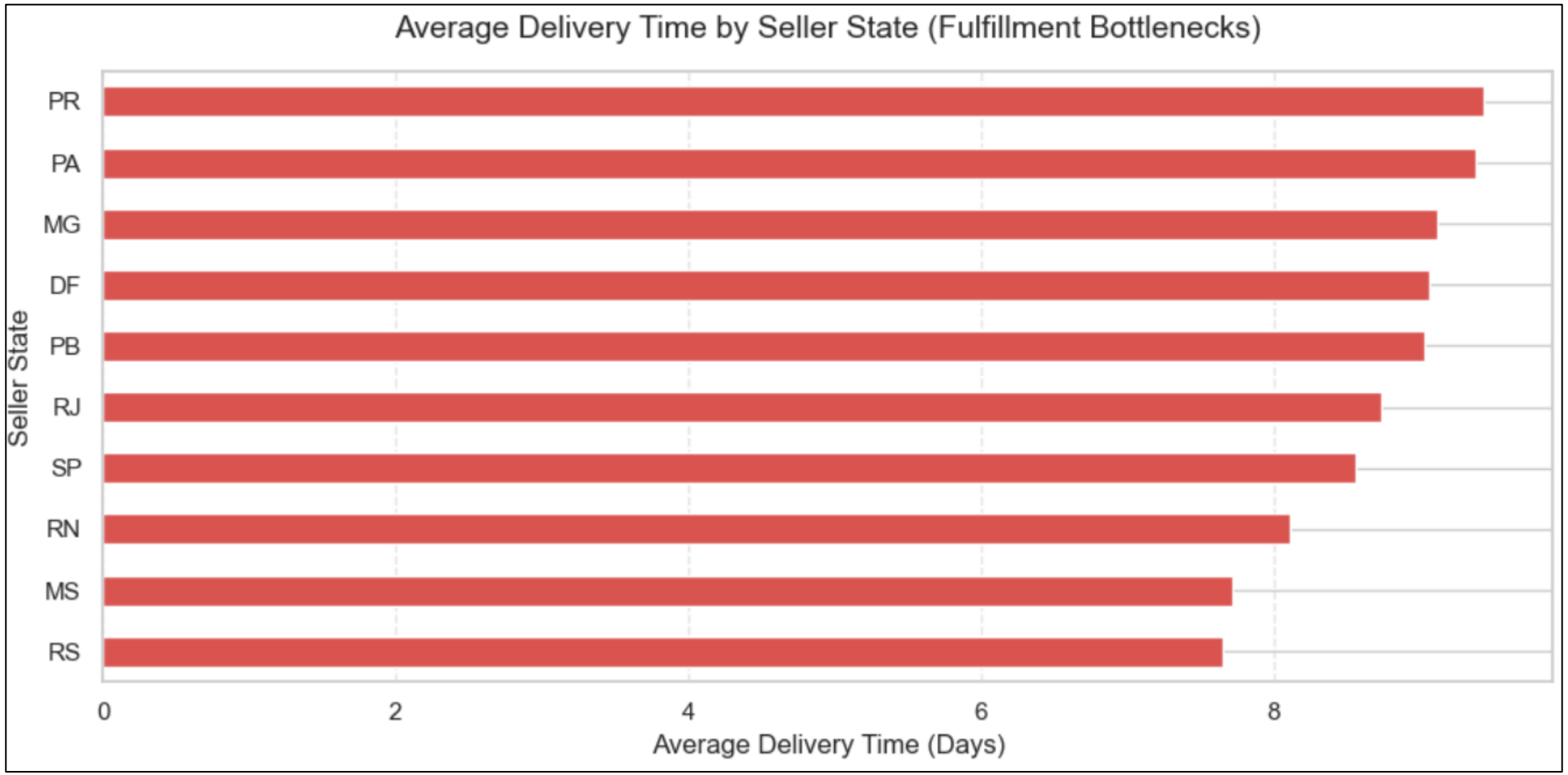
Orders and revenue per seller state



Seller delivery performance state wise (Top 10)

	seller_state	avg_delivery_time	median_delivery_time	total_orders	total_items_sold
0	AM	44.000000	24.0	3	3
1	RO	14.285714	13.0	14	14
2	CE	13.932584	11.0	90	93
3	MA	12.590078	11.0	373	386
4	PI	11.636364	12.0	12	12
5	MT	11.006993	9.0	136	144
6	PE	10.285388	8.0	399	441
7	SC	10.083185	8.0	3606	4006
8	SE	10.000000	11.0	9	10
9	BA	9.932258	7.0	565	639

Seller delivery performance state wise (Top 10)



Observations and Recommendations (Seller Analysis)

- State SP has the most number of sellers and also have the highest orders fulfilled by these sellers. On the same scale states like AM, MA, PA and PI have less number of sellers despite fulfilling more orders than some other states.
- State SP also have the highest revenue generated by sellers followed by MG, PR & RJ.
- Few sellers have order fulfilment and delivery days more than 20 days with maximum being above 6 months. The constitute the worst performing sellers.
- There are certain sellers which have high volume of orders and still manage to keep their median delivery time under 8 days.
- There is only one order from state AC in 2 years.

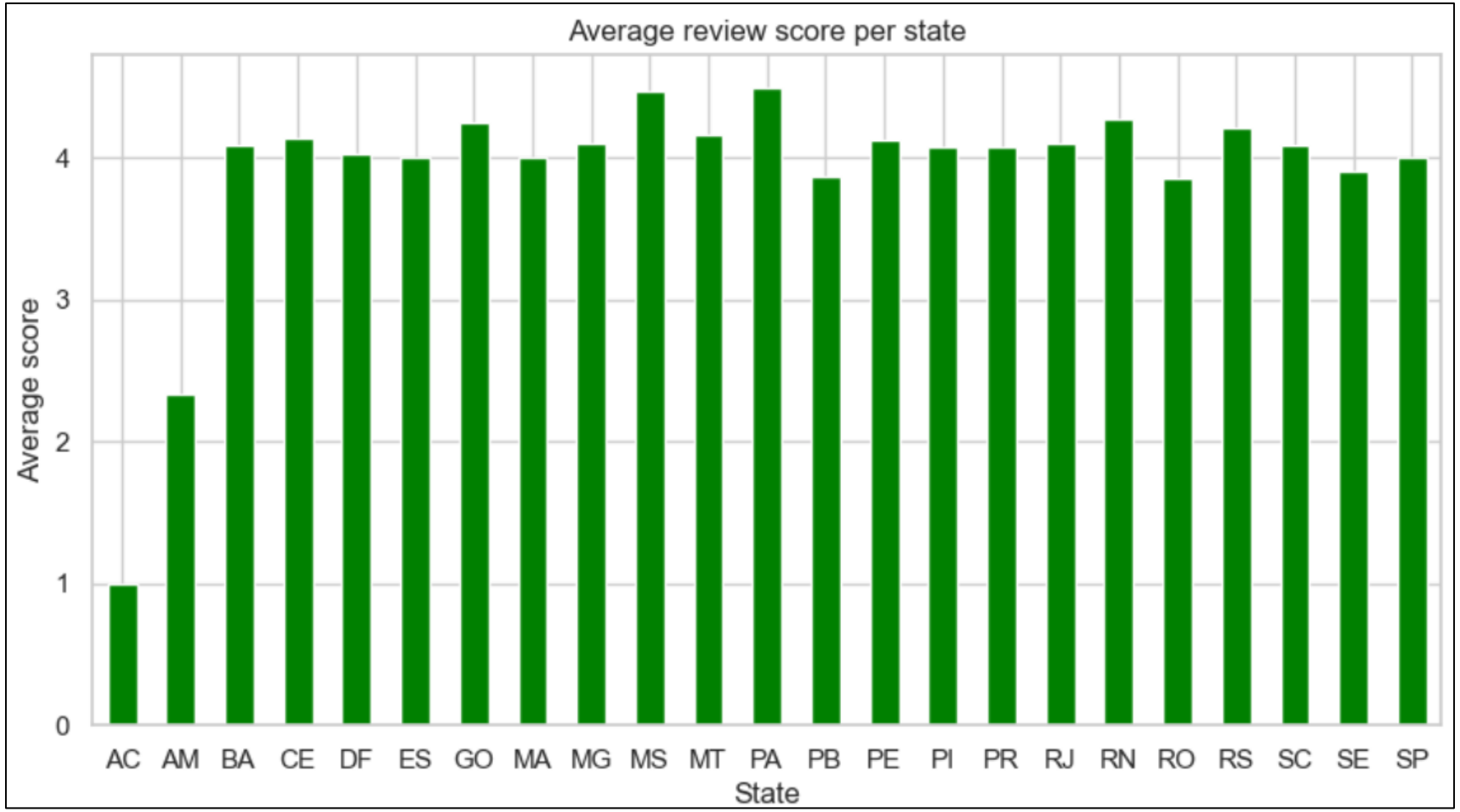
Recommendations (Sellers)

- **Geographic Expansion & Regional Hub Optimization**
  - **Decentralize fulfilment from State SP:** Given that State SP concentrates the highest density of sellers and order volume, establish localized regional fulfilment hubs or warehouses in underrepresented states (such as AM, MA, PA, and PI) where demand outpaces local seller representation. This will reduce shipping distances and transit bottlenecks.
- **Sellers SLA Enforcement & Performance Governance**
  - **Implement Strict Delivery Thresholds:** Identify and penalize the underperforming tail of sellers whose fulfillment and delivery cycles stretch beyond 20 days (and up to 6 months). Establish automated warning flags, penalize low SLA compliance with lower search visibility on the marketplace, or suspend chronically delayed vendors to protect customer satisfaction.
- **Benchmark & Scale High-Performing Vendor Practices**
  - **Study & Replicate Efficiency:** Conduct a best-practice audit on high-volume sellers who successfully maintain a median delivery time under 8 days despite heavy order loads. Create incentive structures or priority placement programs for these high-efficiency vendors and share their operational playbooks across the platform.



Exploratory Data Analysis Customer Satisfaction

Average review score per state



Observation & Recommendation

Observations

- State AC and AM have the least average rating, whereas states MS and PA have the highest rating of 4.5 indicating high customer satisfaction.
- States like SP, SC & RS also show good review score of around 4. showing consistent good score.

Recommendations:

- Prioritize improvement in low-performing states like AC & AM by identifying specific customer dissatisfaction points and implementing targeted efforts to improve customer satisfaction.
- Analyse and replicate the strategies and services contributing to high customer satisfaction in MS and PA to maintain high satisfaction levels and consider implementing similar approaches in other states.
- Enhance customer engagement, service consistency, and introduce customer feedback loops in mid performing states like SP, SC & RS to prevent ratings from falling below the 4 threshold.

Average review score across cities (Top & Bottom 10)

Top 10 review score

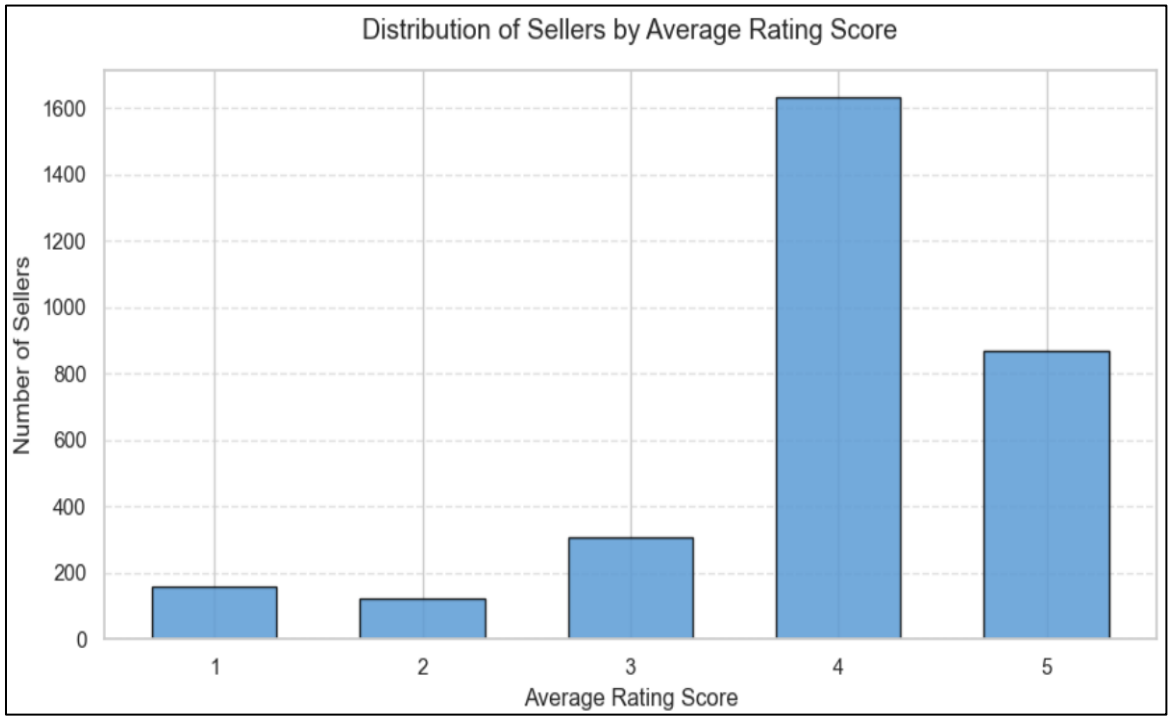
	seller_city	avg review score
0	abadia de goias	5.0
1	uruguaiana	5.0
2	laguna	5.0
3	lambari	5.0
4	minas gerais	5.0
5	messias targino	5.0
6	tocantins	5.0
7	itaporanga	5.0
8	itapui	5.0
9	ivoti	5.0

Bottom 10 review score

	seller_city	avg review score
0	soledade	1.0
1	rio branco	1.0
2	bom jesus dos perdoes	1.0
3	santo antonio da patrulha	1.0
4	pirassununga	1.0
5	orleans	1.0
6	ipua	1.0
7	carmo da mata	1.0
8	marapoama	1.0
9	miguelopolis	1.0

Average review score by Order count & Sellers





Observations & Recommendations

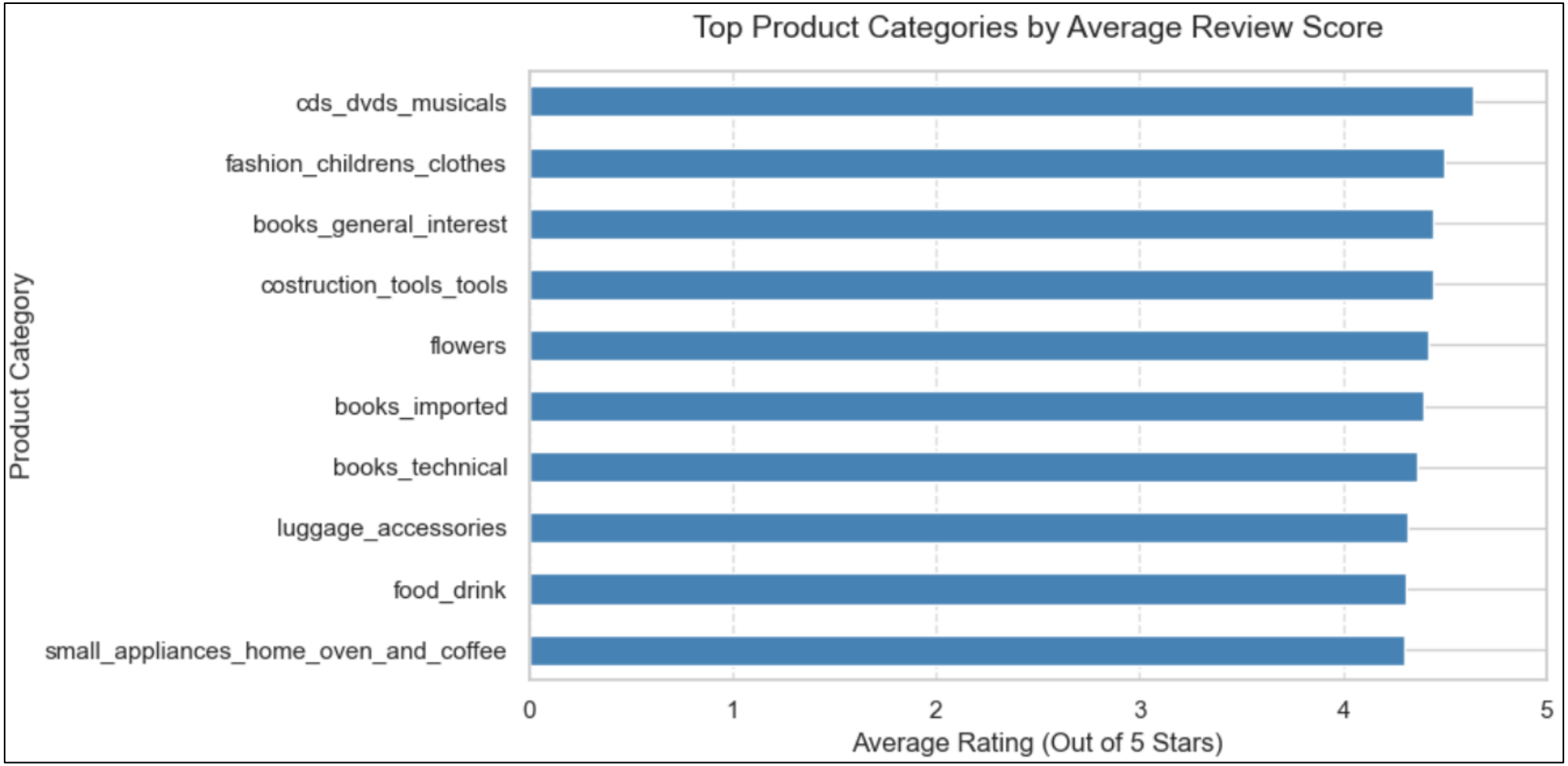
Observations:

- Review score of 5 dominates majority of orders while rating 2 is the least provided review. There are about 60,000 records which were given 5 rating while there were around 11,000 records with review score of 1.
- More than 1600 sellers have an average rating of 4. While around 180 sellers have an average rating of 1. Around 900 sellers have an average rating of 5.

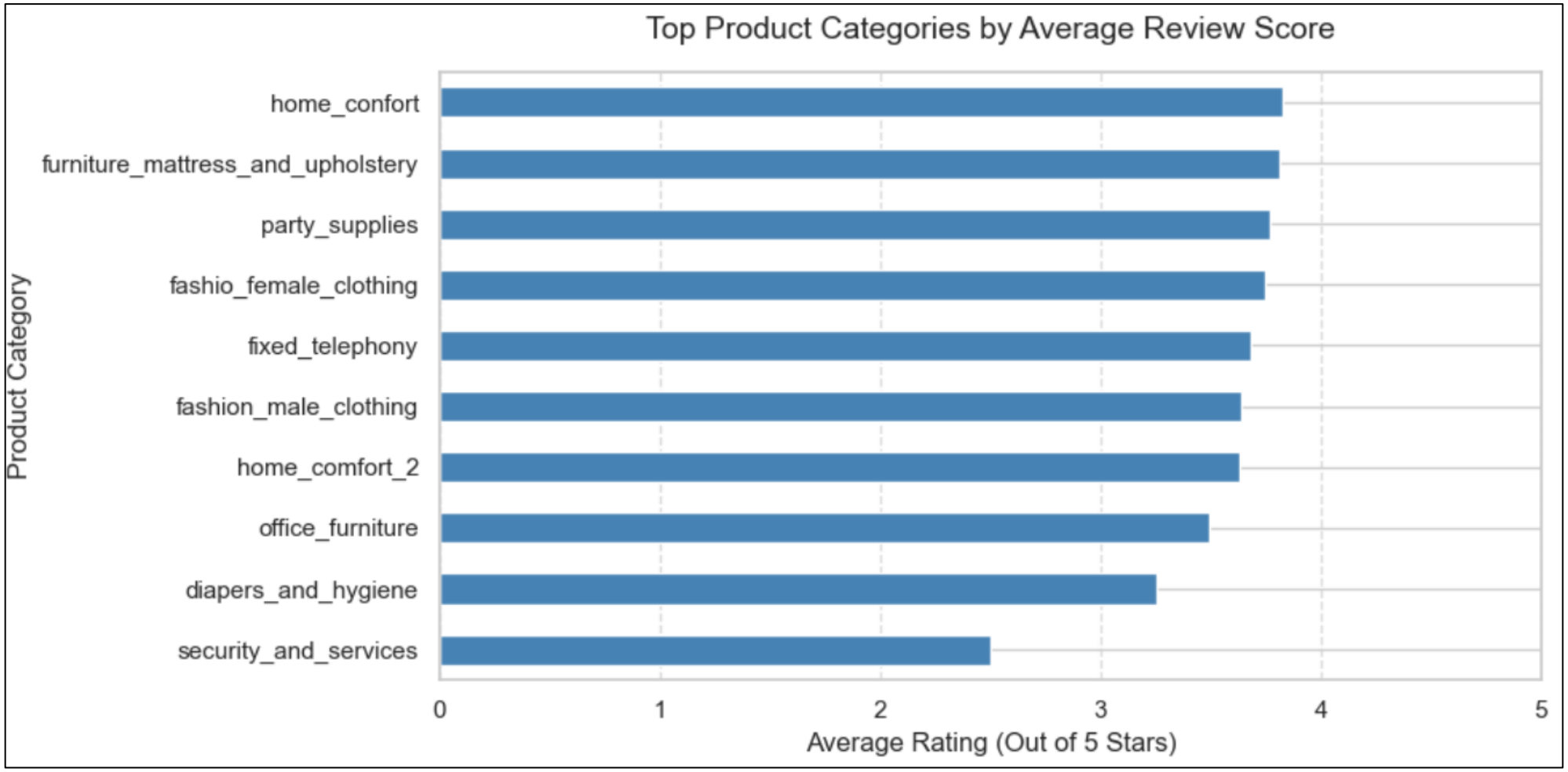
Recommendations:

- Orders with low rating should be analyzed based on the their review title and title description (if provided) to understand the reason for low rating.
- A customer feedback report should be created after asking the customer for the low rating and rectify the issue to prevent future lower ratings.
- For sellers having a low rating, stricter checks should be done on them and they should be provided with measures to improve.

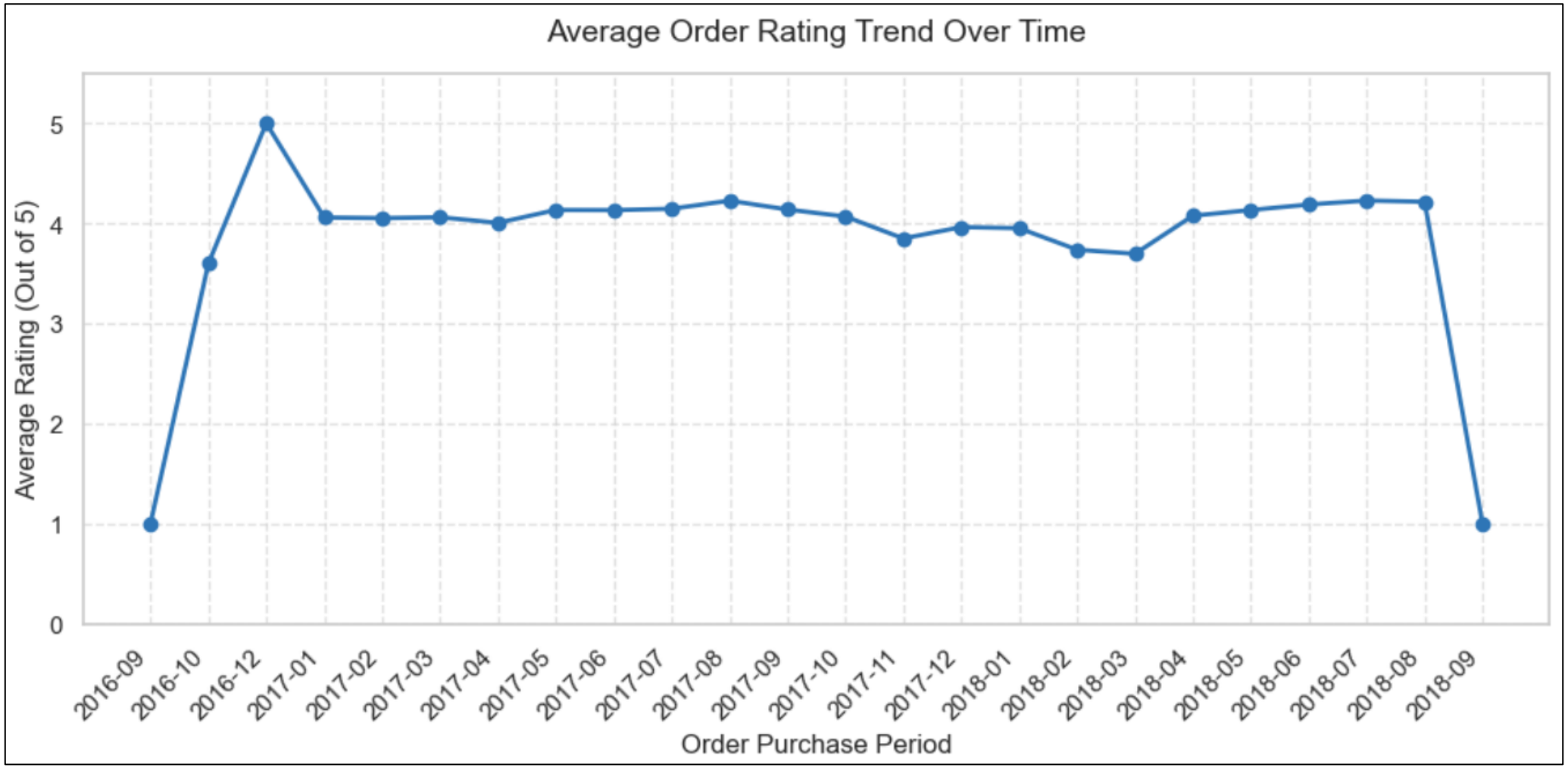
Average rating per category (Top 10)



Average rating per category (Bottom 10)



Average category per time period



Observations and recommendations

Observations:

- “Cds_dvds_musicals” is the highest rated category with an average rating of about 4.5 meanwhile “security_and_services” is the least rated category with average rating of 2.5.
- December 2016 had the highest rating of 5 across all orders while October and November 2016 received the lowest average rating of 1 across all orders. All other months in both years have had an average rating of 4 showing majority of orders across months have a consistent high rating.

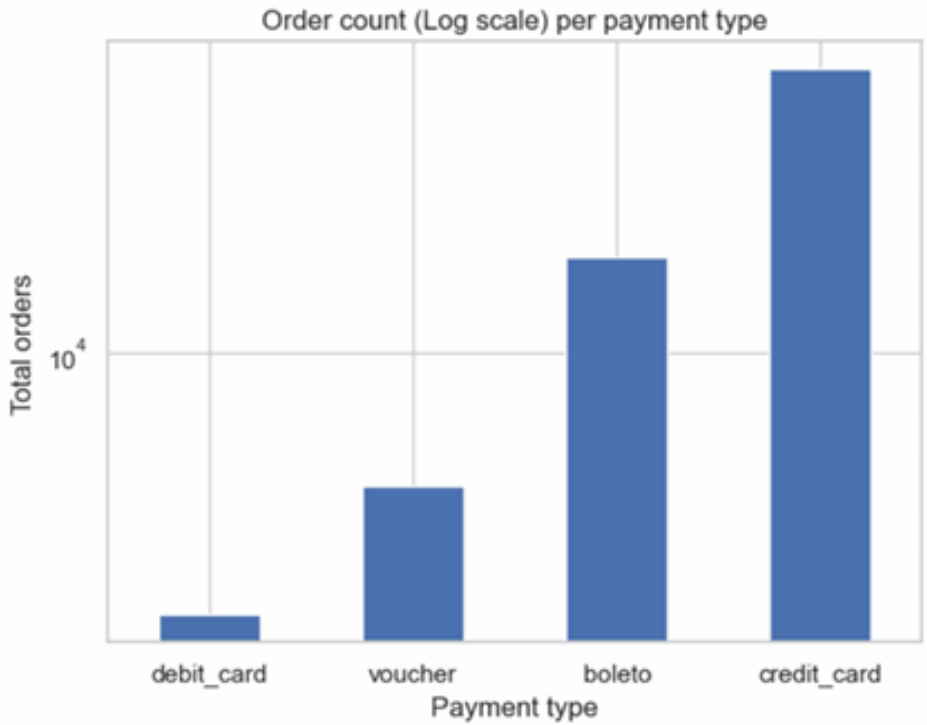
Recommendations:

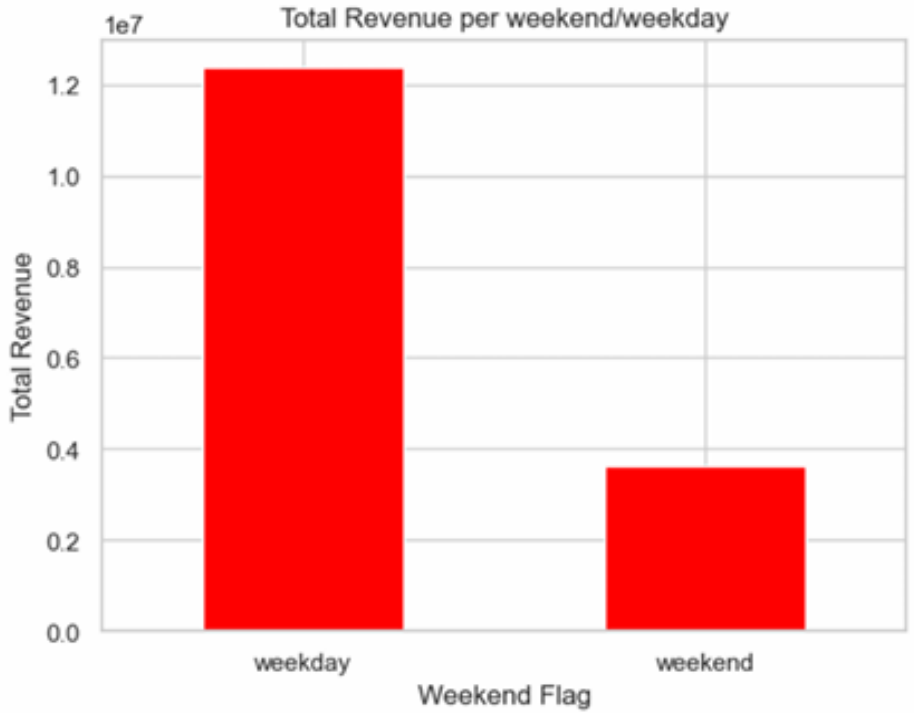
- Consider improving product quality, delivery times, or customer service in the to boost satisfaction.
- Sustain high performance in musicals, fashion and books by rewarding loyal customers with promotions or discounts to strengthen loyalty.
- Launch customer satisfaction improvement programs across all categories with a focus on personalization, better customer service, and optimized shopping experiences.



Dashboard Previews

Brazilian E-Commerce - olist, Executive Summary & Business Insights





Customer level dashboard

The dashboard shows report at customer level. It consists of the following:

- **Relevant KPIs:** Unique Customers, Total states, Total cities & Total zip codes of locations.
- **Dynamic slicers:** Month and cities.
- **Relevant charts:** Total customers by time period, RFM Segmentation, Payment preference by customers, Customers per revenue segment and Total customers per state.
- **Insights shown:** Customer demographics and preferences along with customer categorization.

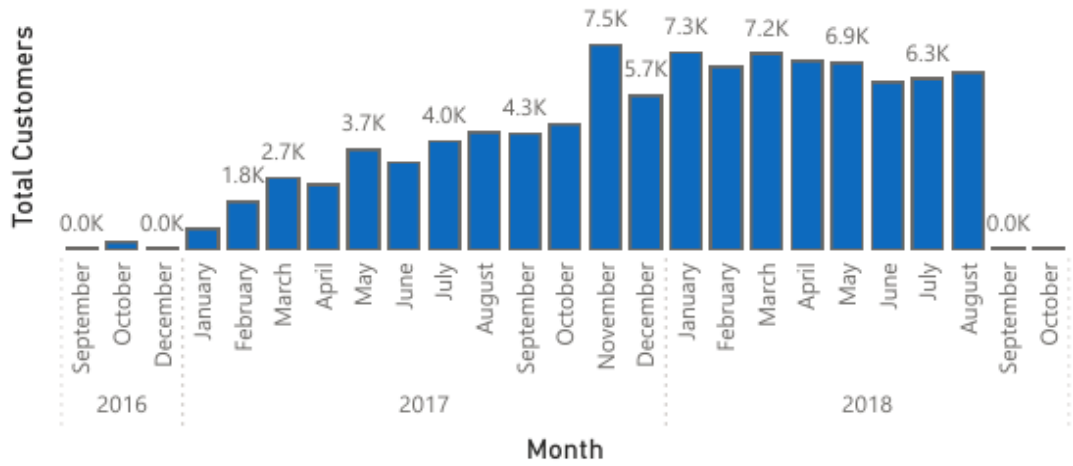
olist Customer Analysis & Behavior

Clear all slicers

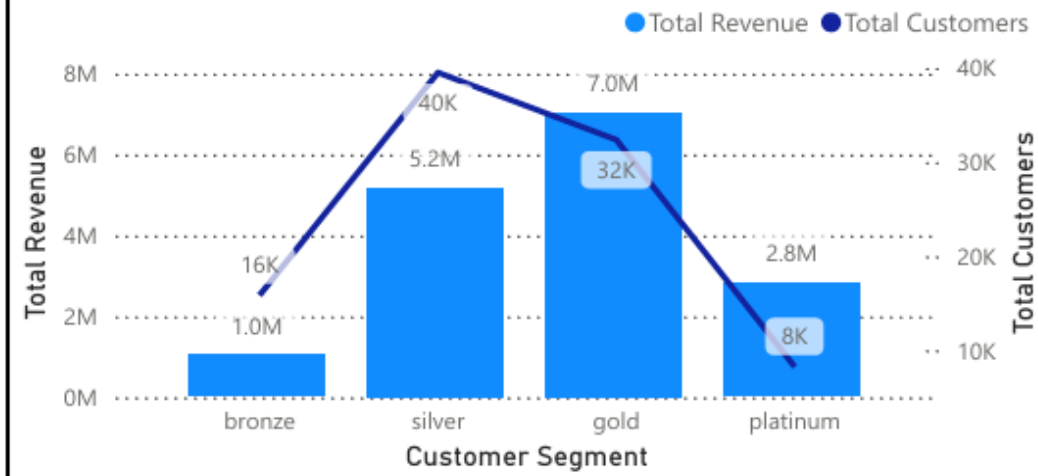
geolocation_city
- abadia dos doura...
- abadiania
- abaete
- abaetetuba
- abaiara
- abaira
- abare
- abatia
- abdon batista
- abelardo luz
- abreu e lima

96.10K	27	4119	14.99K
Total Customers	Total States	Total Cities	Total Zip codes

Total Customers by Year and Month

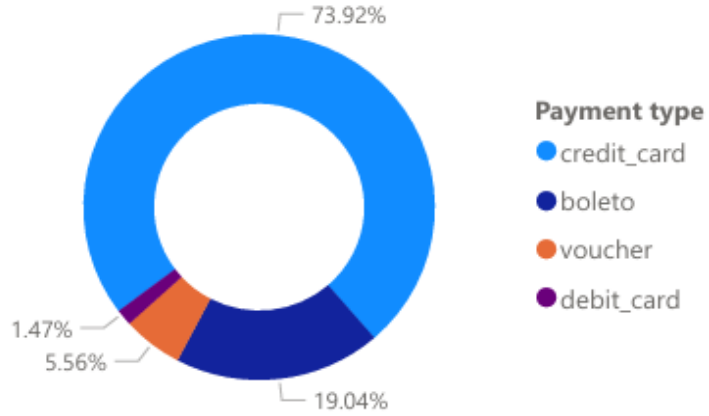


Total customer & Revenue per segment

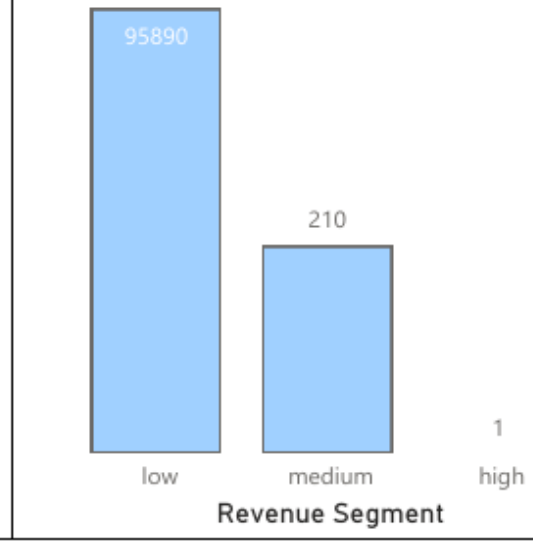


month
- January
- February
- March
- April
- May
- June
- July
- August
- September
- October

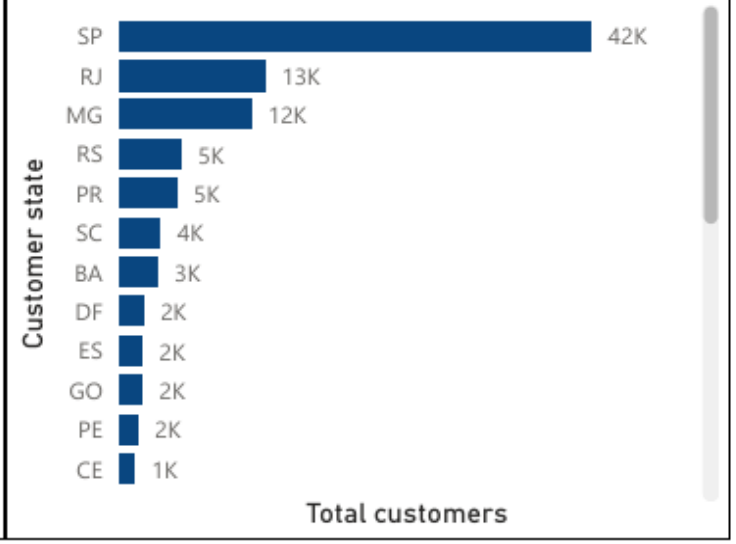
Customer preference by Payment type



Total customers by Revenue Segment



Total customers by Customer state



Order level Dashboard

The dashboard shows report at Orders level. It consists of the following:

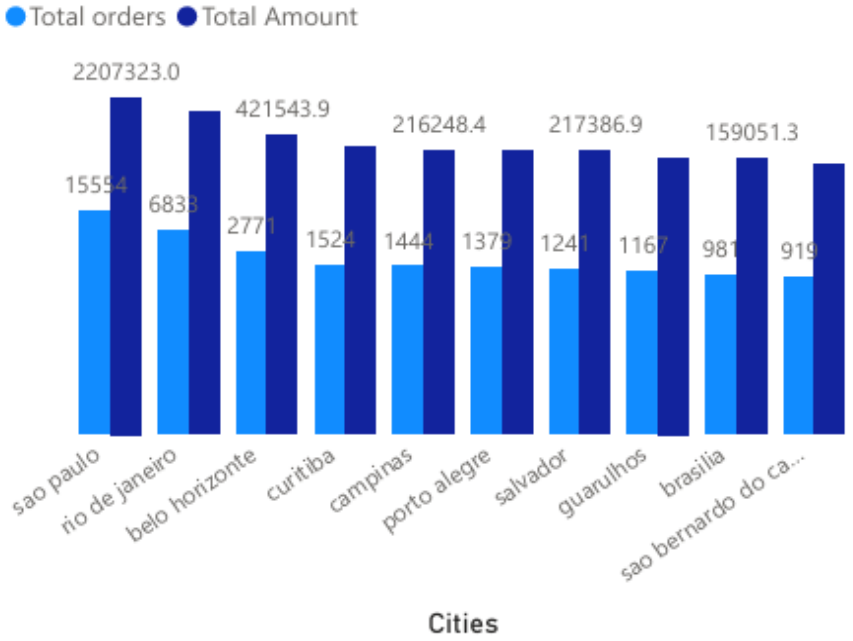
- **Relevant KPIs:** Total orders, Total revenue, Total payment types
- **Dynamic slicers:** Months
- **Relevant charts:** Total orders & Total revenue by cities, Average delivery time by state, Percentage variation of order delivery status, Total orders by state, Month-wise total orders per weekend flag.
- **Insights shown:** Order variation across different demographic sectors, Total revenue generated by orders & how orders vary across time period

olist Order level Analysis

Clear all slicers

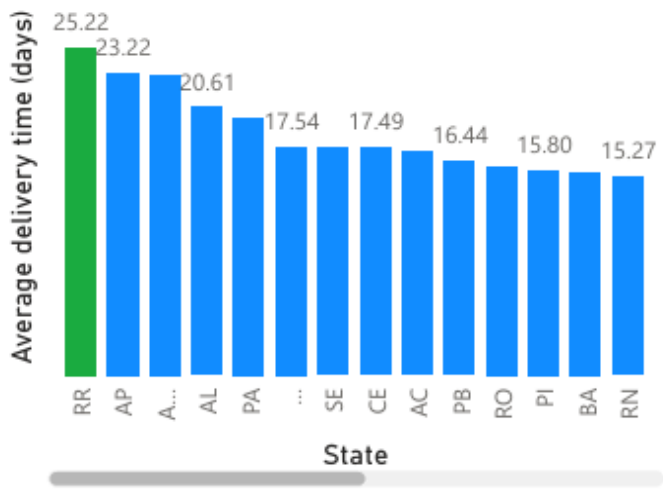
month
- January
- February
- March
- April
- May
- June
- July
- August
- September
- October
- November
- December

Total orders and Total Amount by Cities

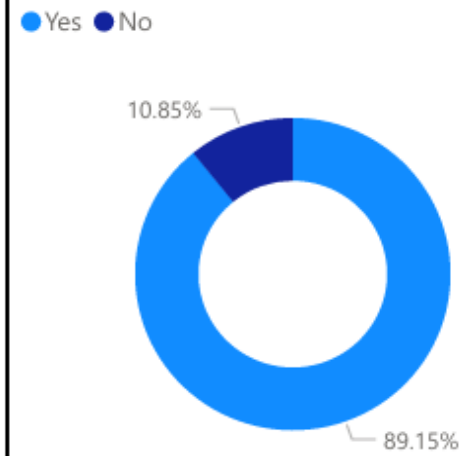


99.44K	16.01M	5
Total Orders	Total Revenue	Total Payment types

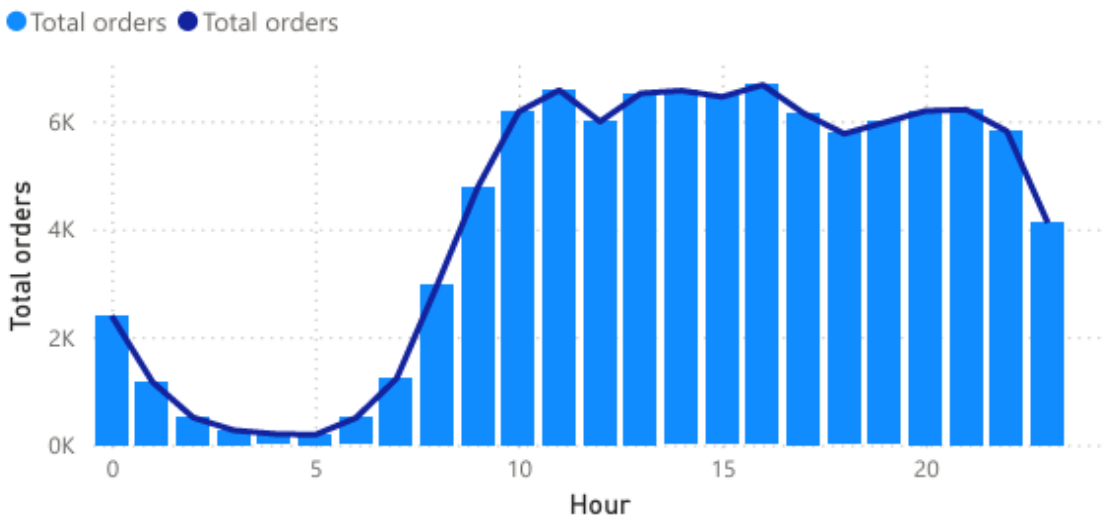
Average delivery time (days) by State



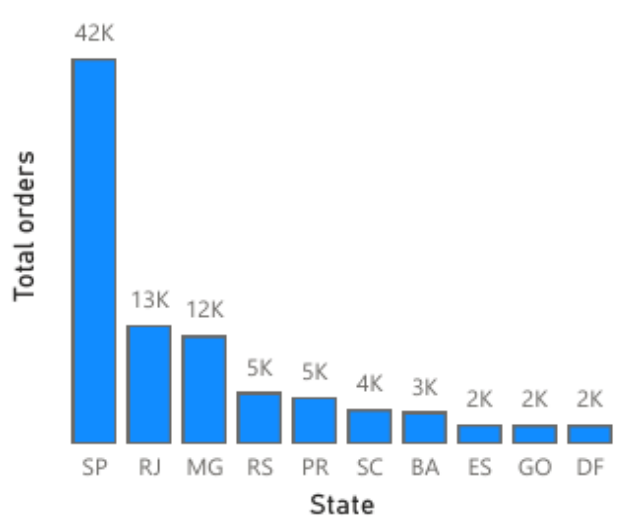
Timely delivered order variation



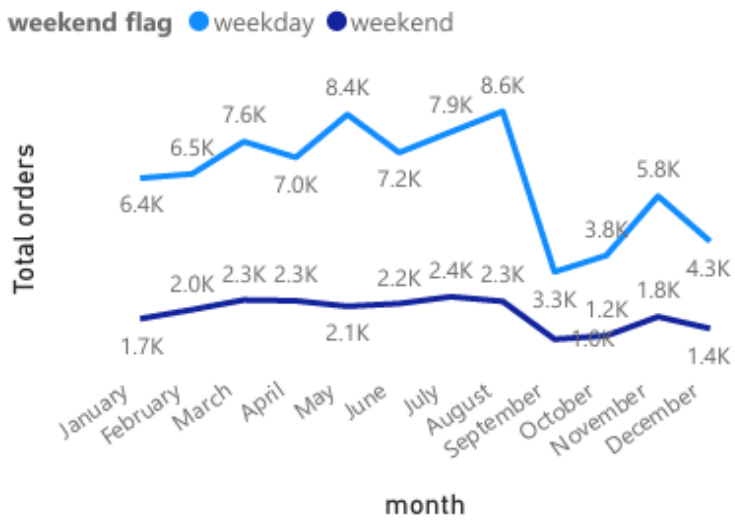
Total orders by Hour



Total orders by State (Top10)



Total orders per weekflag by Month



Sales trend Dashboard

The dashboard shows report of sales trends. It consists of the following:

- **Dynamic slicers:** State, City, weekend flag, timely delivered status.
- **Relevant charts:** Total orders by month, Total orders by day of week, Average delivery days by month, Average delivery days by time period (Month & Years).
- **Insights shown:** This dashboard shows how orders, revenue, delivery days etc vary across time at different levels.

olist Sales Trends Analysis

Clear all slicers

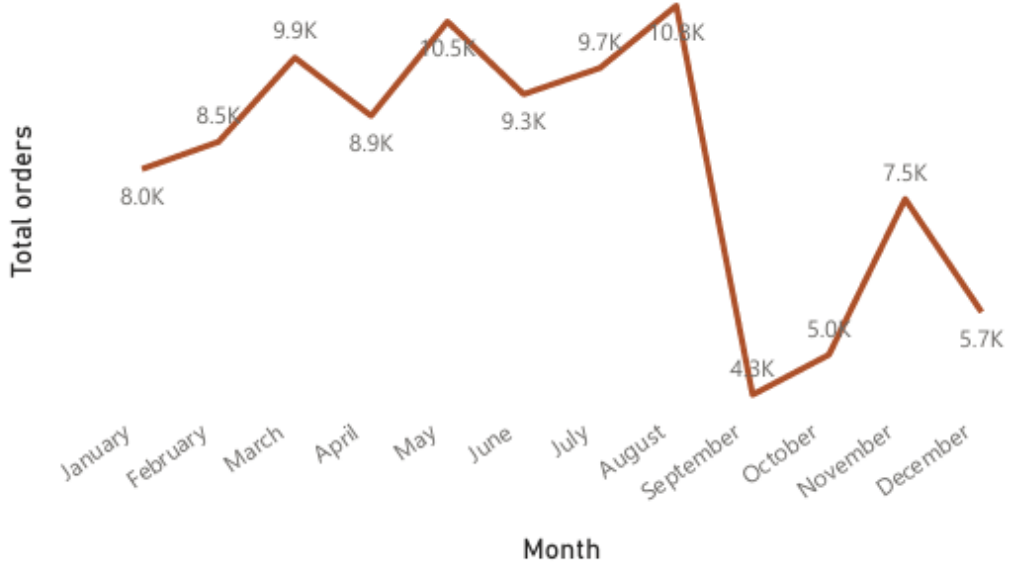
State
- AC
- AL
- AM
- AP
- BA
- CE
- DF
- ES

City
- abadia dos dourados
- abadiania
- abaete
- abaetetuba
- abaiara
- abaira

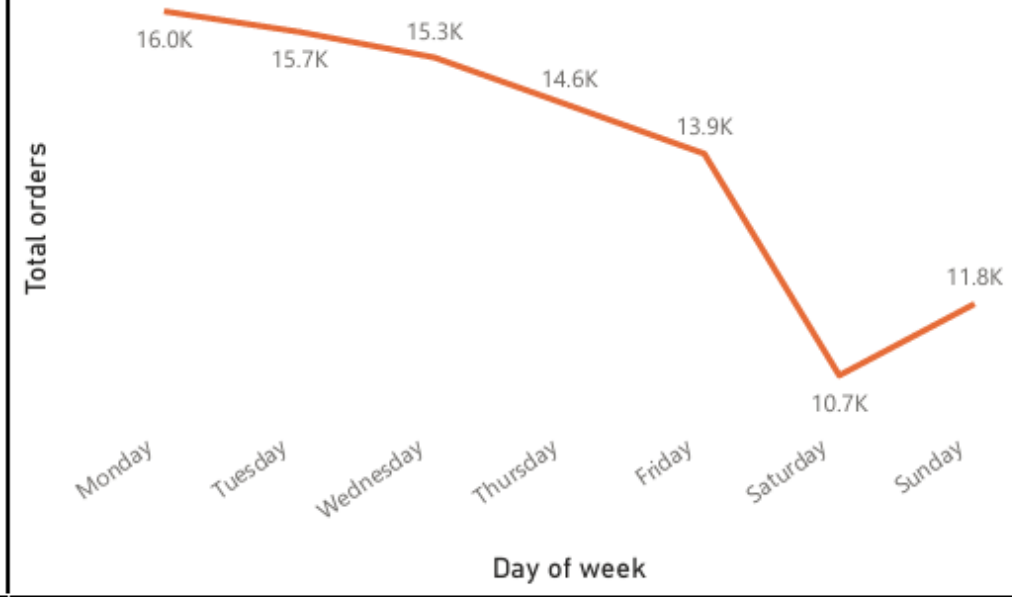
weekend flag
- weekday
- weekend

Timely delivered status
- No
- Yes

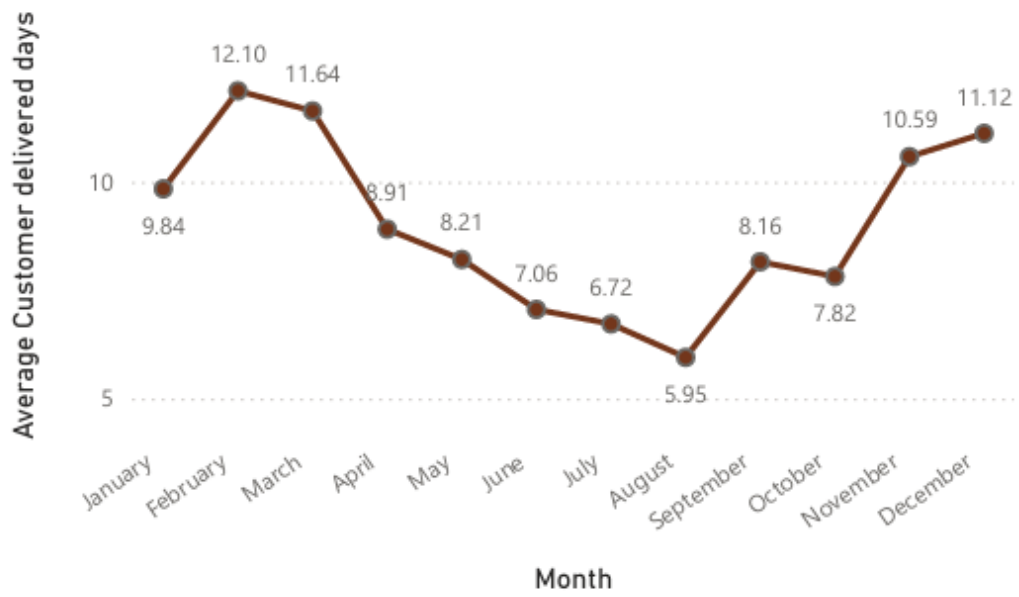
Total orders by Month



Total orders by Day of week



Average Customer delivered days by Month



Average delivered days by Year and Month



Product level Dashboard

The dashboard shows report at Products level. It consists of the following:

- **Relevant KPIs:** Total products, Total product categories, Average product price & freight value
- **Dynamic slicers:** Months
- **Relevant charts:** Average price by product category, Revenue generated by product categories, Total orders per category, Category penetration & Correlation between freight value and product price.
- **Insights shown:** High value products and categories along with which products contribute more to orders and revenue. Category penetration showing percentage contribution of each category to sales and revenue. Also shows products to be bundled together considering that such products have high freight value to price ratio.

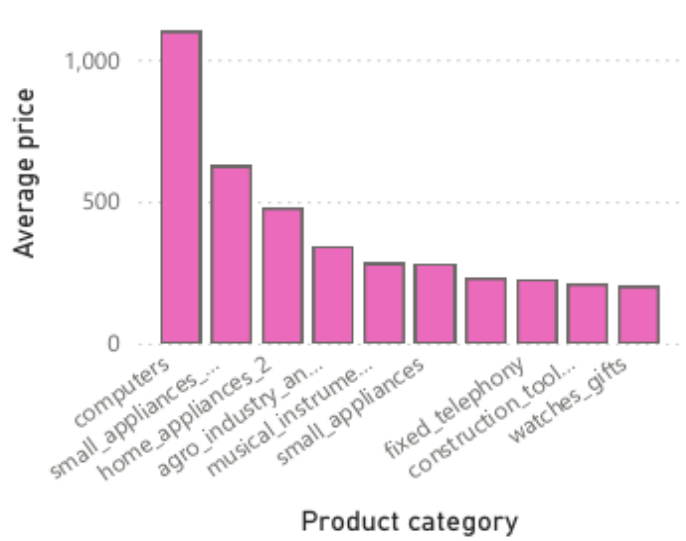
olist Product Analysis & Insights

Clear all slicers

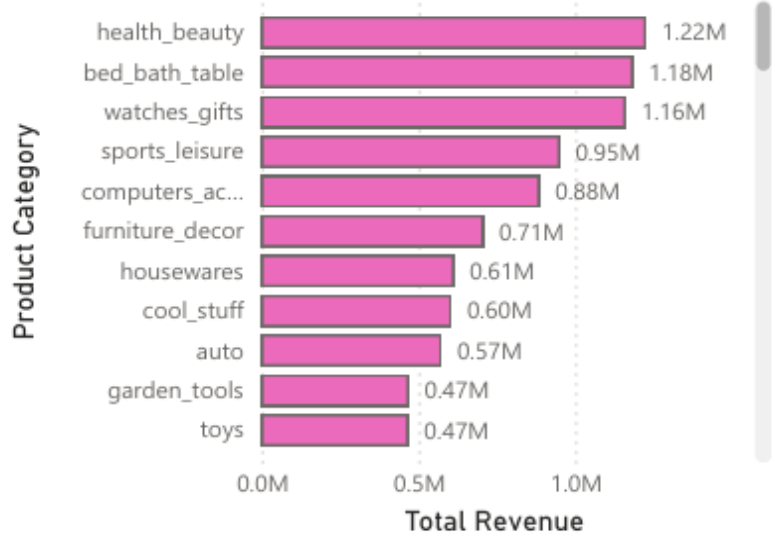
month
- April
- August
- December
- February
- January
- July
- June
- March
- May
- November
- October
- September

32.95K	73	120.65	19.99
Total Products	Total product categories	Average product price	Average of freight_value

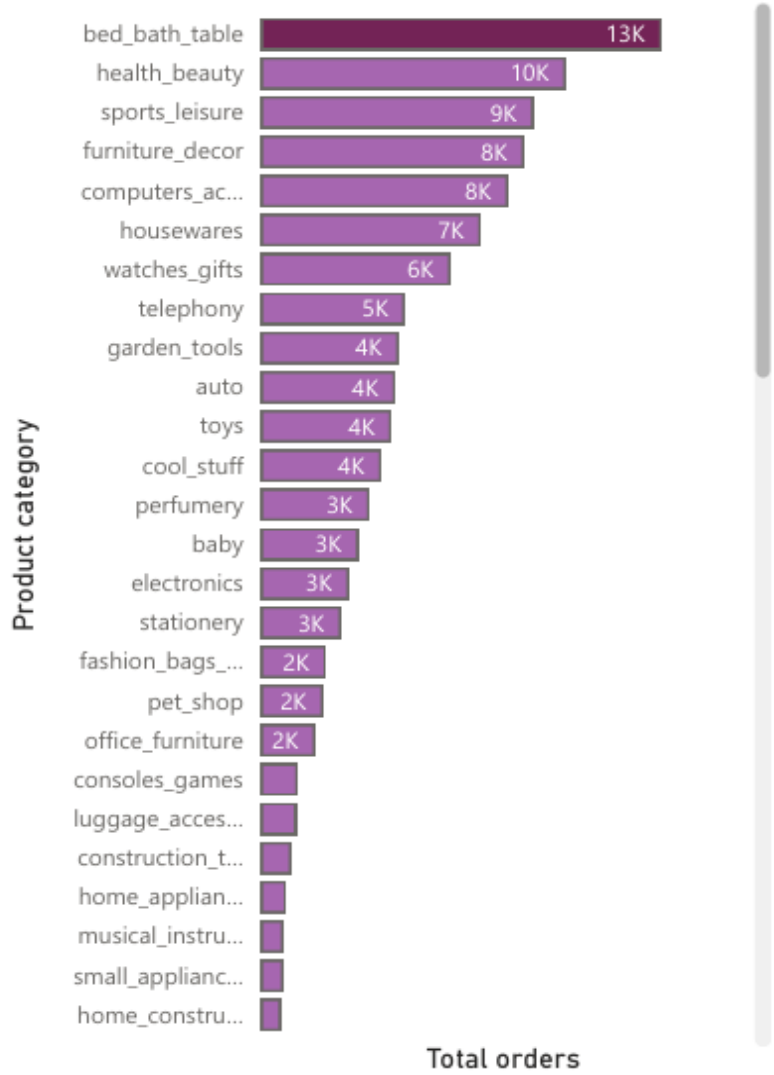
Average price by category (Top 10)



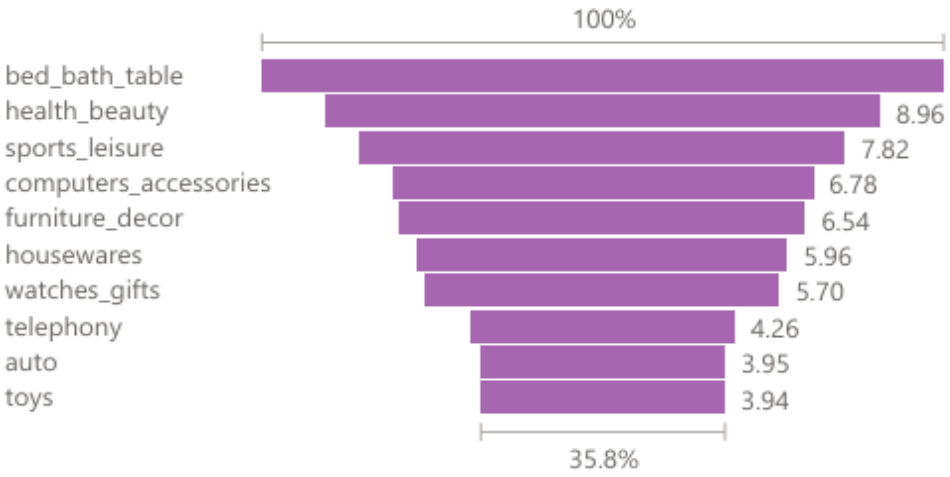
Total Revenue by Product Category



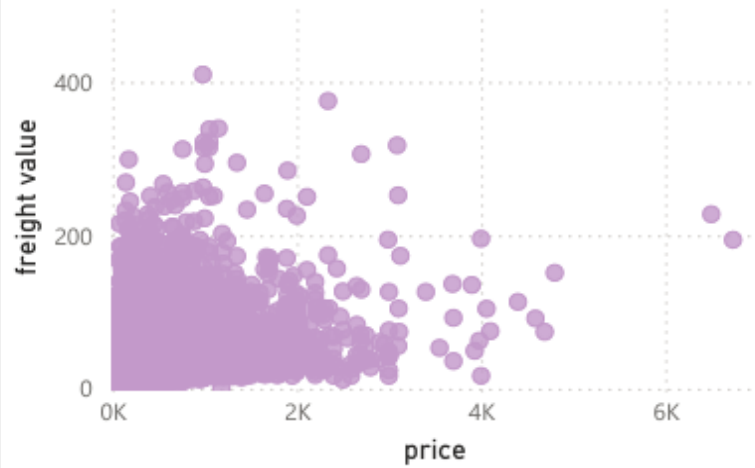
Total orders by Product category



Category Penetration (Top 10 Products)



price and freight value correlation



Seller level Dashboard

The dashboard shows report at Sellers level. It consists of the following:

- **Relevant KPIs:** Total sellers, Median carrier delivery days, Median customer delivery days & Median approval hours
- **Dynamic slicers:** Seller cities and Seller states
- **Relevant charts:** Worst performing sellers by average delivery time, Total orders received and total sellers available per state, Average delivery days by sellers per state & Sellers diversity in terms of unique products and categories they fulfil.
- **Insights shown:** Shows seller performance and number of sellers per location to isolate states that need more sellers and which needs more visibility at the marketplace.

olist Seller Performance Analysis

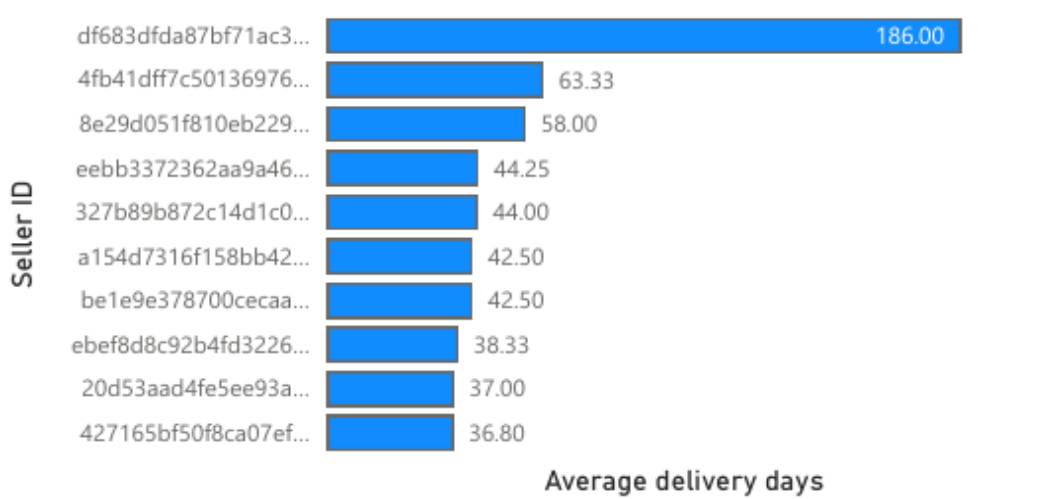
Clear all slicers

Cities
- abadia de goias
- afonso claudio
- aguas claras df
- alambari
- alfenas
- almirante tamand...
- alvares machado
- alvorada
- americana
- amparo
- ampere
- anapolis

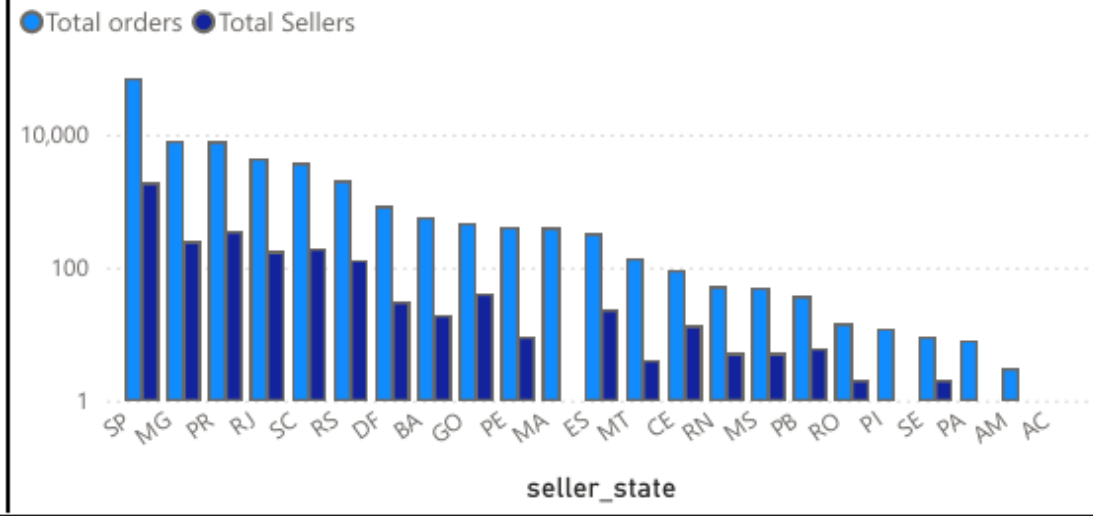
States
- AC
- AM
- BA
- CE
- DF
- ES
- GO
- MA
- MG

3095	1	7	0.33
Total sellers	Median carrier delivery days	Median delivery days	Median approval hours

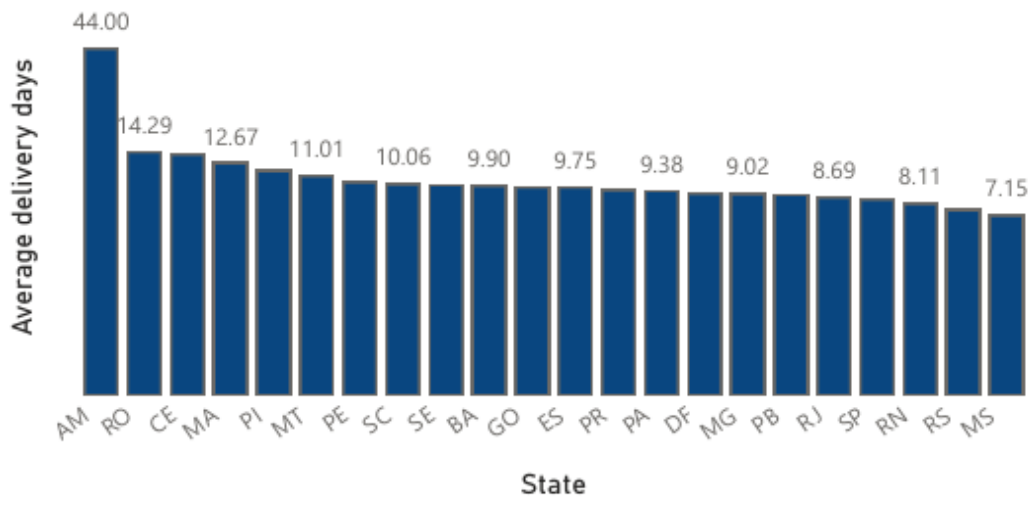
Top 10 worst performning Sellers



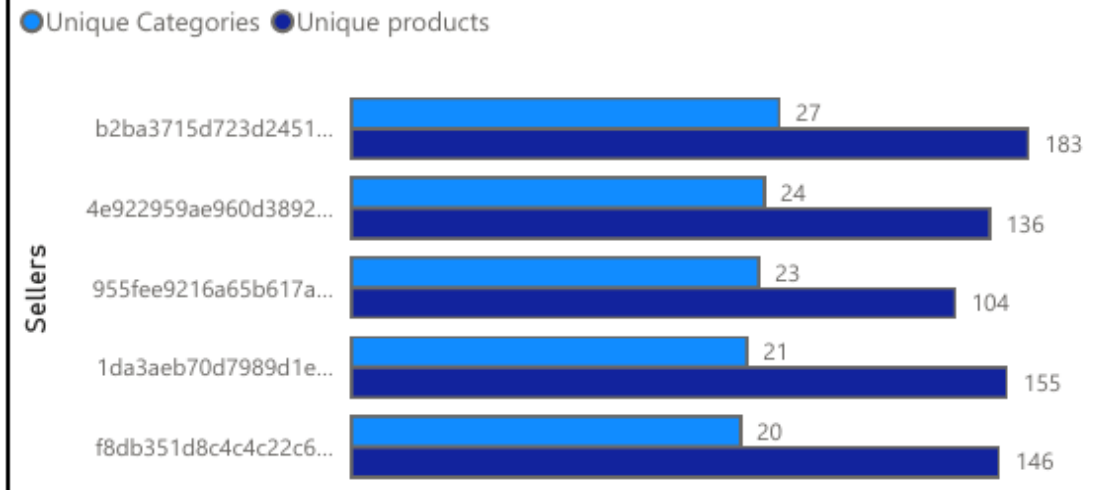
Total orders and Total Sellers by seller_state



Average delivery days by State



Seller product diversity (Top 5)



Customer Satisfaction Dashboard

The dashboard shows report of Customer Satisfaction. It consists of the following:

- **Relevant KPIs:** Total reviews, Average review score, Review time in days.
- **Dynamic slicers:** Order timely delivered status, Cities
- **Relevant charts:** Monthly average review score, Average review score by state, Product category wise average rating, Seller distribution by average rating, Total orders per rating score.
- **Insights shown:** Provides analysis about how products and seller performance affect customer reviews and which products and sellers per state require better checks or rewards. Also analyses which demographic sectors, sellers & products are better or worst performing.

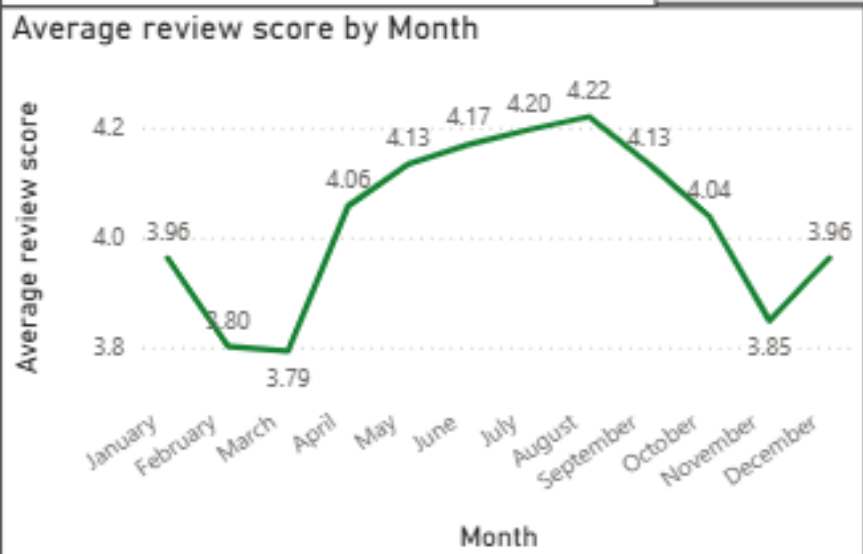
olist Customer Satisfaction Analysis

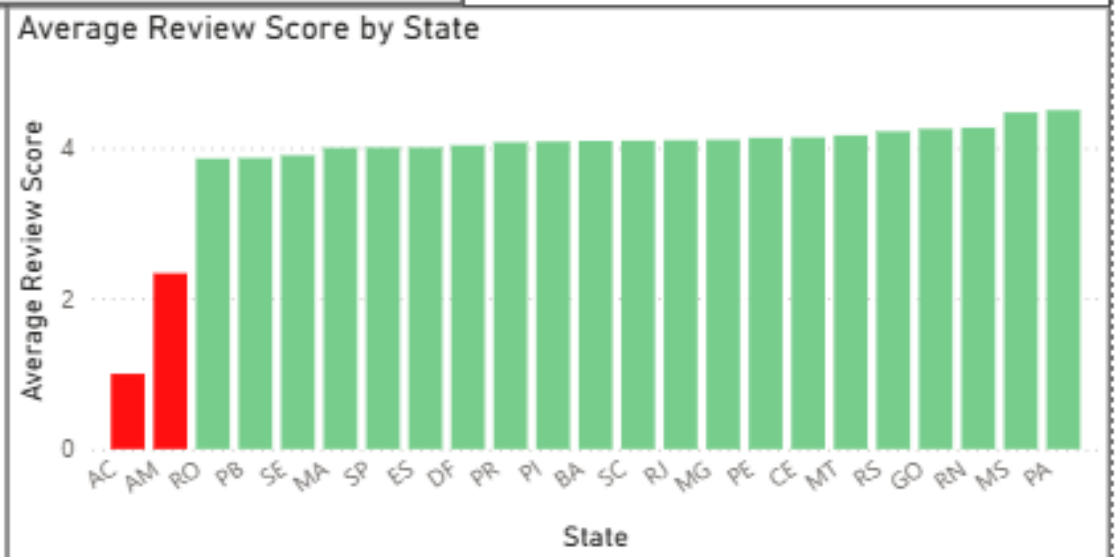
Clear all slicers

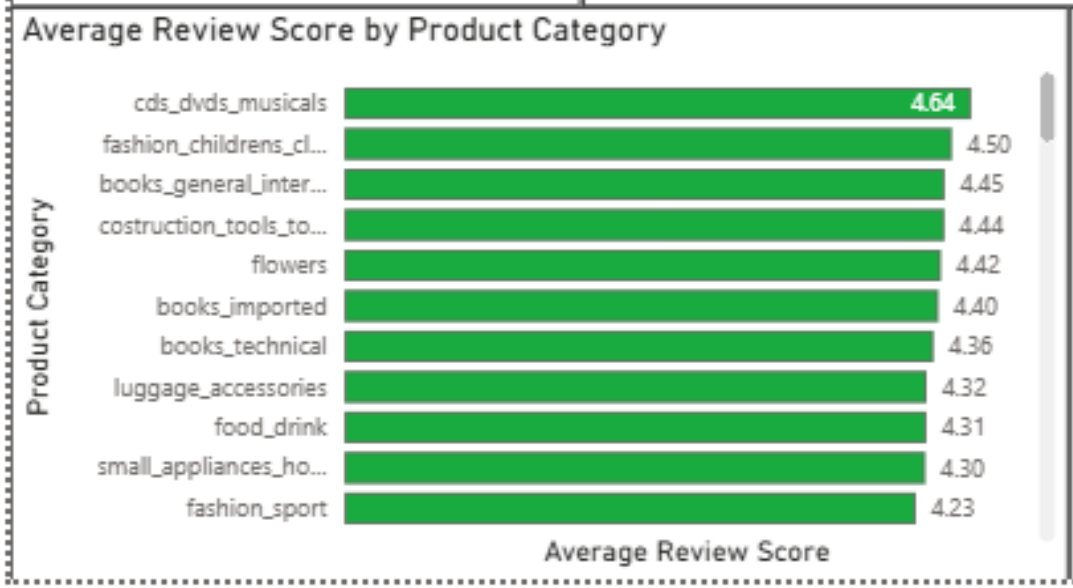
Order timely delivered?
- No
- Yes

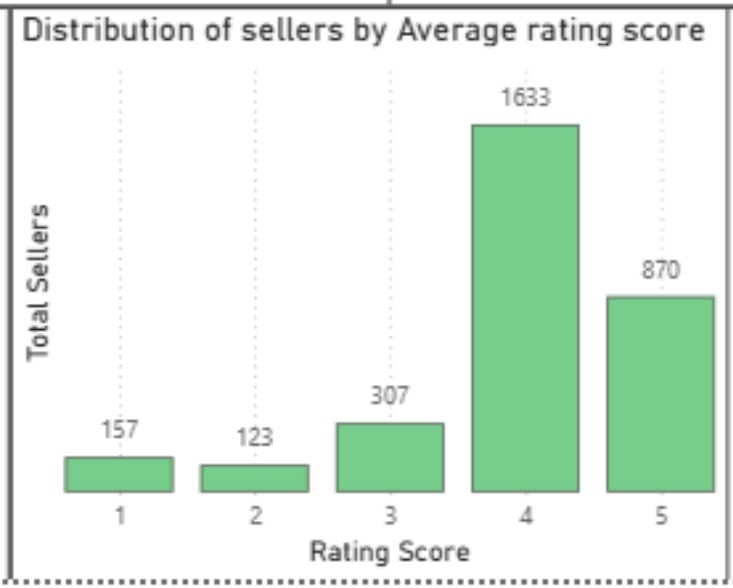
97.40K	4	3.22
Total Reviews	Average Review Score	Average Response time

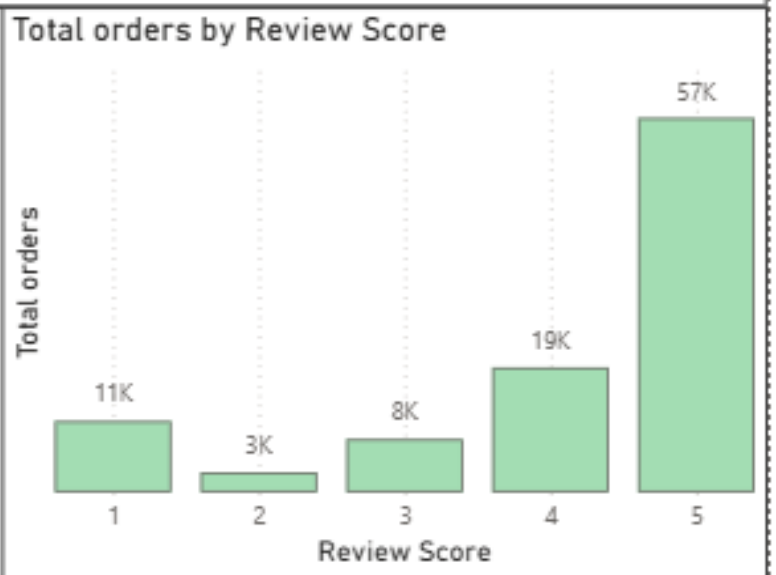
Cities
- abadia de goias
- afonso claudio
- aguas claras df
- alambari
- alfenas
- almirante tamandare
- alvares machado
- alvorada
- americana













Predictive Modelling- Customer Rating Prediction

Predictive Modelling

- **Goal:** Predict customer satisfaction rating from 1 to 5 with 5 being the highest rating showing high customer satisfaction and 1 showing low customer satisfaction.
- **Models Tested:** Logistic Regression, AdaBoost Classifier, Decision Tree Classifier. Random Forest Classifier, Gradient Boost classifier, Multinomial Naïve Bayes.
- **Variable reduction:** Variance inflation factor & Recursive feature elimination.
- **Evaluation metrics:** Accuracy, Precision, F1 score, Recall
- **Highest Accuracy model:** Logistic Regression using Recursive feature elimination.
- **Highest accuracy observed:** 61%

Predictive Modelling

	precision	recall	f1-score	support
1	0.61	0.32	0.42	8266
2	0.00	0.00	0.00	2558
3	0.00	0.00	0.00	6649
4	0.00	0.00	0.00	15503
5	0.61	0.99	0.76	46612
accuracy			0.61	79588
macro avg	0.24	0.26	0.24	79588
weighted avg	0.42	0.61	0.49	79588

Classification Report (Train)

	precision	recall	f1-score	support
1	0.60	0.31	0.41	2066
2	0.00	0.00	0.00	640
3	0.00	0.00	0.00	1662
4	0.00	0.00	0.00	3876
5	0.61	0.99	0.76	11654
accuracy			0.61	19898
macro avg	0.24	0.26	0.23	19898
weighted avg	0.42	0.61	0.49	19898

Classification Report (Test)

Observations:

- Model has low accuracy of 61% and performs poorly in other metrics.
- Reason for that is high class imbalance. Majority of ratings are 5 followed by 1.
- Even with imbalance techniques like SMOTE, the model did not improve with current features.



Final Observations

Customer level

- New customers acquisition showed a positive trend for the majority of analysis but slowed down around August 2018.
- One time buyers were responsible for most of the orders and maximum revenue generated.
- Certain states had more customers who ordered and majority of states had very less number of customers. Same with the cities.
- RFM segmentation showed that majority of customers belong to the silver category and there were far less customers in gold and platinum.
- Most of the customers belong to the low revenue generating segment.
- Customers prefer to pay using credit cards and majority pay with only one method while few paid with vouchers and other payment methods.

Orders level

- Over 73% of orders were paid through credit card and least is debit card.
- Weekday sales contribute to majority of the sales and revenue whereas weekend sales has very less contribution.
- Over 89% orders were timely approved, processed and delivered. Certain states are performing bad in terms of order volume and delivery times. Certain orders in cities have average delivery time above 3 months.
- High volume of orders are received during 10AM and 11PM which marks this time as high productive time for increased revenue.
- Average order delivery time increased substantially in Q4. Christmas, Halloween and New year time shows increased order volume which is likely the reason for extended delivery days.
- Certain cities and states are performing worst while handling order volume and the lowest revenue generators.

Products level

- Certain orders perform well compared to others in terms of order volume and revenue. They hold the majority of market share in the entire dataset.
- A similar pattern was observed across state, regions and product categories.
- Probable Reason: Variations in store management, location demographics, or product availability could influence these differences.
- Category penetration was observed to be higher for a few categories while rest had low category penetration
- Pareto analysis showed that roughly 20% of categories contribute to nearly 78% of all orders.
- Correlation analysis between fright price and product price showed that certain products despite having less price had more freight value. Considering to bundle such orders is recommended.

Seller level

- Out of 3095 sellers, a few of them were performing really bad in terms of order volume and average delivery days with few of them having average delivery days in months.
- Seller diversity in terms of product categories is low for majority of sellers with only a few of them selling more than 5 unique products/product categories.
- Certain states have low number of sellers despite handling high order value meanwhile few states have less number of sellers altogether.
- Few sellers performed worst in terms of order delivery times and revenue generated with sellers from SP outperforming the rest.

Customer satisfaction level

- Majority of states performed well in terms of review rating while a small fraction performed worst in terms of average rating.
- Majority of orders were rated 5, followed by rating 4 and a few intermediatory 2 and 3 rating. Review score 1 has substantial number.
- Certain sellers performed better than others and received higher rating than those sellers.
- Certain product categories receive consistent poor reviews and others receive consistent higher review scores.

Final recommendations (1/3)

For customer retention:

- Retail chain can try personalized marketing strategies to retain customers. They could collect Email address of the customer if they are willing to provide. Thus, retail chain can remind them of any new products or upcoming offers.
- Loyalty rewards can be given for retained customers like coupons, discounts etc.
- Credit points can be given for each purchase and when customer reaches a certain threshold, they can get selected products for free

For less popular categories:

- For categories with fluctuating penetration rates, run targeted promotions or bundle these with top-selling categories to increase their appeal. Offer discounts or perks for purchasing these bundles.
- Stock Optimization: Ensure that frequently cross sold items are well-stocked to meet customer demand and avoid stockouts.
- Warehouse Layout: Arrange the warehouse layout to place frequently cross-sold items closer together to streamline the picking process.

Final recommendations (2/3)

For low sales

- Enhance Weekend Sales: Since weekend sales are lower, consider implementing special weekend promotions or discounts to attract more customers.
- Target low revenue customers and customers with low RFM scores by offering personalised offers and bundles.
- Explore activities or events that can drive more traffic to your business during weekends.
- Analyse the factors contributing to the sharp decline in sales and sudden growth spurt.
- Consider external factors like economic conditions, industry trends, or competitive pressures. Understanding these factors can help to enhance future growth.

For underperforming products

- Bundle underperforming products together and have a clearance sales on specific times when sales are lowest.
- Offer higher profit margin to sellers for such products.

Final recommendations (3/3)

For poor performance of sellers

- Establish localized regional fulfilment hubs per state where demand is more than what the sellers can handle. It will reduce the distance between customers and sellers and ensure higher delivery speed.
- Automated warning flags should be there to penalize seller which are performing less than the SLA compliance by lowering their visibility. Meanwhile high performing sellers should be rewarded with more visibility.
- Create an audit report for such underperforming sellers and offer them implementation practices from high performing sellers.

For better customer satisfaction

- Analyze review score and comments in comparison to customer 360 and understand if reviews are from genuine customers or bots.
- Understand competitions and implement their strategy for better customer satisfaction.
- Send a follow up survey on email or phone number of low rating customers and understand the reason in depth. Reward them for taking this survey by offers for next orders.
- States having consistent lower rating should be audited and the sellers and logistics hub there should be kept in check.

THANK YOU!